

SECTION 3.0

OTOLARYNGOLOGY—HEAD AND NECK SURGERY

SURGEONS

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INTRODUCTION—SURGEON'S PERSPECTIVE

Interdisciplinary cooperation is nowhere more important than where the priority of a secure and sustainable airway maintenance is confronted with adjacent surgical goals. An anesthesiologist skilled in the management of the difficult airway and with an ability to anticipate the issues confronting the surgeon is critical. Similarly, a communicative surgeon fully aware of the problems the anesthesiologist is likely to encounter is essential to minimizing complications. Important issues that require mutual understanding include the following:

1. Airway management: anesthesia induction, endotracheal tube (ETT) size, head range of motion with shared airway space, a well-secured ETT, postop airway edema, and smooth emergence from anesthesia
2. Muscle relaxation, patient immobility, patient positioning
3. Deliberate relative hypotensive anesthesia and management of bradycardia
4. Antiemetics and pain management postop
5. Patient population considerations: frequent cardiac and pulmonary issues associated with older patients, especially with alcohol and tobacco use; degenerative C-spine issues; and prior irradiation affecting neck mobility.

Airway management

An initially compromised airway is not uncommon in many otolaryngology head and neck procedures. Many others may develop airway loss at induction or if premature extubation occurs. Communication between the surgeon and anesthesiologist is essential, as is a discussion of a plan and backup plan should an emergency arise. Availability of a sliding Jackson scope and tracheotomy equipment, as well as plans for fiberoptic intubation, awake intubation, or retrograde intubation, should be discussed as indicated.

For procedures within the airway, an ETT no larger than 6 mm should be adequate and will reduce postop airway edema. An armored tube is helpful when the surgical procedure is intraoral and the tube may be compressed. A nasotracheal intubation should be discussed as an alternative in this situation. If laser surgery is planned, maintaining the FiO_2 at <0.3 is indicated. If fiberoptic bronchoscopy is planned, then a 7.0- or 7.5-mm ETT will facilitate passage of the bronchoscope through the ETT, after which for a longer case, changing to a smaller tube may be warranted.

As the patient is generally turned 90° or 180° away from the anesthesiologist, a very secure airway is important. If the surgeon needs access in the mouth, securing the tube

via a wire to several teeth may work better than tape.

Muscle relaxation and patient positioning

Avoidance of muscle relaxation is important if a motor nerve, such as the facial nerve, is to be dissected. Muscle relaxation is important, on the other hand, in esophagoscopy and tongue surgery. Communicate and coordinate timing of drug administration. Frequently, the surgeon will need to turn the patient's head during surgery. Anticipating this movement when initially securing the ETT and its connections will prevent disconnection. In neck surgery, the neck is often rotated away from the surgeon; overrotation presents the risk of brachial plexus stretch injuries. If a radial free flap is anticipated, then positioning of the arm as well as rotation of the head should be carefully coordinated to avoid injury while still providing needed access and a secure airway.

Deliberate relative hypotensive anesthesia, to a systolic blood pressure (SBP) of <100 mm Hg, depending on individual patient's needs, is often a boon to the surgeon operating in the highly vascular fields of the head and neck or operating with a microscope. For selected cases, the patient also will have had preop embolization of a tumor and its blood supply (e.g., angiofibromas and skull base paragangliomas). **Bradycardia** may occur if the surgeon operates near the vagus nerve or carotid bifurcation. If this occurs, it is usually sufficient for the anesthesiologist to communicate this, and the surgeon can desist for a period of time. Occasionally, topical or locally injected anesthetics will be necessary.

INTRODUCTION—ANESTHESIOLOGIST'S PERSPECTIVE

PATIENT POPULATION

Many ENT patients are smokers and elderly and have $\uparrow$ incidence of coronary artery disease (CAD), hypertension (HTN), chronic renal insufficiency (CRI), and chronic obstructive pulmonary disease (COPD). Careful H&P must be performed to ensure that the patient's functional status is optimized. In patients with significant CAD, HTN and/or associated CRI, controlled hypotension techniques should be avoided. Chronic cigarette smoking and alcohol use also cause an induction of cytochrome P450 multienzyme system $\rightarrow \uparrow$ opioids and $\uparrow$ neuromuscular blocker (NMB) requirements.

AIRWAY

ENT patients often present with H&P findings associated with a difficult airway including (a) anatomic characteristics (e.g., ↓ C-spine ROM, large tongue, receding jaw), (b) Hx of stridor and hoarseness (airway narrowing and possible vocal cord [VC] dysfunction), (c) previous Hx of neck surgery, trauma, or x-ray therapy (XRT) (↓ compliance of the tissues, ↓ neck ROM, ↓ mouth opening), (d) previous Hx of difficult intubation, and (e) infections (e.g., epiglottitis, retropharyngeal abscess, Ludwig angina).

Meticulous examination of the airway must be performed, and there should be a low threshold for an awake intubation if the airway is questionable. The awake versus asleep approach to tracheal intubation should be guided by the presence of predictors of difficult or impossible mask ventilation and their association with difficult direct laryngoscopy (DL) (Tables 3.1 and 3.2). Oxygenation-centered approach to difficult airway management is paramount, and the use of the high-flow nasal oxygen (transnasal humidified rapid-insufflation ventilatory exchange, THRIVE) should be strongly considered when potentially difficult airways are encountered.

Table 3.1 Predictors of Difficult and Impossible Mask Ventilation

Difficult Mask Ventilation	Impossible Mask Ventilation
Mallampati grade 3–4 Decreased mandibular protrusion Presence of beard Obesity (BMI ≥ 30) Age > 57 yo Lack of teeth History of snoring	Mallampati grade 3–4 Male sex Presence of beard OSA* Neck radiation changes

At least three predictors should be present in any category for a reasonably high probability of the event.

*Moderate-to-severe obstructive sleep apnea (OSA), associated with the use of CPAP/BiPAP or a history of OSA surgery.

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Table 3.2 Predictors of Combined Difficulty with Mask Ventilation and Direct Laryngoscopy

Predictors of Difficult/Impossible Mask Ventilation	Additional and Modified Predictors
Mallampati grade 3–4 Decreased mandibular protrusion Presence of beard Obesity (BMI ≥ 30) Male sex OSA*	Age ≥ 46 yo Presence of teeth Neck radiation changes or a neck mass Thick or obese neck Unstable neck or decreased neck extension Decreased thyromental distance

A total of at least five predictors should be present for a reasonably high probability of both, difficult mask ventilation (DMV) and difficult direct laryngoscopy (DL). Class III patients (five risk factors) constitute 5% of all DMV+DL cases, and patients from class IV (six risk factors) and class V (seven to eleven risk factors) constitute the largest group (85%) of these cases.

*Moderate-to-severe obstructive sleep apnea (OSA), associated with the use of CPAP/BiPAP or a history of OSA surgery.

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A team approach to an anticipated difficult airway is essential. Thorough understanding of upper airway anatomy, concurrent pathology, and all steps of the surgical procedure are necessary for devising a safe and rational perioperative airway management strategy. Video laryngoscopy (e.g., GlideScope™) usually improves airway visualization and should be considered for the initial approach to tracheal intubation when difficulty is anticipated. The LMA-Fastrach™ provides rescue ventilation with a 95–100% success rate in difficult airway situations. If a conventional DL was chosen as a first approach, the patient's head positioning should be optimized, and simple intubating aids (e.g., stylets, gum elastic bougie) as well as the ETTs of different sizes made readily available. The number of DL intubation attempts should be limited to two; repeated intubating attempts are associated with increased incidence of airway trauma, unanticipated ICU admission, and “cannot ventilate-cannot intubate” events.

For most ENT surgical procedures, the patient's airway is shared with the surgeon, and immediate access to the airway may be difficult or impossible. The ETT must be secured diligently to prevent an accidental extubation or pullout of the ETT to the subglottic area, which may result in exerting direct pressure on the recurrent laryngeal nerves by the ETT cuff. For certain procedures (e.g., excision of tumors of the base of the tongue [BOT]), nasal intubation may be required to facilitate surgical access, and

the surgeon should be consulted before induction. Specialized types of ETTs are required for laser surgery of the airway and for surgeries requiring monitoring of the recurrent laryngeal nerve function (e.g., thyroidectomy).

ANESTHETIC MANAGEMENT

Some of the essential anesthesia management requirements for ENT surgery include (a) assurance of good intraop and postop analgesia (most of the procedures are performed on highly reflexogenic areas); (b) quiet surgical field (the operated areas are highly vascular), and moderate controlled hypotension is widely employed: $\downarrow$ SBP < 100 mm Hg, MAP = 60–70 mm Hg, unless contraindicated; (c) immobility (for certain procedures profound muscle relaxation may be required, while for others administration of NMB must be avoided); and (d) smooth emergence from anesthesia. The latter constitutes one of the most challenging tasks, as every attempt should be made to avoid or minimize the patient's reaction to the ETT during tracheal extubation. Straining, bucking, or coughing may provoke early postop bleeding ($\uparrow$ venous and arterial pressure), disrupt delicate suture lines (e.g., facial nerve repair), dislodge surgical grafts (e.g., tympanoplasty), or cause additional trauma to the vocal cords after VC surgery. Many ENT surgeons have their own "list" of the procedures where the patient's reaction to the ETT should be completely avoided during extubation, and an anesthesiologist should be familiar with the surgeon's preferences.

Opioid-based anesthetic techniques are especially advantageous and result in significant $\downarrow$ monitored anesthesia care (MAC) while providing sufficient blunting of the tracheal responses to the in situ ETT. In an opioid-naïve patient, the choice of an opioid analgesic depends primarily on the anticipated degree of surgical stimulation and postop pain, duration of surgery, and coexisting medical conditions. High-dose opioids (fentanyl, loading dose 3–10 mcg/kg iv; sufentanil, loading dose 0.5–1.5 mcg/kg iv), followed by either intermittent boluses or continuous infusion, are the author's preferred choice for major ENT surgery. For procedures that may be highly stimulating, but associated with minimal postop discomfort (e.g., laser surgery of the airway), a short-acting opioid is preferred: remifentanyl (loading dose 0.5–1.0 mcg/kg iv, infusion 0.1–0.3 mcg/kg/min iv).

Total intravenous anesthesia (TIVA) is commonly employed, especially for endoscopic procedures. Most studies demonstrate that, compared to inhalational and balanced inhalational techniques, TIVA offers improved hemodynamic stability, quicker recovery times, decreased incidence of postoperative nausea and vomiting (PONV), and

improved patient satisfaction.

Maintenance of moderate **controlled hypotension** (MAP, 60–70 mm Hg) is useful for maintaining optimal operating conditions. A variety of pharmacologic approaches have been successfully employed for this purpose. The use of remifentanyl is particularly effective in that regard and is equally useful with both inhalational and TIVA techniques. The use of esmolol 0.5–1 mg/kg iv bolus, followed by continuous iv infusion 100–300 mcg/kg/min, represents another attractive therapeutic option.

GETA is most widely employed. Use of a **laryngeal mask airway** (LMA) for ENT surgery, when feasible, is beneficial (↓ coughing, ↓ laryngospasm; see [Table 3.3](#)). Furthermore, the use of an LMA does not require NMB and facilitates the resumption of spontaneous ventilation. When used appropriately in selected patients, absence of immediate access to the patient’s airway is not a deviation from the standard of care.

A flexible laryngeal mask airway (FLMA), but not an LMA Classic™, LMA ProSeal™, or LMA Supreme™, is most frequently used in ENT anesthesia, because its shaft can be bent away from the surgical field without causing cuff displacement. Close communication between the surgeon and anesthesiologist is essential, as even a properly placed FLMA can become dislodged during surgical manipulations. The LMA ProSeal/LMA Supreme is preferred whenever central positioning of the LMA can be safely maintained (e.g., otologic surgery). Proper LMA insertion technique is crucial for minimizing the risk of aspiration or gastric insufflation, especially if positive pressure ventilation (PPV) is planned. An increased number of FLMA insertion attempts and low airway seal pressure (ASP) constitute independent risk factors for FLMA failure. We recommend selecting the largest FLMA size suitable for the patient to improve ASP and minimize the air leak during PPV. We also recommend performing tracheal intubation if adequate ASP (≥ 12 cm H₂O in patients without associated gastroesophageal reflux disease [GERD] and ≥ 15 cm H₂O in patients with GERD) cannot be achieved within three FLMA insertion attempts. Gastric insufflation is further minimized by ↓ Vt to 6–10 mL/kg and by keeping PIP < 20 cm H₂O. The absence of gastric insufflation during PPV when LMA is used should be documented in the anesthesia record after auscultating the epigastric area.

Table 3.3 Advantages of the LMA Compared with the ETT			
Adverse Event ETT/LMA	ETT %	LMA %	Ratio
Clinically significant problems	3.4	0.9	3.8

Laryngeal spasm	0.38	0.12	3.2
Aspiration	0.017	0.02	0.85
Sore throat	50	10	5
Laryngeal trauma	6.2	<1	>6
Coughing on emergence	60	2	30

Modified from Brimacombe JR, Brain AJ: The Laryngeal Mask Airway. A Review and Practical Guide, 1st edition. WB Saunders, Philadelphia: 1997. Copyright © 1997 Elsevier. With permission.

Smooth emergence from anesthesia constitutes a particularly challenging task when GETA is used. Deeper stages of anesthesia are usually required until the very end of the procedure to blunt laryngotracheal responses. A low-dose remifentanyl infusion to blunt the tracheal responses and promote smooth extubation may be helpful. Current data indicate that EC₉₅ of effect site concentration of remifentanyl for blunting tracheal reflexes after balanced desflurane or sevoflurane anesthesia is 2.3–2.9 ng/mL, which corresponds to an infusion rate of 0.08–1.0 mcg/kg/min. This is higher than that observed with TIVA, where a target concentration of 1.5–2.1 ng/mL, corresponding to an infusion rate of 0.05–0.07 mcg/kg/min is usually sufficient.

Antiemetic prophylaxis should be routine and is most commonly achieved by iv administration of a 5-HT₃ blocker. This is further facilitated by 4–8 mg dexamethasone iv, which is commonly requested by the surgeon to reduce postop airway edema. Multimodal antiemetic prophylaxis should be employed for patients at high risk for PONV. The addition of metoclopramide (10–20 mg iv) may be beneficial for the patients who had undergone the procedures resulting in accumulation of the passively swallowed blood in the stomach (e.g., nasal or intraoral surgery).

Suggested Readings

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LARYNGOSCOPY/BRONCHOSCOPY/ESOPHAGOSCOPY

SURGICAL CONSIDERATIONS

Description

Laryngoscopy is used for inspection of the pharynx, hypopharynx, or larynx for diagnostic and/or therapeutic benefit. The patient is supine with cervical spine flexed and atlantoaxial joint extended (this position is best achieved with a headrest), and the teeth are protected with a mouth guard. The laryngoscope is introduced ([Fig. 3.1](#)); then, with a lifting motion, a thorough examination of the oropharynx, hypopharynx, and larynx is carried out, with biopsies if indicated. Bleeding typically is minor, easily controlled with cotton pledgets soaked in epinephrine or cautery. Laryngoscopy often is combined with flexible esophagoscopy and bronchoscopy to survey the aerodigestive tract for malignancy. If the procedure is diagnostic, the surgeon may need to see the airway before intubation and/or muscle relaxation. If a laser is to be used, a special

laser ETT, <30% FiO₂, and avoidance of N₂O are required.

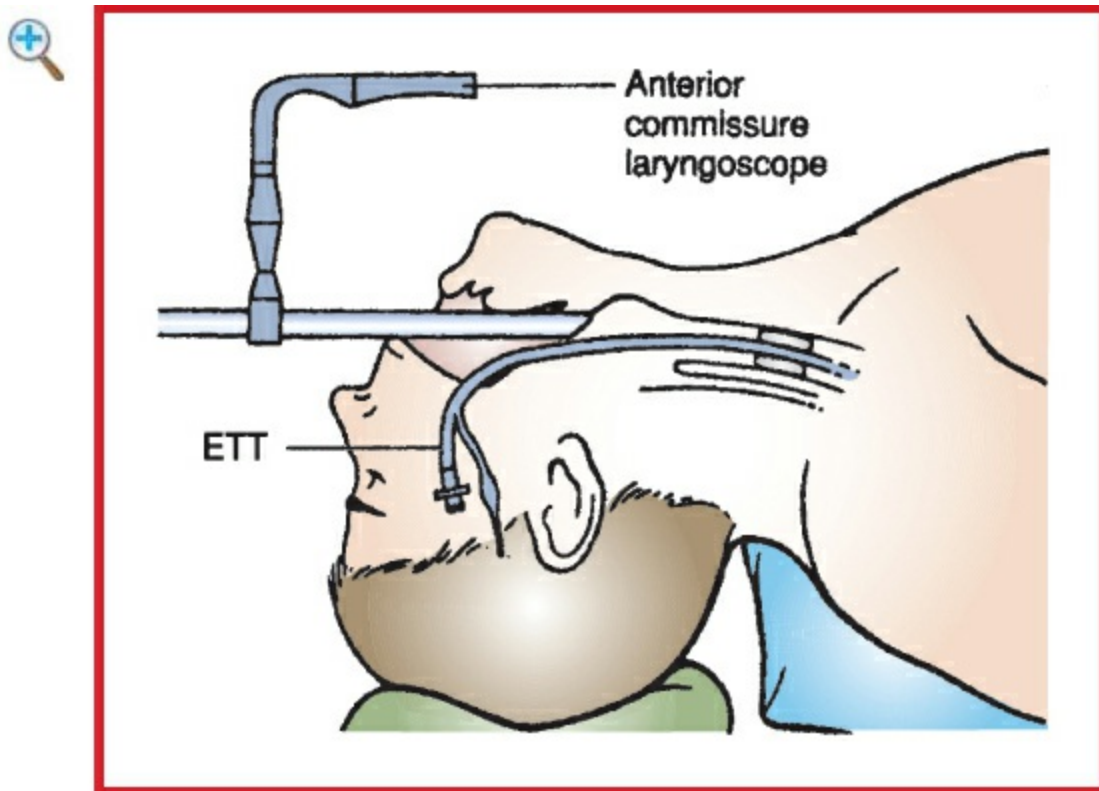


Figure 3.1 Placement of anterior commissure laryngoscope for laryngoscopy.

Usual preop diagnosis

Oropharyngeal, hypopharyngeal, or laryngeal tumors

Description

Operative microlaryngoscopy. Many laryngeal lesions (such as papilloma, cysts, polyps, selected malignancies) can be removed endoscopically. Specialized endoscopes are deployed transorally to allow the surgeon a binocular view of the vocal fold and the target lesion. They can be suspended from a Mayo stand to free both of the surgeon's hands for operating. Using the microscope, a variety of specialized endoscopic instruments, and lasers, the surgeon may be afforded excellent unobstructed access to the vocal folds for diagnostic and therapeutic purposes. Operative microlaryngoscopy may necessitate several hours of work. Because of the precision involved in such procedures and the high degree of stimulation to the patient, general anesthesia ($\pm$ jet ventilation) with muscle relaxation is required. Intubation with a small-caliber microlaryngeal or laser-safe tube (5 or 6 mm) may be required for these procedures. In cases where jet ventilation is to be performed, an endoscope suitable for this technique should be available. Intermittent apneic ventilation is also a possibility,

although this involves periodic interruption of surgery, which can be cumbersome and distracting.

Usual preop diagnosis

Vocal fold neoplasm; vocal fold paralysis

Description

Bronchoscopy inspects the tracheobronchial tree for diagnostic and therapeutic purposes. The patient is supine with head elevated and neck extended at the upper cervical level. The bronchoscope is directed along the right side of the tongue forward toward the midline to visualize the epiglottis. Next, the bronchoscope tip is used to lift the epiglottis and advance the bronchoscope through the VCs, into the trachea and bronchus (Fig. 3.2). With the aid of telescopes, the bronchoscope can be directed for inspection of the carina, main bronchi, and the segmental bronchi. Rigid bronchoscopes provide a large working channel through which to introduce grasping and biopsy forceps. As such, rigid bronchoscopes may provide a more stable platform for removal or retrieval of foreign bodies, tumors, and stents than flexible fiberoptic bronchoscopes.

Flexible fiberoptic bronchoscopy is more commonly performed than rigid bronchoscopy. The endoscope with biopsy channels is connected to a monitor, suction, and irrigation. Spontaneous ventilation can be maintained, which may allow the procedure to be performed without an ETT in place (or the bronchoscope may be introduced through a large enough ETT, such as a 7.5-mm one), affording unobstructed inspection of the upper airway. A bite block is placed to protect the endoscope from dental trauma and to facilitate advancement through the oropharynx into the larynx.

Usual preop diagnosis

Head and neck squamous cell carcinoma; foreign body (FB) in the bronchus

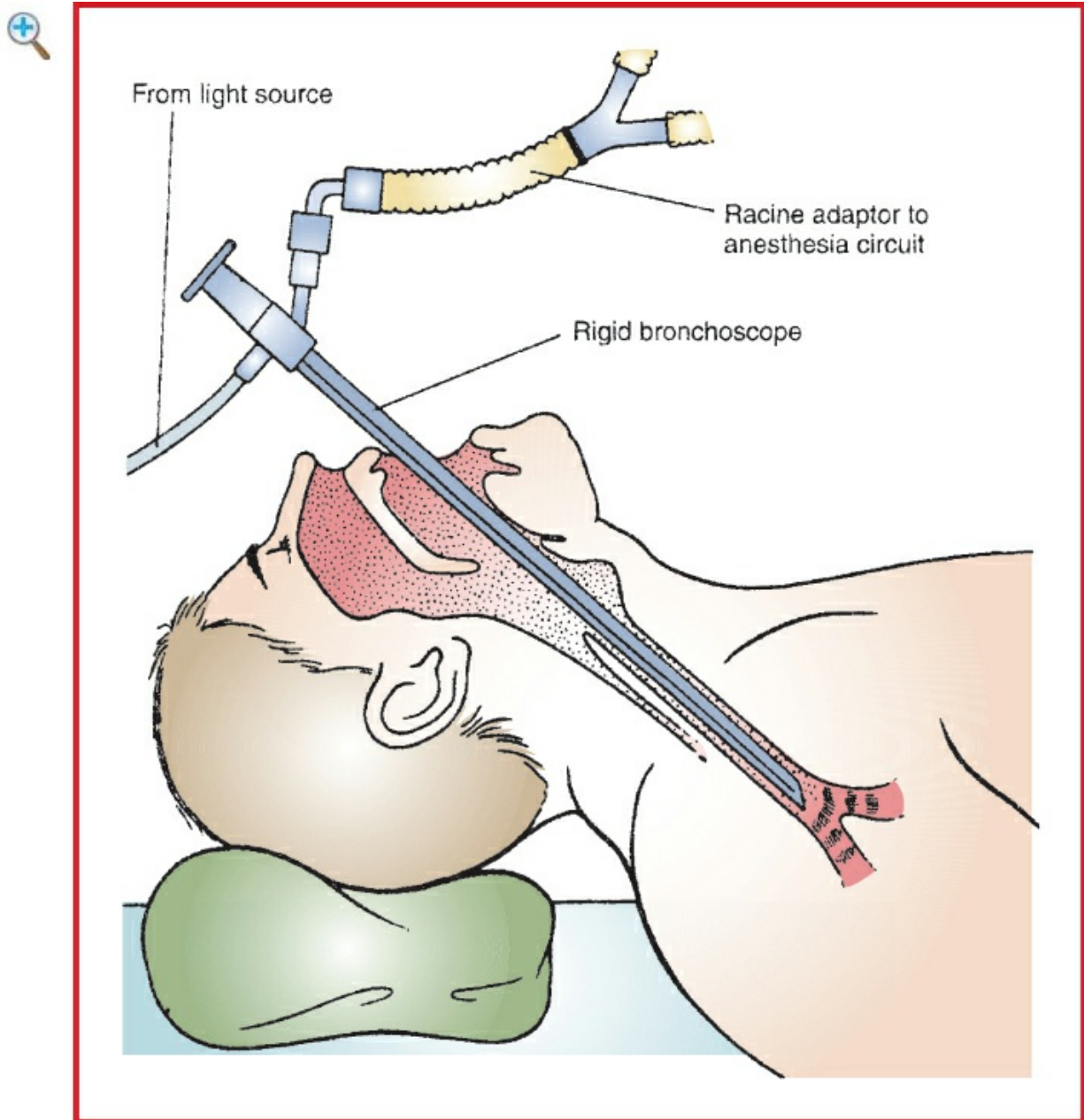


Figure 3.2 Rigid bronchoscopy showing adaptor (Racine) for anesthesia machine. Note neck flexion and head extension to align oropharyngeal and tracheal axes.

Description

Esophagoscopy inspects the esophagus for either diagnostic or therapeutic benefit. The patient is supine with head elevated and neck extended at the upper cervical level. The esophagoscope (held in the dominant hand) is advanced through the mouth behind the arytenoids, gently using the thumb of the nondominant hand. The bevel of the scope is then used to advance through the cricopharyngeal muscle (upper esophageal sphincter) with an upward lifting movement, entering the cervical esophagus. As the scope advances, the head may have to be lowered or the neck extended and the scope directed slightly toward the left. The scope is advanced to the gastroesophageal junction with

great care to ensure a visible lumen is seen at all times to avoid inadvertent perforation.

Flexible fiberoptic esophagoscopy today is far more common because of superior optics and is performed in an essentially identical manner. Biopsies may be taken and percutaneous gastrostomy tubes may be placed.

Usual preop diagnosis

Head and neck squamous cell carcinoma; FB ingestion

Description

Panendoscopy, or triple endoscopy, is the combination of laryngoscopy, bronchoscopy, and esophagoscopy. It is usually performed as part of the evaluation of patients with newly diagnosed cancer of the head and neck and may be done just prior to planned resection, assuming no unexpected findings. Goals of panendoscopy include (a) evaluating extent of a primary tumor and assessing resectability, (b) checking for synchronous second tumors in other locations within the upper aerodigestive tract, and (c) searching for the source of the primary lesion in patients who present with neck squamous cell carcinoma in whom a primary has not been identified by clinical examination or imaging including MRI and PET/CT. In the latter case, endoscopy may be supplemented with transoral robotic surgery (TORS) to resect the lingual tonsils or one or both palatine tonsils.

Usual preop diagnosis

Head and neck squamous cell carcinoma

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting nothing by mouth (NPO) duration and alcohol and smoking cessation. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

SUMMARY OF PROCEDURES

	Laryngoscopy	Bronchoscopy	Esophagoscopy	If Therapeutic Procedure Added
Position	Supine	⇐	⇐	⇐
Special instrumentation	Endoscopes, video monitors, suction	⇐	⇐	Laser, laser-safe ETT, gastro-scope and G-tube (abdominal prep)
Unique considerations	Dexamethasone 4–8 mg if airway edema	⇐	⇐	⇐ if laser ablation, keep O ₂ <30%
Antibiotics	None	None	None	Cefazolin 2 g and metronidazole 500 mg or clindamycin 600 mg
Surgical time	10–30 min	⇐	⇐	30–120+ min
EBL	Minimal	⇐	⇐	Typically minimal
Postop care	PACU; rarely 23 h stay 2° airway issues	⇐	⇐	⇐; depends on specific intervention; if G-tube, monitor for abdominal signs
Mortality	<1%	⇐	⇐	⇐
Morbidity	With MDL, incisor trauma <1%	⇐	Perforation <1%	⇐; if G-tube placement, <2% Abdominal bleed or free air; if MDL with laser or laser bronchoscopy, <1% pneumothorax, <5% airway compromise
	Airway obstruction or postprocedure edema requiring reintubation: <1%	⇐	⇐	
	Laryngospasm: <5%	⇐	⇐	
Pain score	1–2	1–2	1–2	1–2

PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–90+	⇐	⇐	⇐
Male:female	1:1	⇐	⇐	⇐
Incidence	Common	⇐	⇐	⇐
Etiology	Tumors, foreign bodies, congenital webs	⇐	⇐	⇐; nutritional need (G-tube)
Association conditions	Other congenital issues; if malignancy, prior RT or chemotherapy and the conditions associated with that malignancy	⇐	⇐	⇐

ANESTHETIC CONSIDERATIONS

(Procedures covered: bronchoscopy, esophagoscopy, panendoscopy, operative microlaryngoscopy, Zenker diverticulectomy, laser surgery of the airway.)

PREOPERATIVE

Many of these patients are elderly and have a history of smoking and alcohol abuse with corresponding implications for intraop management. Patient fluid and nutritional status may be further compromised by preexisting malignancy. Meticulous attention to airway management is paramount in these procedures, and close communication with the surgeon is essential. Some patients presenting for esophagoscopy may have obstructing lesions of the esophagus or **Zenker diverticulum** or active gastrointestinal bleeding or require the removal of an FB, putting them at increased risk of aspiration.

Airway

Although rare, the airway may be compromised in these patients at both the upper and lower airway levels; thus, a thorough preop airway assessment is essential. A clear backup plan for securing the airway should be devised and discussed with the surgeon before induction of GA. Stridor at rest suggests an airway narrowing ≤ 4.5 mm, although the airway diameter can be seriously reduced even without stridor being present.

Patients with lesions in the mediastinum may have involvement of the recurrent laryngeal nerve, presenting with hoarseness and potential airway management problems (e.g., difficult mask ventilation, difficult intubation, $\uparrow$ aspiration risk). (See also Anesthetic Considerations for Laryngectomy.)

Respiratory

These patients may have a high incidence of COPD and $\downarrow$ respiratory reserve. The nature and characteristics of a productive cough must be noted. Wheezing on exam must be treated with bronchodilators before induction. Patients presenting with vocal cord (VC) paralysis or achalasia may have had repeated episodes of pulmonary aspiration.

	Patients with significant lung disease may not be the candidates for specialized ventilation techniques, including jet ventilation (JV) or apneic intermittent ventilation (AIV). Tests: CXR; other tests as indicated from H&P
Dental	Physical exam should include a careful dental assessment and documentation of any missing, loose, or damaged teeth. Patients (or parents, as appropriate) should be informed that dental trauma may occur as a result of surgical instrumentation of the patient's mouth during the procedure.
Cardiovascular	Adrenergic responses during endoscopy may be associated with up to a 4% incidence of myocardial ischemia. Careful preop Hx and thorough physical exam should be undertaken in patients with Hx of CAD and CHF or those with cardiac risk factors (including age >40 yr, male, HTN, hypercholesterolemia, long Hx of smoking, obesity, and family Hx). The volume status of debilitated patients who are unable to eat because of obstructing lesions of the esophagus should be assessed by measuring orthostatic BP changes. Tests: ECG; other tests as indicated from H&P
Neurologic	Some patients may have Hx of ETOH abuse, which may result in ↑ anesthetic requirements 2° hepatic enzyme induction. Symptoms of alcohol withdrawal (e.g., tremulousness, ↑ sympathetic activity, and altered mental status), if present, should be controlled before surgery.
Hematologic	Patients with malignancy or chronic disease may have evidence of anemia or coagulopathy. Tests: CBC; coagulation studies as indicated from H&P
Gastrointestinal	In some patients (e.g., with malignant tumors), significant electrolyte abnormalities may be present 2° malnutrition. Hypokalemia and hypomagnesemia

	should be corrected preop. In patients with Hx of ETOH abuse, liver disease and cirrhosis may be present. Tests: Electrolytes; BUN; Cr; LFT; coagulation studies, as indicated from H&P
Premedication	An antisialogogue (e.g., glycopyrrolate 0.2 mg iv) may be desired by the surgeon to facilitate panendoscopy, especially in patients with copious secretions. Sedative premedication is routine but should be minimized or avoided in the elderly and avoided completely in patients with symptoms of upper airway obstruction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Airway management requires careful planning and continuous communication with the surgeon. Surgical requirements include adequate muscle relaxation (movement, coughing, or bucking during endoscopy may have disastrous consequences) and immobile VCs for VC surgery. Cardiovascular stability is important: laryngeal, tracheal, and cranial reflexes may provoke severe $\uparrow$ BP and $\uparrow$ HR, which can be detrimental in some patients. Adequate depth of anesthesia is essential, but the requirements for rapid awakening and return of laryngeal reflexes present additional challenges in anesthetic management. Short-acting β -blockers (e.g., esmolol) may be indicated to treat breakthrough sympathetic responses, especially in patients with cardiac disease. Remifentanyl or alfentanil (see Introduction) is useful for these procedures, which, although highly stimulating, are characterized by minimal or absent postop pain, and the majority of patients are discharged home from the recovery room.

Special considerations

If **flexible bronchoscopy** is planned, the patient's trachea is intubated with a large-diameter ETT (7.5–8.0 mm ID) to accommodate a flexible video bronchoscope. The ETT is usually taped midline. As an alternative, in selected patients, flexible

bronchoscopy can be performed without tracheal intubation through the **Patil-Syracuse mask** during manual bag-mask ventilation. Placement and manipulation of the flexible bronchoscope by the surgeon will be facilitated by the concomitant use of one of the hollow oral airways used for the fiberoptic intubation (e.g., **Williams airway**). The patient can then be tracheally intubated with a smaller diameter ETT (e.g., 6.0-mm ID) to facilitate surgical access.

Flexible esophagoscopy is rarely performed as an isolated procedure but, if done, would also be facilitated by tracheal placement of a small-diameter (e.g., 6.0-mm ID) ETT. The ETT is usually moved over to the left corner of the mouth and taped to the lower jaw, to provide more room for esophagoscope advancement and manipulation.

If **rigid bronchoscopy** is planned first, GA is induced, and the patient is hyperventilated through the face mask with $\text{FiO}_2 = 1.0$. Following muscle relaxation, the surgeon may proceed, without securing an airway. Quick and gentle DL for application of topical lidocaine to the larynx and upper trachea (lidocaine tracheal anesthesia [LTA], 4% lidocaine 3–4 cc) before rigid bronchoscopy may help blunt hemodynamic responses to the subsequent surgical manipulation. DL with LTA should be avoided in patients with tumors at the base of the tongue ($\uparrow$ risk of bleeding; see Anesthetic Considerations for Glossectomy) or when preexisting copious secretions or bleeding are present in the upper airway 2° $\uparrow$ risk of pulmonary aspiration on emergence from anesthesia. It is advisable to demonstrate to the surgical team and document in the chart absence of dental damage after DL has been performed.

After the rigid bronchoscope is introduced into the patient's trachea, it is connected to an anesthesia circuit through a flexible side port adapter (Racine, [Fig. 3.1](#)), and the patient is ventilated manually. High flows are usually required because of the leak around the bronchoscope. Close communication with the surgeon is essential for adjusting ventilation when the bronchoscope is introduced into the mainstem bronchus to avoid high inflating pressures and to ensure complete exhalation ($\downarrow$ risk of barotrauma). After rigid bronchoscopy is completed, mask ventilation with $\text{FiO}_2 = 1.0$ before intubation is advisable to ensure adequate oxygenation and normocapnia. A reinforced ETT is useful for **rigid esophagoscopy** to avoid possible compression of the ETT. The ETT should be moved to the left side of the patient's mouth to facilitate introduction of the surgical instruments and taped to the lower jaw because full opening of the patient's mouth is required.

For **operative microlaryngoscopy**, a small-diameter (usually 5.0-mm ID), cuffed long microlaryngeal tube (MLT) is most commonly used to facilitate visualization of the

larynx but will interfere with posterior glottis lesions, for which jet ventilation (see below) is most commonly employed. The CO₂ laser, which can precisely vaporize superficial tissue, is widely used for **VC laser surgery**, whereas the Nd:YAG (neodymium-yttrium-aluminum-garnet) laser is usually employed for airway tumor debulking because of its ability to coagulate deeper lesions. The Nd:YAG can be used through the suction channel of the fiberoptic or video bronchoscope, whereas the CO₂ laser must be aimed directly at the targeted tissue. Both the CO₂ and Nd:YAG wavelengths lie outside the visible spectrum, and a separate, lower-energy visible beam is used for aiming. The patient must be motionless, the patient's eyes must be protected with tape and moistened gauze, and OR personnel must wear protective goggles.

An **intermittent apnea technique** involves hyperventilation, followed by intermittent tracheal extubation for 1–5 min, during which the laser is used. This approach is time consuming and may be associated with a higher incidence of airway trauma and edema^{2°} repeated intubations. Careful pulse oximeter monitoring is essential for this technique.

Manual low-frequency jet ventilation (Fig. 3.3) or **automated high-frequency jet ventilation** (HFJV) may be necessary when an ETT cannot be used (e.g., some posterior glottic and subglottic lesions). For supraglottic manual jet ventilation, the operating laryngoscope is most commonly employed. The axis of the jet should be in line with the trachea, and full egress of air (complete chest deflation) should be ensured between the jet ventilator “puffs.” Close communication with the surgeon is essential. The jet should be triggered during pauses between laser firings to keep the VCs immobile. If jet ventilation is used for laser resection of papillomas, there is a risk of spreading the virus to OR personnel, and special face masks should be worn in the OR. Jet ventilation generally provides adequate ventilation without introducing flammable material into the airway or obstructing the surgical field. Its use, however, may be associated with potentially severe complications, including barotrauma, pneumothorax, and gastric distension (risk of regurgitation), and is hindered by ↓ chest wall, airway or lung compliance. Full muscle relaxation is usually required to facilitate precise laser targeting. Automatic HFJV is preferred to manual JV and can be delivered through either an operating laryngoscope or specialized subglottic catheter (e.g., LaserJet or Hunsaker tube).

A recent introduction of the **THRIVE technique** (see Introduction, above) has made it possible to conduct short laryngeal surgical procedures using high-flow nasal oxygen insufflation as the sole means of oxygenation and ventilation.

For laser cases, precautions must be taken to prevent **airway fire**, including the following:

- Use special laser ETT (e.g., Mallinckrodt Laser-Flex), although none of them provides 100% protection from all types of lasers. Small (5.0 mm ID and smaller) diameter laser ETTs are preferred to facilitate surgical access.
- Use the lowest possible FiO_2 (≤ 0.3) that will ensure adequate oxygenation (dilute O_2 with air, N_2 , or helium) and avoid N_2O (both O_2 and N_2O promote combustion).
- Use colored (methylene blue-tinged) NS in the ETT cuff (will immediately alert the surgeon in case of a laser hit).
- Place the ETT sufficiently deep into the trachea to reduce the risk of laser contact with ETT cuff.

Similar considerations apply in patients undergoing endoscopic **Zenker diverticulectomy**, when the CO_2 laser is used.

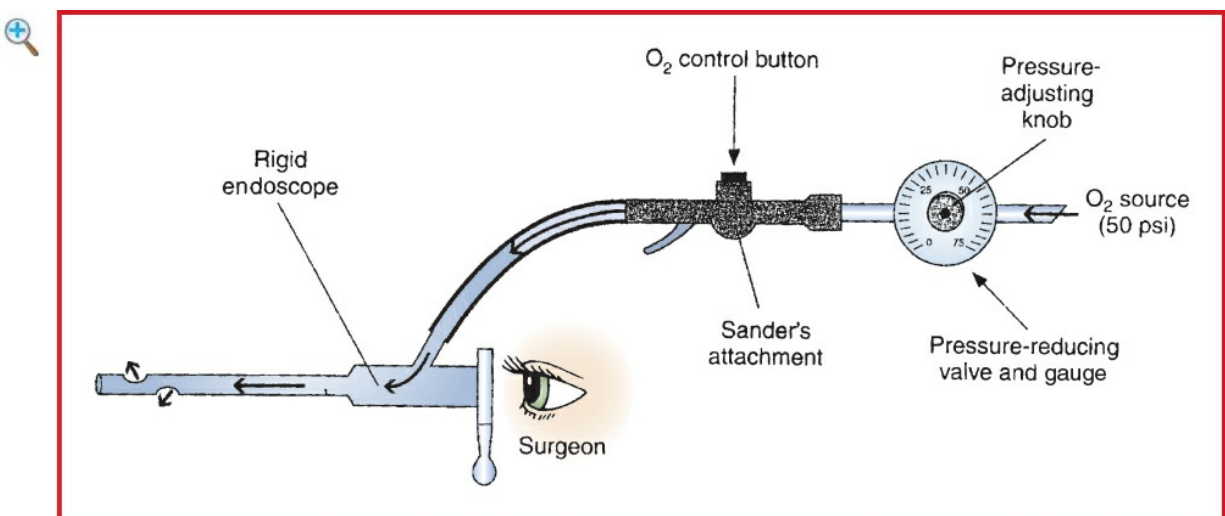


Figure 3.3 Rigid endoscope with modified Sanders jet ventilation technique. The oxygen supply at 50 psi is connected to a reducing valve that allows the pressure to be adjusted from 0 to 50 psi. The side port of the endoscope is used as the Venturi injector site, and the open end can be used for continuous viewing by the endoscopist.

Induction

Patients at risk for aspiration may require RSI with cricoid pressure (CP) if symptoms of active gastroesophageal reflux disease are present. Otherwise, anatomical location of the diverticulum (above the cricoid cartilage) makes application of CP largely ineffective. With the possible exception of endoscopic Zenker diverticulectomy, fentanyl use should be minimized (1–2 mcg/kg) or even

completely avoided in favor of the short-acting opioids (remifentanyl or alfentanil iv; see Introduction), 2° minimal postop pain. Propofol (1–2 mg/kg) is the ideal induction agent. Patients with mobile, floppy, supraglottic tumors (e.g., papillomatosis, epiglottic cancer, large vocal cord polyps) are at ↑ risk for complete airway obstruction after induction of GA with or without neuromuscular blockade. Anesthetic management of these patients requires careful planning and meticulous preparation (see Anesthetic Considerations for Laryngectomy). The surgeon should be consulted about the extent of the disease and the potential need for a tracheostomy. If the disease is severe, the surgeon should be present during induction for the possible need to perform urgent or emergent rigid bronchoscopy or cricothyrotomy/tracheostomy.

ERAS

Maintain normovolemia and normothermia.

Maintenance

TIVA (e.g., propofol: 80–180 mcg/kg/min; remifentanyl 0.1–0.25 mcg/kg/min) is widely used, especially when rigid bronchoscopy or jet ventilation/intermittent apnea techniques are planned. Compared to alfentanil, the use of remifentanyl during TIVA is associated with the increased hemodynamic stability, reduction in propofol dose and faster respiratory recovery. When the ETT is used throughout the case, delivery of anesthetic gases may be used instead of a propofol infusion, especially if a bronchodilating effect is desirable. Desflurane and sevoflurane are the preferred agents because of their low blood-to-gas solubility; sevoflurane may be favored in patients with preexisting cardiac disease to minimize the risk of a dose-dependent tachycardia. Breakthrough sympathetic responses can be managed safely by administration of β -blockers iv or small boluses of a short-acting opioid (see Introduction). With intermittent apnea or jet ventilation techniques, it

may be difficult to avoid hypercapnia and hypoxemia, which may provoke intraop dysrhythmias. Full muscle relaxation is routine: vocal cord immobility is essential for microlaryngeal and laser surgery. Monitoring the neuromuscular blockade may be more accurate at the facial nerve (orbicularis oculi muscle), than the ulnar nerve (adductor pollicis muscle) because the recovery of the former better reflects the recovery of the laryngeal muscles. Care should be taken to avoid inadvertent prolonged neuromuscular blockade by coordinating administration of additional doses of muscle relaxants with the surgeon. As the surgery approaches its end, deepening the level of anesthesia may be the preferred approach to minimize VC movement. A succinylcholine infusion (2–6 mg/min iv) can be used safely in selected patients (limiting the total dose of succinylcholine $\leq 4\text{--}6$ mg/kg will help avoid phase 2 block).

Emergence

Patient should have full return of protective airway reflexes prior to extubation. The patient's stomach may need to be decompressed following jet ventilation. Smooth emergence from anesthesia is obligatory to avoid additional trauma to the vocal cords. For this reason, some surgeons request deep extubation after vocal cord surgery, which presents extra challenges to the anesthesiologist (see Emergence for Tonsillectomy, Adenoidectomy).

Blood and fluid requirements

IV: GI bleeder: 14–16 ga $\times$ 2 Others: 20 ga $\times$ 1	Blood loss is usually minimal; however, in patients with $\uparrow$ risk of GI bleed, blood loss may be massive. See Goal-Directed Fluid Management, Appendix K, p. K-4.
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	BSS @ 2–3 mL/kg/h	T&C these patients for 2 U PRBC with blood immediately available in OR.
Monitoring	Standard monitors ± arterial line	In patients with ↑ risk of GI bleed, an arterial line is desirable.
Positioning	OR table rotated 90° or 180° away from the anesthesiologist ✓ and pad pressure points ✓ eyes	A shoulder roll is placed and the patient's neck is extended to facilitate endoscopy. Special attention should be paid to prevent compression of the patient's shoulder by the vertical bar of the Mayo stand, which is frequently employed for suspending the operating laryngoscope or endoscope.
Complications	Inadequate ventilation Loss of airway Perforation of airway Dysrhythmias Pneumothorax	Hypoxia and hypercarbia may be difficult to avoid with jet ventilation. Mechanical or laser perforation of the airway may → bronchospasm or uncontrollable hemorrhage. Dx: ↑ RR, ↓ BP, ↑ CVP, wheezing, ↓ O ₂ saturation, ↑ PIP, SOB, chest pain, ↓ breath sounds, ↓ ECG amplitude, dullness on percussion. Rx: chest tube or needle aspiration, FiO ₂ = 1.0, ventilation and volume

	Eye trauma Bleeding postbiopsy	expansion. Eye trauma from surgical instruments used during endoscopy may require ophthalmology consult.
Airway fire (steps taken simultaneously by the anesthesiologist and surgeon)	Disconnect the inspiratory limb of the anesthesia circuit Remove ETT Extinguish fire with oral NS Remove all burning material Resume ventilation by face mask with 100% O₂. Consider reintubation Flexible fiberoptic bronchoscopy Save ETT for later examination	This is an acute, life-threatening emergency. NB: turning ventilator off only may not be sufficient due to continued delivery of fresh gas to the patient's lungs. Rigid or flexible bronchoscopy is required to ✓ extent of damage, presence of ETT fragments, and airway edema. All inhaled gases should be humidified. Patients are usually reintubated and monitored in the ICU. To assess lower airway damage and to remove ETT fragments.

POSTOPERATIVE

Complications	Dental trauma Bleeding Eye trauma	Dental trauma may result from surgical manipulation of the airway, and the anesthesiologist must check the teeth immediately after airway access is gained. If dental trauma is detected, notify
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		the surgeon to discuss management options, both immediate and subsequent.
	Postop airway compromise	Has a higher incidence than in general surgery patient population. Patients with T3 lesions and those with associated laryngeal biopsy (↑ airway swelling) after panendoscopy may be at high risk for reintubation in PACU. Delayed upper airway edema after laser surgery has been described, when the coagulation (Nd:YAG) laser is used.
	Esophageal perforation	Complaints of painful swallowing as well as fever, tachycardia, neck/chest/back pains postop may suggest an esophageal perforation.
Complications (cont'd)	Pneumothorax Pneumomediastinum Hemothorax Aspiration	Pneumothorax (see above), mediastinal air or hemothorax from esophageal perforation may present as ↓ BP and cardiovascular collapse.
Multimodal pain management	Short-acting opioid	Fentanyl 25–50 mcg iv prn is usually sufficient.
Tests	✓ CXR	For evidence of pneumothorax, hemothorax, mediastinal

ERAS	PONV protocol (see Appendix B, p. B-1)	air. Encourage early enteral nutrition and mobilization.
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TRACHEOTOMY/TRACHEOSTOMY AND CRICOTHYROIDOTOMY

SURGICAL CONSIDERATIONS

Description

When an airway needs to be obtained right away and intubation is not an option, a cricothyrotomy is indicated. A vertical incision is made in the midline through the cricothyroid membrane, an initial influx of air is facilitated if the patient is spontaneously breathing, and a small ETT is placed. It is wise to convert this to a tracheostomy as soon as it is convenient to do so as this reduces the subsequent incidence of subglottic stenosis and cricoid chondritis.

A tracheotomy is generally done in a controlled setting, under general anesthesia in an intubated patient or under local anesthesia; using a short transverse incision 1–2 cm inferior to the cricoid or a midline vertical incision beginning at the same location. Strap muscles are retracted laterally, the anterior tracheal wall inferior to the thyroid isthmus identified (divided the isthmus if necessary), and, in adults, an inferiorly based tracheal flap consisting of the 2nd or 3rd tracheal ring is made and secured to the skin inferiorly. In small children, it is better to make only a vertical midline incision to reduce the risk of subsequent tracheal stenosis; left and right stay sutures are instead placed to assist in reintubation in the event of accidental dislodgment of the tracheotomy tube. The tracheotomy tube is sutured in all patients to the skin. Trach ties supplement this securing of the tube unless these circumferential ties would interfere with venous drainage of a microvascular free flap used in head and neck reconstruction.

When prolonged need for a tracheotomy without mechanical ventilation is anticipated, there are specialized silicon tracheotomy tubes with minimal intraluminal plastic and may be associated with fewer intraluminal potential complications. Extended length tracheotomy tubes are available and may be needed in obese patients or those with tracheal narrowing distal to the tracheotomy site.

Usual preoperative diagnosis

Indications for tracheostomy are numerous but share the common theme of securing a safe airway either in anticipation of postop airway edema or inability to protect the airway from aspiration or as an urgent need to obtain an upper airway in pending obstruction. The fastest way to obtain an airway in an outright emergency when intubation is not an option is a cricothyrotomy. Another third general indication is to protect the larynx from injury if prolonged intubation is anticipated, such as in a prolonged ICU setting or in paralysis associated with cervical spinal cord trauma. Rarer indications are bilateral vocal cord paralysis or a history of recurrent allergy associated with laryngospasm.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Tracheostomy	Cricothyrotomy
Position	Supine, head extended	←

Incision	Transverse or vertical	Stab incision through cricothyroid membrane
Special instrumentation	None (most institutions have a “trach set”)	Scalpel is all that is initially needed in this emergency setting
Unique considerations	Local or general, intubated or not. Specialized equipment and silicon trach cannulas selectively arranged for	⇐
Antibiotics	None	⇐
Surgical time	3–20 min	30 sec–2 min
EBL	Minimal	⇐
Postop care	Warm humidification; suctioning; changing inner cannula; skin care; monitor for bleeding and dislodgement; in emergency remove entire tracheotomy tube	⇐; convert to tracheotomy as soon as convenient
Mortality	<1%	⇐
Morbidity	Bleeding <5% cellulitis/tracheitis <5% tracheal stenosis <1%	Subglottic stenosis risk if not converted
Pain score	1–2	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	All	⇐
Male:Female	Dependent on underlying condition	⇐
Incidence	Common	⇐
Etiology	Anticipated or actual airway obstruction, prolonged	Trauma

Associated conditions	intubation Dependent on underlying condition, but typically cardiopulmonary or neurologic conditions Trauma
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ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Typically, there are three patient populations presenting for tracheostomy: (a) critically ill intubated patients in chronic respiratory failure or following major trauma, (b) patients for whom tracheostomy is part of a scheduled procedure (e.g., laryngectomy), and (c) patients with impending or total upper airway obstruction (e.g., Ludwig angina, retropharyngeal abscess). If the latter constitutes life-threatening emergency, tracheotomy/cricothyroidotomy may be the preferred approach. Aside from an occasional otherwise healthy patient in the third category, all patients presenting for tracheostomy are usually debilitated, have associated cardiac or pulmonary disease, and frequently present with neurologic and metabolic abnormalities.

Respiratory	Patients in group One usually require mechanical ventilation with PEEP to maintain adequate oxygenation. The continued application of PEEP may be an important consideration during their transport from ICU to OR. Patients in group Two require a careful airway evaluation (see Anesthetic Considerations for Neck Dissection, and Laryngectomy). Based on the assessment, the anesthesiologist must decide whether to choose a DL, an awake FOI, or a tracheostomy under local anesthesia to secure the airway. Patients in group Three will usually require semiemergent or emergent tracheostomy. Patients in all three groups should be evaluated for the presence of possible recurrent aspiration. Tests: CXR; ABG, as indicated from H&P
Cardiovascular	All patients may have significant cardiac risk factors, including smoking, ETOH abuse, male gender, ↑

	cholesterol, family Hx, and HTN. Some patients may have recently undergone major cardiac surgery and may be on pharmacologic inotropic support, with full invasive monitoring, including PA line. Tests: ECG, and other tests as indicated, in patients of the second group.
Neurologic	Preop neurologic deficits should be fully documented.
Hematologic	In cases of malignancy or chronic disease, coagulopathies or anemia may be present. Tests: CBC; coagulation studies
Laboratory	Other tests as indicated from H&P
Premedication	Standard premedication in elective cases. Premedication is best avoided in critically ill patients and patients with symptoms of upper airway obstruction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of superficial cervical plexus nerve blocks for postop pain management. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA in intubated patients or in patients who pose no problems for conventional DL and tracheal intubation. In the presence of significant airway compromise or anticipated very difficult intubation, local anesthesia may be required. Most tracheostomies are elective or semiurgent and are performed under GA. Patients with ↑ mucosal swelling (e.g., generalized edema, prolonged intubation) and ↑ tissue fragility (e.g., chronic steroid therapy) are at risk for tracheal mucosal separation → creation of a false passage during tracheostomy. This will constitute a true emergency (see Postoperative Complications, below).

Induction	If already intubated (group One, above), convert preexisting
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	sedation to GA, using a carefully titrated inhalational agent. intubated, and no airway problems are anticipated, a standard induction may be appropriate. If airway problems are anticipated, an awake FOI or tracheostomy performed under local anesthesia may be the techniques of choice (see Anesthetic Considerations for Neck Dissection, and Laryngectomy). In any event, the anesthesiologist should be prepared to deal with a failed intubation and have a surgeon immediately available to perform cricothyrotomy/tracheostomy if ventilation proves impossible. Use of THRIVE during intubation attempts, and even during conversion of a failed tracheal intubation into the surgical airway, may be advisable to maximize oxygenation and prolong the window.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. Full muscle relaxation is required. FiO₂ 1.0 usually is required before insertion of the tracheostomy tube. This high O₂ concentration, combined with electrocautery, may precipitate airway fire (see Postoperative Complications, below). A cuffed ETT should be used to prevent O₂ from leaking into the surgical site. In order to avoid inadvertent ETT cuff puncture, consider advancing the ETT closer toward the carina, before the trachea is opened. The trachea is usually opened above the cricoid, and the ETT is slowly retracted cephalad, under direct vision by the surgeon. (Note: Do not remove the ETT completely.) Spontaneous gas sampling tubing, extra ETTs, and anesthesia circuit should be readily available (see Maintenance for Laryngectomy). After the tracheostomy tube is secured, it should be suctioned and connected to the anesthesia circuit. Verify the ETCO₂ tracing, the inflation pressures, and remove the ETT.
Emergence	Considerations for group Two and Three patients are described above. Patients in the first group will continue on ventilatory support in the ICU. The tracheostomy tube must be carefully suctioned, and the delivered O₂ should be humidified. Opioid sedation will minimize reaction to suctioning in the early postoperative period. Tracheostomy tubes should not be removed for at least

	first 5–7 d until a track is formed. Discuss with surgeon local wound infiltration or nerve block (e.g., superficial cervical plexus block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PO prophylaxis should include dexamethasone and ondansetron	
Blood and fluid requirements	IV: 18 ga × 1 BSS @ 2–3 mL/kg/h	EBL is typically minimal, when the tracheostomy is performed as an isolated procedure. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	Avoid ECG pad placement in the preop area. Invasive monitoring may be appropriate depending on the patient's condition.
Positioning	Shoulder roll ✓ and pad pressure points ✓ eyes	A shoulder roll is usually placed, and patient's neck extended, which may reduce ETT cuff migrating cephalad, closer to incision site (danger of inadvertent perforation by the surgeon on entering trachea). Make sure the ETT is positioned at the sufficient depth and taped securely.
Complications	Pneumothorax Pneumomediastinum Hemorrhage Aspiration of blood False tracheal passage/tracheal disruption Difficult ETT insertion/reinsertion	Pneumothorax may occur with a low neck dissection, or if a false passage (see Potential Complications, below) has been created during the tracheostomy tube insertion and Rx—see Anesthetic Considerations: Laryngoscopy/Bronchoscopy/Esophageal Intubation. This is an airway emergency and must be quickly recognized (absent CO ₂ , ↑ PIP, absent or very distant breath sounds). Reintroduction of the existing ETT from the proximal trachea should be attempted. Rigid bronchoscope should be available to reestablish the airway, in case of a failed reintubation. An alternative approach is to insert a large-bore iv catheter into the tracheal lumen distal to the tracheostomy site and jet ventilate the patient. Inse

Airway fire

of an airway exchange catheter through bronchoscopy elbow adapter attached to existing ETT has been advocated prior to tracheostomy tube insertion in patients at high risk for this complication. Tracheostomy fires are usually not catastrophic, possibly because the tracheostomy acts as a vent. Rx: immediately disconnect the patient from the anesthesia machine, extinguish the fire with NS, and ventilate the lungs with room air using a self-inflating bag. Removing or changing the existing ETT at this point may be more risky than leaving it in, especially if the patient was previously difficult to intubate or the airway has become edematous.

POSTOPERATIVE

Complications	Pneumothorax	See Postoperative complications after Laryngoscopy/Bronchoscopy/Esophagoscopy
	Pneumomediastinum	
	Hemorrhage	This could be due to secretions, mucus, blood, or positioning of the tube in the mainstem bronchus or against the tracheal wall.
	Occlusion of the tracheostomy tube	
	Tracheostomy tube displacement	Reintubate orally or through the tracheostomy site.
Multimodal pain management	PCA IV opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
Tests	CXR	✓ tracheostomy tube position and ✓ 1 evidence of either pneumothorax or pneumomediastinum.

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INTUBATION FOR EPIGLOTTITIS

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Epiglottitis is an acute inflammation and swelling of the epiglottis, associated with a generalized systemic toxicity, usually due to *Haemophilus influenza* type B. It also can be caused by β -hemolytic streptococci, staphylococci, pneumococci, or unusual pathogens among immunocompromised individuals and drug/alcohol abusers. It occasionally results in total laryngeal obstruction and death 2° asphyxia. At one time, the typical patient was a previously healthy child 3–5 yr old; however, since the advent of the H-flu vaccine, epiglottitis is more common in adults (predominantly males). The most common presenting symptoms are sore throat, dysphagia/odynophagia, fever, respiratory difficulty, and drooling. Pediatric patients may appear toxic on presentation.

Airway

Enlarged epiglottis seen frequently on neck x-ray. Neck tenderness or swelling also observed in some patients. Patients with stridor are at high risk for upper airway obstruction.

Respiratory	The patient, typically sitting upright, may display hoarseness, muffled voice, dyspnea, and chest wall retractions. NB: Rapid treatment should be instituted instead of performing time-consuming investigations. The sudden development of respiratory obstruction in the x-ray suite could have a disastrous outcome.
Hematologic	A high leukocyte count usually is found in these patients. Tests: Blood drawing before the airway is secured may be inadvisable.
Premedication	None

INTRAOPERATIVE

Anesthetic technique

GETA: Patients presenting with imminent or actual airway obstruction should be intubated immediately. Airway management for **adult** patients presenting with **mild-to-moderate** symptoms is controversial. Although routine prophylactic intubation of these patients may not be necessary, 18% subsequently develop complete airway obstruction; thus, close monitoring is mandatory if intubation is deferred. It is imperative to realize that total airway obstruction can occur suddenly and without warning.

An experienced anesthesiologist should be present for this procedure. In the pediatric patient, it is critical that neither visualization of the epiglottis nor other maneuvers be attempted to confirm the Dx before anesthesia. In adults, it has been suggested that indirect laryngoscopy or fiberoptic bronchoscopy (FOB) may be performed without the risk of precipitating complete airway obstruction, although this approach is controversial.

Induction	If an iv is in place, atropine 0.02 mg/kg may be given to the pediatric patient to prevent vagal reflexes resulting from manipulation of the inflamed epiglottis. If no iv access is available, inhalation induction (sevoflurane in $\text{FiO}_2 = 1.0$) with the patient in the sitting position should be used. Intramuscular injection should be avoided to prevent agitation, crying, and subsequent total airway obstruction (iv line is placed after induction of anesthesia). An
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	experienced ENT surgeon, ready to perform an emergency tracheostomy, must be present at induction. Early application of CPAP is essential to maintain the upper airway patency. Inhalation induction may be prolonged and intubation extremely difficult. Different sizes of ETTs must be available.	
ERAS	Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance.	
Emergence	The patient will remain intubated for 24–48 h and should be kept sedated and restrained to prevent accidental extubation.	
Blood and fluid requirements	IV: 18–20 ga (adult) 22–26 ga (child) BSS @ 3–5 mL/kg/h (adult); @ 1.5 mL/kg/h (child)	Minimal blood loss Fluid requirements may be ↑ 2° fever and dehydration. See Goal-Directed Fluid Management, Appendix K , p. K-4.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	POPE (postobstructive pulmonary edema)	A short-lived POPE occasionally occurs after relief of the obstruction and must be treated with IPPV.

POSTOPERATIVE

Complications	Accidental extubation ETT blockage	These complications can prove fatal. The blockages are sometimes
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		due to crusting of the ETT 2° insufficient humidification.
Multimodal pain management	PCA IV opioids	Postop sedation and manual restraints are often required.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition and mobilization.

Suggested Readings

1. Chroboczek T, Cour M, Hernu R, et al: Long-term outcome of critically ill adult patients with acute epiglottitis. PLoS One 2015; 10(5):e0125736.
2. Feldman MA, Patel A: Anesthesia for eye, ear, nose, and throat surgery. In: Miller RD, ed. Miller's Anesthesia, 7th edition. Elsevier, Philadelphia: 2010, 2357–88.
3. Ovnat Tamir S, Marom T, Barbalat I. et al: Adult supraglottitis: changing trends. Eur Arch Otorhinolaryngol 2015; 272(4):929–35.
4. Senior BA, Radkowski D, MacArthur C, et al: Changing patterns in pediatric supraglottitis: a multi-institutional review, 1980 to 1992. OHNS 1994; 110:203–10.
5. Verbruggen K, Halewyck S, Deron P, et al: Epiglottitis and related complications in adults. Case reports and review of the literature. B-ENT 2012; 8(2):143–8.
5. Westerhuis B, Bietz MG, Lindemann J: Acute epiglottitis in adults: an under-recognized and life-threatening condition. S D Med 2013; 66(8):309–11, 313.

ZENKER DIVERTICULECTOMY (OPEN APPROACH)

SURGICAL CONSIDERATIONS

Description

Zenker diverticulum is a herniation of mucosa through the posterior hypopharyngeal wall immediately above the cricopharyngeus muscle and below the inferior constrictor in an area called Killian dehiscence. This is an acquired disorder, usually seen in the sixth to ninth decades of life and felt secondary to tonic spasm or achalasia of the cricopharyngeus muscle. Clinically, patients may experience dysphagia, globus, coughing, and regurgitation of undigested food. Weight loss and aspiration pneumonia can be seen in severe cases. Barium swallow is usually diagnostic. Endoscopy will reveal the presence of a pouch of variable size posterior to the cricopharyngeus muscle

often filled with undigested debris. Cricopharyngeal achalasia may present in the absence of a well-formed diverticulum with much the same symptom complex. Treatment is usually aimed at division of the cricopharyngeus muscle and eradication of the pouch.

Variant approaches

Open: Under GA with muscle relaxation, an incision is made in the left side of the neck to expose the diverticulum and cricopharyngeus muscle. The muscle is cut, the diverticulum resected, and the hypopharyngeal defect closed. In patients with pure cricopharyngeal spasm, the muscle alone is cut. A drain is placed, and the wound is closed. A nasogastric tube may be placed for postop feeding. Aspiration precautions should be observed at both induction (consider awake intubation in severely symptomatic patients) and reversal of anesthesia, as this is a common comorbidity in patients with Zenker diverticulum. **Endoscopic:** Selected patients may be candidates for endoscopic treatment of a Zenker diverticulum. This is better described as diverticulotomy rather than diverticulectomy because the redundant hypopharyngeal mucosa is not removed, and the cricopharyngeus muscle is not divided in its entirety. Rather, the common wall separating the diverticulum from the esophagus is reduced to prevent food and debris from collecting within the confines of the diverticulum. With the patient under general anesthesia and muscle relaxation, an esophagodiverticuloscope is placed transorally and advanced into the hypopharynx. The endoscope is deployed to visualize clearly the opening into the diverticulum. The common wall separating the diverticulum from the true esophageal inlet is then cut, either with a stapler or with a CO₂ laser.

Usual preop diagnosis

Dysphagia; Zenker diverticulum; cricopharyngeal spasm

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Open	Endoscopic
Position	Supine; table 90°; head extended	Supine

Special instrumentation	Anode or laser ETT; microscope	⇐; endoscopic stapler or CO ₂ laser
Unique considerations		Keep O ₂ < 30% and avoid N ₂ O if lasering
Antibiotics	Cefazolin 2 g and metronidazole 500 mg; clindamycin 600 mg	
Surgical time	1–2 h	30 min–1 h
EBL	<25 mL	0–25 mL
Postop care	Reflux precautions should be observed; patients may require nasogastric tube feeding for several days	Reflux precautions should be observed
Mortality	<1%	<1%
Morbidity	Fistula formation <5% Vocal cord paralysis Hematoma 1–5% Infection 1–5% Aspiration	Perforation <5% Aspiration
Pain score	3–5	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	60–90
Male:Female	1.5:1
Incidence	1 in 50,000; Caucasians
Etiology	GERD; neurodegenerative disorders
Associated conditions	GERD

ANESTHETIC CONSIDERATIONS

(See [p. 204](#))

Suggested Readings

1. Dzeletovic I, Ekbom DC, Baron TH: Flexible endoscopic and surgical management of Zenker's diverticulum. *Expert Rev Gastroenterol Hepatol* 2012; 6(4):449–65.
2. Jackson B, Ahmad Z, Morton RP: Utility of transcervical management of Zenker's diverticulum. *J Laryngol Otol* 2016; 130(Suppl 1):S16–9.
3. Jones D, Aloraini A, Gowing S, et al: Evolving management of Zenker's diverticulum in the endoscopic era: a North American experience. *World J Surg* 2016; 40(6):1390–6.

LARYNGEAL FRAMEWORK SURGERY (THYROPLASTY, ARYTENOID ADDUCTION, INJECTION LARYNGOPLASTY)

SURGICAL CONSIDERATIONS

Description

Thyroplasty is a surgical technique of medializing a paralyzed vocal fold via placement of an implant inserted through the cartilage of the thyroid ala. Under local anesthesia with intravenous sedation, a skin incision is made at the level of the larynx and the thyroid cartilage exposed. Using a small saw, drill bit, or knife, a window is cut in the cartilage to the level of the inner perichondrium. Lateral pressure is then applied to the paralyzed side to gauge the amount of medialization necessary to improve phonation. Because the shape and size of the implant are created based on the location and degree of medialization needed to improve the voice, it is important for the patient to be awake and able to phonate during the procedure. Often, the patient is kept in a state of deeper sedation in the beginning of the case and then is lightened as the case proceeds to allow the patient to be responsive and interactive with the surgeon. This can be a challenge for the anesthesiologist to strike a happy medium between patient comfort and coherence. After the desired degree of medialization is obtained, the implant is secured in place and the wound closed over a drain.

Description

Arytenoid adduction is often performed in conjunction with thyroplasty. This involves placement of a suture around the muscular process of the arytenoid cartilage, which, when tightened, causes posteromedial rotation of the vocal process and adduction of the vocal fold. Usually, the technique is employed when there is a persistent gap between the vocal folds posteriorly.

Description

Injection laryngoplasty refers to medialization of a paralyzed vocal fold by means of injection, whether percutaneous or endoscopic. Its minimally invasive nature is its chief advantage over thyroplasty or arytenoid adduction. However, the longevity of the injected material and its side effect profile are major determinants as to whether or not this procedure should be considered. Traditionally, Teflon was the injectable material of choice. However, because of the potential for an FB reaction, Teflon has been replaced by more biocompatible materials, such as acellular human dermis, fat, Gelfoam™, and calcium hydroxylapatite to name a few. Because of the low viscosity of some of these injectables, many patients are candidates for percutaneous injection, which can be performed in the office or procedure room under simple topical anesthetic. The patient is seated upright and the nasal cavity topically anesthetized with lidocaine 2% and Neo-Synephrine. A fiberoptic rhinolaryngoscope attached to a camera and television monitor is then passed transnasally to the level of the glottis. Through a 25- or 27-gauge needle, the desired material is injected either transcartilagenously, through the cricothyroid membrane, or under the thyroid ala into the paraglottic space. Observing on the monitor, the material is injected until the desired degree of medialization is obtained. This is well tolerated by patients with minimal pain. Airway obstruction or bleeding is exceedingly rare.

For those who request or require injection under GETA, a 5.0 MLT is desirable. Jet ventilation and apneic techniques can also be used. An operating laryngoscope is advanced to the level of the glottis and suspended from a Mayo stand. The paralyzed vocal fold is then injected, usually at the mid and posterior aspects, until adequate medialization is seen. Usually, this entails slight overcorrection such that the medialized fold will now be slightly across midline. Hemostasis if needed is usually obtained with epinephrine-soaked pledgets placed directly on the vocal fold. The scope is then withdrawn and the procedure terminated.

It is important for the anesthesiologist to be aware if a patient has undergone a prior medialization procedure. This is not a contraindication to orotracheal intubation for subsequent procedures requiring GETA. However, traumatic intubation or the use of a tube larger than 6 mm may cause trauma to the adducted fold. A laceration of the vocal fold can potentially expose or dislodge a thyroplasty implant, making it more likely to become infected and potentially extrude.

Usual preop diagnosis

Vocal fold paralysis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Thyroplasty	Arytenoid Adduction	Injection Laryngoplasty
Position	Supine	⇐	⇐
Special instrumentation	Head and neck set; headlights	—	Operating laryngoscope; laryngeal injection needles; jet ventilation setup
Unique considerations	Overmedialization of the vocal fold can potentially cause stridor and dyspnea; coordination between the surgeon and anesthesiologists is essential to time lightening of anesthesia; stimulation can cause coughing	If GETA is requested, intubate with 5.0 microlaryngeal tube	Overmedialization of the vocal fold can potentially cause stridor and dyspnea
Antibiotics	Cefazolin 2 g	⇐	⇐
Surgical time	2 h	⇐	30 min
EBL	<10 mL	10–20 mL	Minimal
Postop care	Decadron intravenously for 24 h; racemic epinephrine; cool	⇐	Racemic epinephrine; cool mist as needed

	mist		
Mortality	Rare	⇐	⇐
Morbidity	Need for revision: 10–20%	⇐	Need for repeat injection common
	Transient dyspnea or laryngospasm	⇐	5–10%
	Infection: <5%	⇐	Rare
	Airway obstruction requiring tracheostomy: 1%		⇐
	Extrusion: rare		
Pain score	3–4	4–5	1–2

■ PATIENT POPULATION CHARACTERISTICS

Age range	18 and over
Male:Female	1:1
Incidence	Common
Etiology	Surgical trauma to recurrent laryngeal nerve; cancer; trauma; idiopathic
Associated conditions	Cancer; coronary artery disease; lung disease; tumors of skull base

Suggested Readings

1. Costello D: Change to earlier surgical interventions: contemporary management of unilateral vocal fold paralysis. *Curr Opin Otolaryngol Head Neck Surg* 2015; 23(3):181–4.
2. Crolley VE, Gibbins N: One hundred years of external approach medialisation thyroplasty. *J Laryngol Otol* 2017; 131(3):202–08.
3. Damrose EJ, Berke GS: Advances in the management of glottic insufficiency. *Curr Opin Otolaryngol Head Neck Surg* 2003; 11(6):480–4.
4. Mizoguchi K, Hatakeyama H, Yanagida S, et al: Perioperative complications and safety of type II thyroplasty (TPII) for adductor spasmodic dysphonia. *Eur Arch Otorhinolaryngol* 2017; 274(5):2215–23.
5. Rosow DE, Al-Bar MH: Type I thyroplasty in previously irradiated patients assessing safety and efficacy. *Otolaryngol Head Neck Surg* 2015; 153(4):582–5.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The laryngeal framework surgery (LFS) is the surgery of choice for dysphonias resulting from incomplete glottic closure or inadequate vocal fold tension. With thyroplasty, direct surgery on the vocal folds is avoided, eliminating the possibility of scarring and voice aggravation. Type I (medialization) thyroplasty is most commonly performed for the unilateral VC paralysis or bowing (in elderly patients) and involves placement of a small silicone implant within the larynx on the affected side.

Airway	No airway problems are usually encountered in these patients.
Cardiovascular	Elderly patients may present with advanced cerebrovascular, cardiovascular, or pulmonary disease. Preop pulmonary aspiration due to glottic incompetence is uncommon but may be seen when the VC paralysis is of central origin, affecting both motor and sensory components of the VC innervation.
Tests	EKG; other tests as indicated from H&P.
Laboratory	Tests, as indicated from H&P.
Premedication	Standard premedication with iv midazolam can frequently be reduced or avoided, especially in the elderly.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of superficial cervical plexus nerve blocks for postop pain management. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Most commonly, thyroplasty is performed under MAC, which allows the surgeon to achieve optimal improvement in the patient's voice quality (sometimes facilitated by visualization of the VC movement through a nasally introduced flexible endoscope) before locking the laryngeal implant in place. It can also be performed under GA, where the concomitant use of the LMA and flexible scope allows a real-time assessment of VC positioning. **Arytenoid adduction** and **injection laryngoplasty** are usually performed under GETA with the use of a small (5.0-mm ID) MLT ETT.

The surgical requirements for **thyroplasty** performed under MAC include patient immobility during injection of local anesthetic; different levels of sedation depending on the phase of surgery (deep during the surgical approach to the cartilage, light for implant placement, and deep again during the surgical closure); absence of straining/bucking/coughing during the implant placement. Manipulation of the thyroid cartilage may necessitate the use of iv opioids.

GETA for **arytenoid adduction** and **injection laryngoplasty** requires full muscle relaxation with quick return of protective airway reflexes and smooth emergence from anesthesia.

When providing MAC for thyroplasty, a quick patient transition to the "light" state for the implant placement is required. A combination of a small dose (1–2 mcg/kg) of iv fentanyl with the iv propofol infusion, or iv midazolam with iv remifentanyl infusion (typically, 0.05–0.1 mcg/kg/min to avoid severe respiratory depression), will achieve the stated objective. For a thyroplasty performed under GA, as well as for the GETA employed for an **arytenoid adduction** and **injection laryngoplasty** surgery, the use of TIVA (see Introduction, Anesthesiologist's Perspectives) has been shown to be highly beneficial. **Injection laryngoplasty** procedures are short and can be effectively and safely performed under THRIVE in selected patients.

Tracheal intubations in patients who have undergone thyroplasty should be performed with special care: the use of an ETT stylet should be avoided, a smaller ETT size should be used, and tracheal extubation should be performed smoothly and atraumatically.

Blood and fluid Requirements	IV: 20 ga × 1 LR @ 2 cc/kg/h	EBL is minimal during all the LFS procedures.
Monitoring	Standard monitors ± arterial line	Even in patients with preexisting significant CV disease, A-line monitoring

		is usually not necessary.
Positioning	Table may be turned 90–180° ✓ and pad pressure points ✓ eyes	To prevent claustrophobia, inform the patient presenting for thyroplasty under MAC that his/her face will be partially covered with surgical towels.
Complications	Overmedialization Stridor and dysphonia	Can be quickly corrected by the surgeon.

▼ POSTOPERATIVE

Complications	Wound hematoma Airway obstruction Stridor Extrusion of the prosthesis	
Multimodal pain management	Short-acting opioids	For example, fentanyl 25–50 mcg bolus is usually sufficient. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6)
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition and mobilization

Suggested Readings

1. Hoffman H, McCabe D, McCulloch T, et al: Laryngeal collagen injection as an adjunct to medialization laryngoplasty. *Laryngoscope* 2002; 112(8 Pt 1):1407–13.
2. Lundy DS, Casiano RR, Xue JW: Can maximum phonation time predict voice outcome after thyroplasty type I? *Laryngoscope* 2004; 114:1447–54.
3. Remacle M, Lawson G, Jamart J, et al: Treatment of vocal fold immobility by injectable homologous collagen: short-term results. *Eur Arch Otorhinolaryngol* 2006;

263:205–9.

4. Remacle M, Lawson G, Mayné A: Use of a laryngeal mask during medialization laryngoplasty. *Rev Laryngol Otol Rhinol (Bord)* 2003; 124:335–8.
5. Rosero EB, Ozayar E, Mau T, et al: A sequential anesthesia technique for surgical repair of unilateral vocal fold paralysis. *J Anesth* 2016; 30(6):1078–81.

TRACHEAL AND CRICOTRACHEAL RESECTION

SURGICAL CONSIDERATIONS

Description

Prolonged endotracheal intubation and complications from tracheotomy account for most cases of subglottic or tracheal stenosis. Wegener disease commonly affects the subglottic airway, and secondary dyspnea may be the first presenting sign of the disease. In females in the fourth to sixth decades of life, subglottic stenosis can be idiopathic. Whatever the etiology, subglottic and tracheal stenosis can be a major source of morbidity for afflicted patients, resulting in dyspnea, dysphonia, and tracheostomy dependence. In selected individuals, resection of the stenotic region with primary repair can often produce marked and sustained relief of symptoms in a single-stage procedure, often without need for a temporary tracheostomy.

Tracheal resection is the treatment of choice in patients with isolated tracheal stenosis. Up to four to five rings of cartilage can be resected, more if releasing maneuvers to mobilize the larynx and intrathoracic trachea are performed. Induction of anesthesia is simplest in patients with preexisting tracheostomies. For those patients who present without a tracheostomy, a tracheostomy under local anesthesia can be performed at or immediately below the stenotic segment. If possible, it is preferable to dilate the stenosis and proceed to endotracheal intubation to avoid further injury to the trachea with a fresh tracheotomy. With the patient in the supine position, a transverse cervical incision is made. The thyroid isthmus is divided and separated from the cervical trachea. Careful blunt dissection on the lateral aspects of the diseased segment is performed to avoid injury to the recurrent laryngeal nerves. A vertical incision is made along the face of the stenotic segment until a healthy complete ring of cartilage is identified above and below the stenotic segment. The anesthesiologist withdraws the ETT up into the subglottis, and the diseased segment of the trachea is resected. At this time, the surgeon may intubate the distal segment of trachea to afford better access to the superior aspect of the diseased segment. In this case, the breathing circuit will need to be switched from the orotracheal tube to the distal tube. The distal trachea can be more

easily mobilized superiorly following blunt finger dissection along the anterior tracheal wall into the mediastinum. The orotracheal tube is then advanced inferiorly into the distal healthy segment. The healthy distal and proximal segments of trachea are then reanastomosed over the ETT using interrupted sutures. The wound is closed and a drain placed. If the patient is to remain intubated overnight, it is important to ensure that the cuff of the ETT remains inferior to the suture line so as not to put tension on the repair. If the patient is to be extubated in the OR, care should be taken to minimize coughing or bucking so as not to rupture the repair site. Endoscopy can be performed prior to transport from the operating room to ensure that at least one vocal fold is mobile. If neither vocal fold is mobile, a bilateral recurrent laryngeal nerve injury should be suspected and a tracheostomy performed. Regardless of whether or not the patient is extubated immediately or later, fiberoptic laryngoscopy to assess vocal fold motion postop is standard of care. Voice quality is not an adequate assessment of vocal fold function. A paralyzed vocal fold can be midline with complete compensation by the mobile fold and no overt dysphonia. In bilateral vocal fold paralysis, the voice is often normal, and the patient's only symptoms will be dyspnea and stridor.

Cricotracheal resection allows single-stage repair of subglottic or a combined subglottic/tracheal stenosis. The procedure is similar to that of tracheal resection with several caveats. It is important to carefully gauge the relationship of the stenosis to the vocal folds. Stenosis that involves the vocal folds is a contraindication to cricotracheal resection. The anterior arch of the cricoid cartilage is usually resected, along with the subglottic soft tissue component of the stenosis, preserving the cricoid plate. This creates a cradle into which the trachea will slide superiorly. No more than one-third of the inferior aspect of the cricoid plate can be resected. More than this will disrupt the posterior cricoarytenoid muscles and prevent vocal fold abduction during inspiration. As resection proceeds, the ETT will be withdrawn above the glottis. It is helpful at this point to attach an umbilical tape to the distal tip of the ETT to allow the surgeon to assist the anesthesiologist in advancing the tube back through the glottis and into the trachea when the resection is complete. The trachea is sutured to the thyroid cartilage anteriorly and the cricoid ring laterally; the wound is closed; and a drain may be placed. The patient is extubated in the OR and vocal fold mobility checked. Tracheostomy is only required in the setting of bilateral vocal fold paralysis and should otherwise be avoided. As with tracheal resection, a preop assessment of vocal fold motion is critical in planning surgery. If unilateral paralysis is present preop, great care is needed to minimize potential injury to the contralateral recurrent laryngeal nerve.

Usual preop diagnosis

Subglottic stenosis; tracheal stenosis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Tracheal Resection	Cricotracheal Resection
Position	Supine	Supine
Incision	Transverse cervical	Transverse cervical
Special instrumentation	Major head and neck tray; headlights	⇐
Unique considerations	Intubation may be difficult owing to degree of stenosis; dilation may be required prior to intubation; awake tracheostomy under local anesthesia may be necessary in select cases	⇐
Antibiotics	Cefazolin 2 g iv with metronidazole 500 mg; clindamycin 600 mg; vancomycin 1 g; piperacillin/tazobactam 3.375 g	⇐
Surgical time	3 h	⇐
EBL	50–100 mL	⇐
Postop care	Most patients can be extubated at the conclusion of surgery; coughing and stimulation should be minimized; if significant edema is expected, patients may remain intubated for 24–48 h and then extubated; patients may require temporary NG tube feedings if laryngeal releasing maneuvers performed; monitoring in the ICU setting for airway	⇐

	compromise or subcutaneous emphysema required postop	
Mortality	1–2%	⇐
Morbidity	Restenosis: 10%	10–15%
	Vocal cord paralysis: 2–3%	3–5%
	Dehiscence: 5%	<5%
	Subcutaneous emphysema: 1–2%	1–2%
	Airway edema requiring tracheotomy: 2%	5%
Pain score	5–6	5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages
Male:Female	2:1
Incidence	1% of all patients requiring intubation
Etiology	Prolonged intubation; prior tracheostomy; Wegener disease; external trauma; neoplasm; radiation; gastroesophageal reflux; idiopathic
Associated conditions	Atherosclerosis; asthma; gastroesophageal reflux

Suggested Readings

1. Akhtar N, Ansar F, Baig MS, et al: Airway fires during surgery: Management and prevention. *J Anaesthesiol Clin Pharmacol* 2016; 32(1):109–11.
2. Chee WK, Benumof JL: Airway fire during tracheostomy: Extubation may be contraindicated. *Anesthesiology* 1998; 89:1576–8.
3. Hobai IA, Chhangani SV, Alfille PH: Anesthesia for tracheal resection and reconstruction. *Anesthesiol Clin* 2012; 30(4):709–30.
4. McGuire G, El-Beheiry H, Brown D: Loss of the airway during tracheostomy. Rescue oxygenation and re-establishment of the airway. *Can J Anesth* 2001; 48:697–700.
5. Rogers ML, Nickalls RWD, Brackenbury ET, et al: Airway fire during tracheostomy. Prevention strategies for surgeons and anaesthetists. *Ann R Coll Surg Engl* 2001; 83:376–80.

LARYNGECTOMY: PARTIAL AND TOTAL

SURGICAL CONSIDERATIONS

Description

Radiation, chemoradiation, and surgery are commonly employed as primary treatment alternatives for laryngeal squamous cell carcinoma. Selected tumors can be removed endoscopically, as reviewed above. Open procedures may be primary or following recurrence after irradiation and are tailored to fit tumor extent. If at least one cricoarytenoid unit (innervated posterior cricoarytenoid muscle and working cricoarytenoid joint) is uninvolved by tumor, the patient may be a candidate for less than a total laryngectomy (TL).

A **vertical partial laryngectomy** (VPL) involves removal of the affected true and false vocal folds and up to one-third of the contralateral folds including the anterior commissure. The contralateral cricoarytenoid unit is preserved, and reconstruction often includes a pedicled sternohyoid flap and thyroid cartilage perichondrium. Exposure and anesthetic considerations are similar to that of a TL (discussed below) other than the fact that a temporary tracheotomy is necessary for any partial laryngectomy.

A **supraglottic laryngectomy** (Figs. 3.4 and 3.5) or **anterior horizontal partial laryngectomy** (AHPL; [Fig. 3.6](#)) involves removal of laryngeal structures superior to the true VCs. Both cricoarytenoid units remain. The resection may include some of the base of the tongue. Exposure and anesthetic considerations are the same as for a VPL including a temporary tracheotomy.

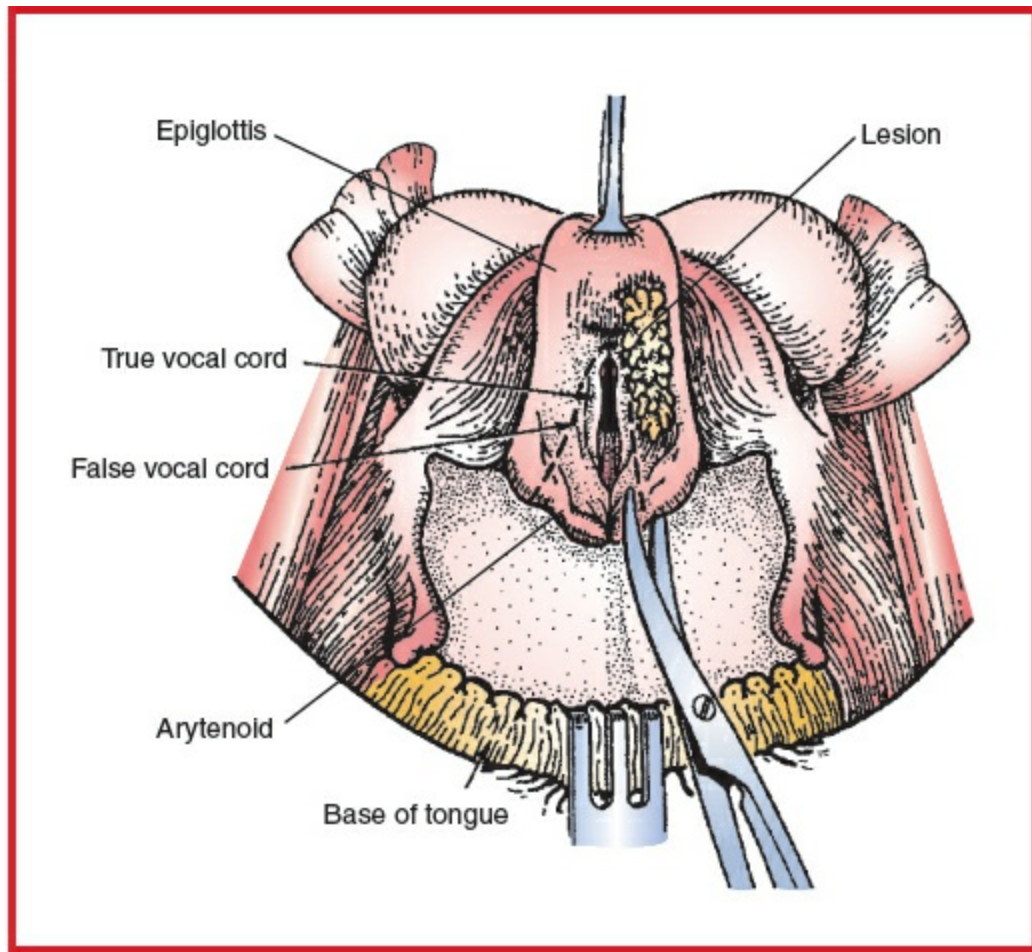


Figure 3.4 The larynx is viewed from the midline, as seen by the surgeon standing at the head of the operating table. Unless the lesion extends posteriorly to the arytenoid, the aryepiglottic fold is transected on each side by placing one blade of the dissecting scissors into the laryngeal ventricle or above the false vocal cord and the other blade in the pyriform sinus. The arytenoid on one side can be resected if the tumor extends posteriorly to involve this structure. (Reproduced with permission from Montgomery WW: *Surgery of the Upper Respiratory System*, 3rd edition. Williams & Wilkins, Baltimore: 1996.)

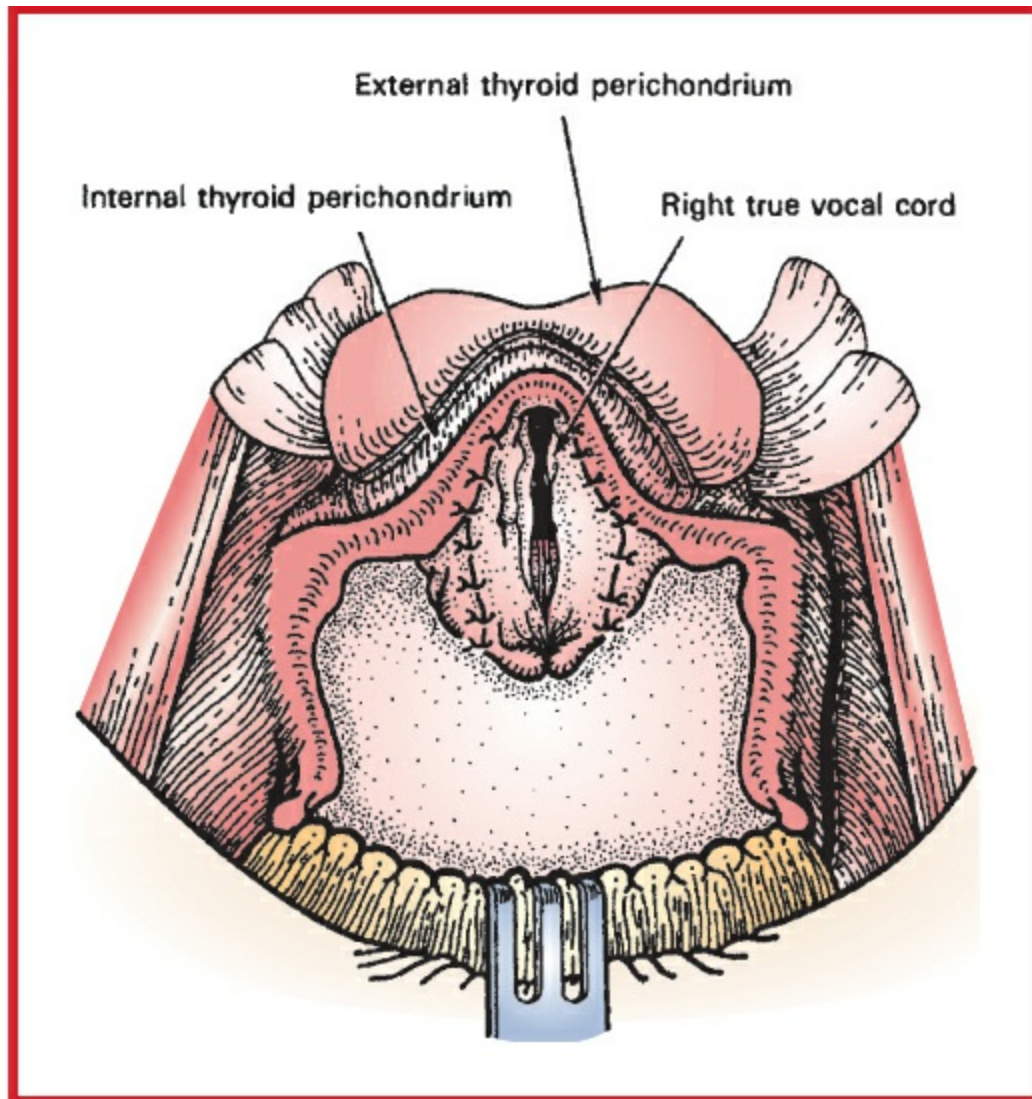


Figure 3.5 The repair following supraglottic partial laryngectomy begins by carefully approximating the margin of the mucous membrane of the pyriform sinus to the lateral margin of the laryngeal ventricle or to the margin of resection above the false vocal cord. There is usually some distortion of the true vocal cord when the repair is accomplished, as is shown on the patient's right side. The repair is continued anteriorly by placing multiple interrupted 3-0 chromic catgut sutures. (Reproduced with permission from Montgomery WW: Surgery of the Upper Respiratory System, 3rd edition. Williams & Wilkins, Baltimore: 1996.)

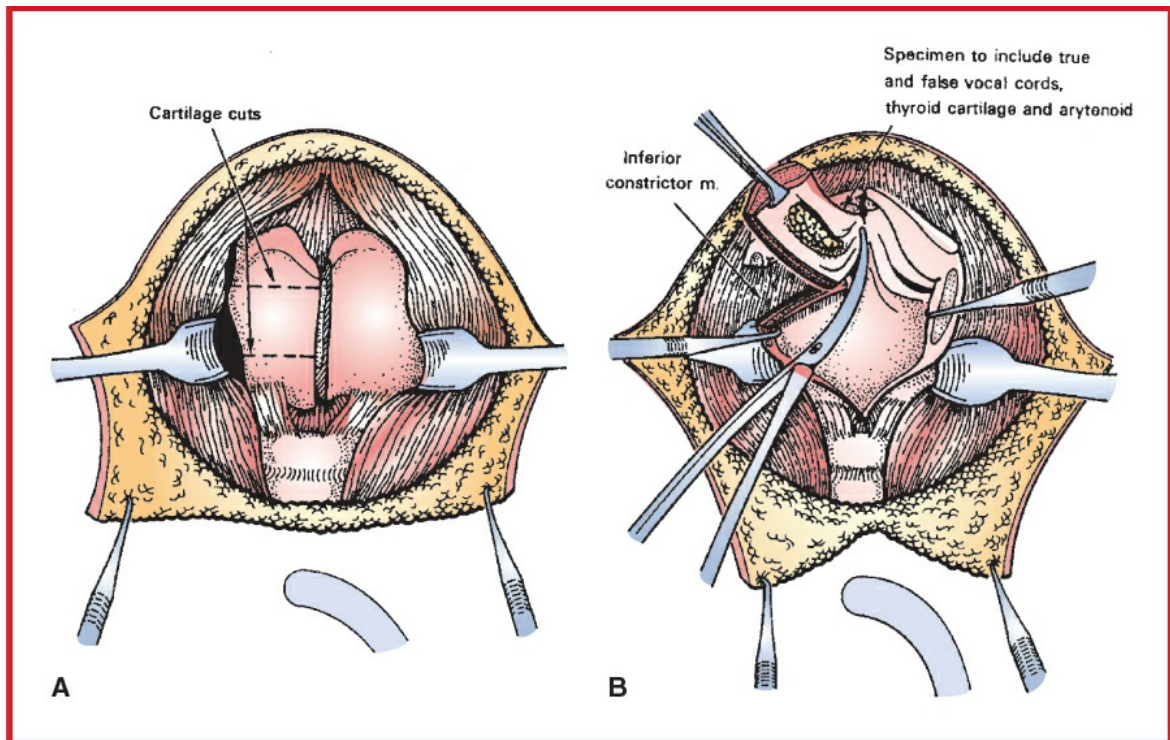


Figure 3.6 **A:** Horizontal incisions, corresponding to the mucosal incision, are made through the thyroid lamina. **B:** The specimen—including true and false vocal cords, the arytenoid, and a portion of the thyroid lamina—is resected en bloc. (Reproduced with permission from Montgomery WW: *Surgery of the Upper Respiratory System*, 3rd edition. Williams & Wilkins, Baltimore: 1996.)

A **supracricoid laryngectomy** involves removal of the larynx from the top of the cricoid ring to the hyoid bone with preservation of one arytenoid. A temporary tracheostomy is required. Cuts are made superior to the thyroid ala, through the cricothyroid membrane, and anterior to the arytenoid cartilages. The epiglottis may be included in the resection, depending upon the extent of tumor. Blunt dissection anterior to trachea into the mediastinum facilitates superior tracheal mobilization. A cricohyoidopexy (CHP), involving suturing the cricoid to the hyoid bone, is then done with three heavy sutures. If the epiglottis has been preserved, a cricohyoidoepiglottopexy (CHEP) is done. The strap muscles are then used to reinforce the closure, and drains placed.

A **total laryngectomy** involves the resection of the entire larynx, via a transverse 8-cm low midline neck incision, extended via apron incision toward a mastoid tip if a level 2–5 neck dissection is planned. The thyroid gland is often preserved, pedicled on its superior and inferior vasculature after dividing the isthmus; but a partial thyroidectomy may be included. The neopharynx is then closed (after the tracheoesophageal puncture [TEP] [see below], if that is to be done). A nasogastric tube is used for nutrition. As the remaining trachea is sutured to the anterior skin in a true tracheostoma, no tracheotomy tube or ETT is required postop unless there is marked stomal edema or mechanical

ventilation is required, which is uncommon.

A **tracheoesophageal puncture** is often done at the time of a TL. This involves the creation of a tract between the trachea and the esophagus for placement of a voice prosthesis (a one-way valve that allows airflow from the trachea into the pharynx for alaryngeal speech). The voice prosthesis may be placed at the time of the laryngectomy or later. If performed later, it is often placed using the rigid esophagoscopy (see previous section). If TEP is performed at the time of laryngectomy, the valve is generally placed simultaneously. Some surgeons prefer to place a red rubber catheter initially instead, allowing the patient to be fed via this route in lieu of a nasogastric tube, and insert the voice prosthesis later once peroral nutrition obviates the need for a tube. If a rubber catheter is used, the tube will protrude from the stoma, and care must be taken not to dislodge it during suctioning or while removing or replacing the laryngectomy tube if one is temporarily used.

A TL can be extended to include part of the hypopharynx or oropharynx. If flap reconstruction is necessary because of the extent of the tumor, options include a pectoralis major myocutaneous flap or a free flap, such as a radial free flap, to reconstruct less than a circumferential defect. If a circumferential defect following resection exists (e.g., after resection of the superior cervical esophagus and the larynx), options for reconstruction include use of a laparoscopically harvested jejunal free flap, gastric pull-up, or a tubed radial free flap possibly reinforced with a pectoralis major myogenous flap.

Postop care

Inpatient admission for 5–10 d, monitored bed for tracheostomy or free flap care, and a cuffed tracheostomy tube (and a temporary tracheotomy) will usually be required for partial laryngectomy patients or if the patient will require postop mechanical ventilation. A short tracheostomy tube following a TL may be needed if there is significant peristomal edema or if mechanical ventilation is required. General and drain management is similar to that of a neck dissection.

Usual preop diagnosis

Cancer of the larynx; intractable aspiration with resultant pneumonia unresponsive to other techniques

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Total or Partial	Pharyngolaryngectomy
Position	Supine, head extended	⇐
Special instrumentation	Major head/neck set; headlights	⇐
Unique considerations	Laryngeal tumors may distort airway. Fiberoptic intubation or other management of the difficult airway should be considered. May be combined with neck dissection	⇐
Antibiotics	Cefazolin 2 g and metronidazole 500 mg; clindamycin 600 mg	⇐
Surgical time	2–4 h; 4–8 if neck dissection added	8+ h; reconstruction required +/- free flap
EBL	50–150 mL With neck dissection: up to 250 mL Transfusions: rare	⇐ ⇐ ⇐
Postop care	Tracheotomy care, drain management, tube feedings	⇐
Mortality	<1%	⇐
Morbidity	Fistula formation: 5%: 15–20% if prior irradiation Hematoma: <4% Infection: <5% DVT/PE: <1%	1% ⇐ ⇐ ⇐
Pain score	4–6	4–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–80, occasionally older and younger
Male:Female	3:1
Incidence	Common
Etiology	Smoking; alcohol
Associated conditions	COPD; atherosclerosis; diabetes

■ ANESTHETIC CONSIDERATIONS

■ PREOPERATIVE

See preop considerations for Neck Dissection.

Airway	If a stridor at rest is present, its severity must be carefully evaluated. Patients with symptoms and signs of severe obstruction (complaints of difficulty breathing, nocturnal symptoms, sleeping upright, use of accessory muscles on inspiration at rest) will likely be difficult to ventilate and intubate.
Respiratory	Patients with severe stridor may manifest baseline hypoxemia (with or without hypercarbia) 2° ↓ airway diameter and atelectasis caused by ↓ ability to effectively handle secretions.

■ INTRAOPERATIVE

Anesthetic technique

GETA: Although tracheostomy is universally performed in TL patients, smooth emergence is still important 2° delicate suture lines. Full muscle relaxation is essential. Moderate ↓ BP (see Introduction) is desirable but may be limited by concomitant cardiovascular disease.

Induction	Slow-growing upper airway tumors can cause a high-grade airway obstruction without significant associated symptoms. Accommodated respiratory
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muscles can sustain adequate alveolar minute ventilation at rest through a 3-mm orifice, but acute deterioration occurs when a critical narrowing is reached. A preop endoscopic airway examination (PEAE) should be performed in all ENT patients with symptoms of upper airway obstruction. It is the only technique that allows visualizing the lesion and degree of obstruction directly, and assessment of respiratory mechanics and tumor mobility.

There is no universal recipe for management of advanced airway obstruction occurring at supraglottic, glottic, or subglottic levels. An awake flexible scope intubation, inhalation induction, and intravenous induction with muscle relaxants have been used successfully, but none should be considered fail-safe. The patient with a partially obstructed airway requires a team approach with cooperation of the ENT surgeon for devising optimal airway management strategies. The use of THRIVE during intubation attempts should be strongly considered in these patients.

If stridor is **absent** and airway assessment is normal, standard induction followed by DL may be appropriate. The use of VL is strongly recommended to maximize laryngeal exposure. A combined video intubation technique can be highly effective in these patients, as it permits continuous visualization of the intubation procedure both from above and below the lesion, minimizing the chance of tumor disturbance. An awake flexible scope intubation may fail with obstructing glottic lesions, either because of the inability to displace the tumor or because ETT simply would not pass. A forceful advancement of ETT off the flexible scope may cause tumor fragmentation, bleeding, and contamination of the lower airway and may precipitate complete airway obstruction.

Patients with **moderate stridor** and a significant lesion seen on PEAE, but who are considered

	possible to intubate, can be managed with an awake flexible scope intubation or an inhalation induction. Inhalation induction is best reserved for noncollapsing lesions that can be rescued by mask ventilation. A sufficiently deep and stable plane of anesthesia is essential to avoid loss of the airway (e.g., cough, laryngospasm). The ENT surgeon should be present in the OR, ready to perform rigid bronchoscopy or emergent tracheostomy. If the airway cannot be secured after two attempts, tracheostomy must be performed under controlled conditions with the patient anesthetized and breathing spontaneously. In patients with severe stridor and low probability of successful intubation, a preliminary tracheostomy should be performed under local anesthesia without any sedation. Other anesthetic considerations are similar to those in neck dissection.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	See Anesthetic Considerations for Neck Dissection. After tracheostomy, a sterile reinforced ETT (6.0–7.0-mm ID) is inserted by the surgeon and connected to an anesthesia breathing circuit. Proper ETT placement should be confirmed by presence of ETCO₂ equal bilateral breath sounds and normal airway compliance. Following tracheostomy, ↑ FiO₂ to at least 50% because the surgeon will continue working around the ETT, occasionally removing it from the patient's trachea for suture application. The position of the ETT in the tracheostoma should be monitored closely to prevent the ETT from slipping into the right mainstem bronchus.
Emergence	See Anesthetic Considerations for Neck Dissection. Discuss with surgeon about local wound. Consider iv

acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

For further discussion, see Intraoperative Considerations for Neck Dissections.

POSTOPERATIVE

See Postop Considerations for Neck Dissections.

Suggested Readings

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8. Patel A, Nouraei SA: Transnasal Humidified Rapid-Insufflation Ventilatory Exchange (THRIVE): a physiological method of increasing apnoea time in patient with difficult airways. *Anaesthesia* 2015; 70:323–9.
9. Patel A, Pearce A: Progress in management of the obstructed airway. *Anaesthesia* 2011; 66(Suppl 2):93–100.
10. See Suggested Readings for Neck Dissection.

TONSILLECTOMY AND/OR ADENOIDECTOMY

SURGICAL CONSIDERATIONS

Description

The dissection for **tonsillectomy** is carried out with the patient supine, shoulders elevated on a small pillow (Fig. 3.7). A mouth gag is inserted, and if an **adenoidectomy** is being done concurrently, adenoids are removed first with a curette, and the nasopharynx packed. The palatine tonsil is dissected from underlying constrictor and removed. Hemostasis is secured with various techniques, such as electrocautery. Tonsillectomy may be combined with **palatopharyngoplasty** for obstructive sleep apnea (OSA). If an oropharyngectomy (“radical tonsillectomy”) is being done for oncologic reasons, TORS may be used, the resection is more extensive, and the procedure may be combined with a level 2–4 neck dissection.

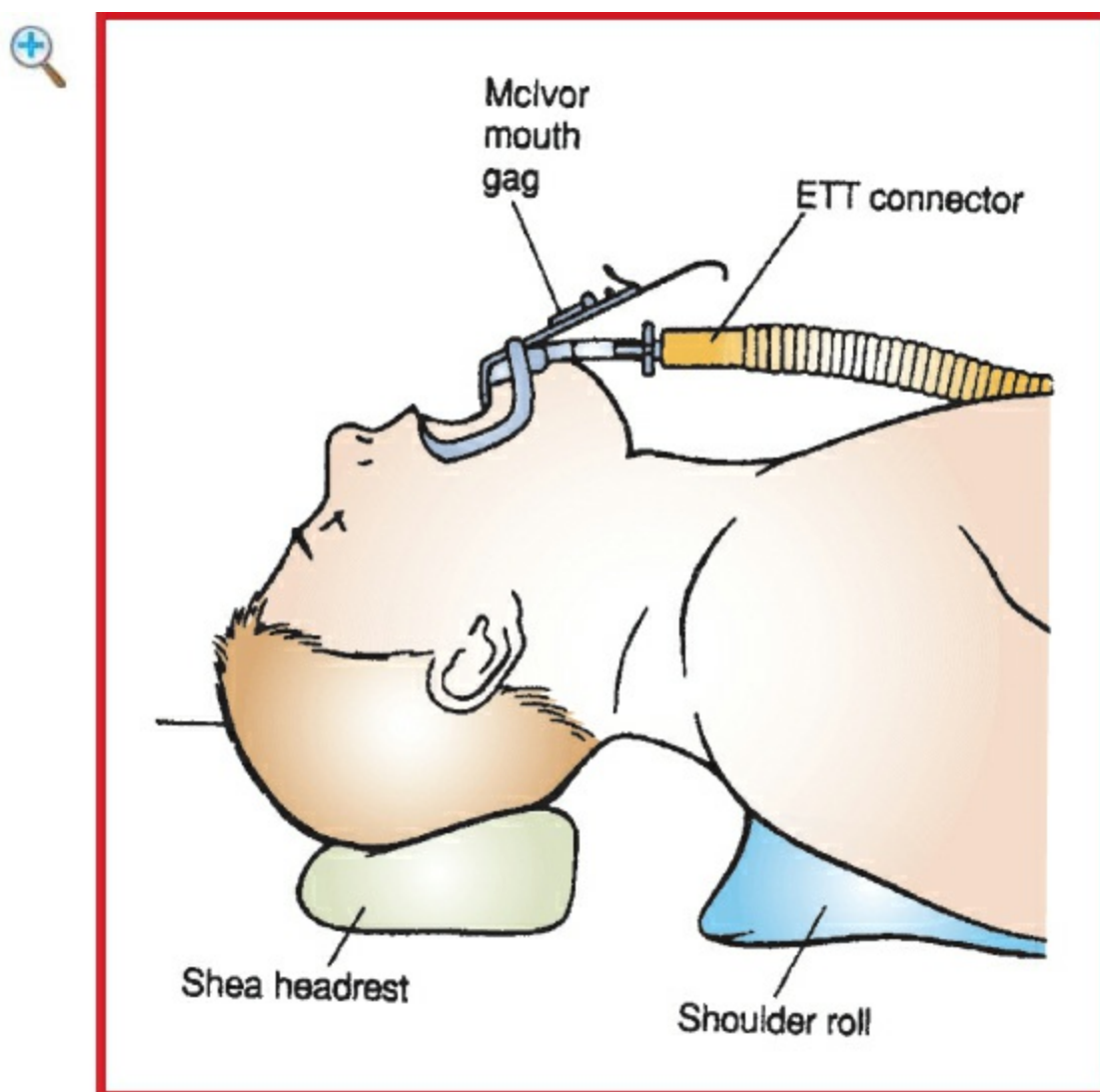


Figure 3.7 The “Rose” position for tonsillectomy.

Variant procedure or approaches

Guillotine technique (rarely used)

Usual preop diagnosis

Chronic tonsillitis and/or adenoiditis (most common); OSA; asymmetric enlargement of tonsils (to r/o cancer); nasal airway obstruction; snoring; peritonsillar abscess

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	“Rose” (supine, shoulder roll, head extended); surgeon at head of table (turned 90–180°)
Incision	Intraoral mucosal
Special instrumentation	Mouth gag (McIvor)
Unique considerations	Use of armored ETT prevents compression of tube by mouth gag. Tube should be secured to lower lip in midline
Antibiotics	Amoxicillin po
Surgical time	30–60 min
Closing considerations	If patient has OSA, a heightened sensitivity to narcotics and sedatives may make emergence more difficult. Avoid hypercapnia on emergence to prevent vasodilation and resultant bleeding. Awake extubation will provide maximum airway protection
EBL	25–200 mL. Monitor suction bottle contents and irrigation as an indication of blood loss
Postop care	Lateral position; head down; gentle suctioning
Mortality	Rare
Morbidity	Bleeding: 4% Infection: 4% Delayed bleeding: 3.2% Aspiration: Rare

	Tooth damage: Rare
Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	2+ yr
Male:Female	1:1
Incidence	50 per 10,000 children in the United States
Etiology	Chronic infection; OSA (most common indication); peritonsillar abscess; snoring; cancer
Associated conditions	Nonspecific

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Most patients are young and otherwise healthy; however, a subset of both pediatric and adult patients may present with OSA or URI. For many children, this is their first anesthetic; therefore, it is imperative to ✓ family Hx for anesthetic problems. Most adult and pediatric patients are discharged from the hospital on the day of surgery. If TORS is being done, a neck dissection may also be planned.

Airway	In patients with Hx of snoring or OSA, the probability of difficult mask ventilation and/or intubation may be high, 2° airway characteristics peculiar to these patients (see Introduction and Anesthetic Considerations for Reconstructive Surgery for Sleep-Disordered Breathing). In children with short-stature syndrome, including achondroplastic dwarfism and selected cases of Down syndrome, atlantoaxial subluxation and stenosis of the spinal canal may be present, and neck extension in these patients should be avoided.
Respiratory	In patients presenting with Sx of acute URI (purulent sputum or nasal secretions, fever, etc.), the general recommendation is to postpone any elective

	procedure until symptoms have abated, usually within 7–14 d. Tests: As indicated from H&P
Dental	The presence of chipped, loose, or broken teeth should be documented preop, and the patient (or parents, if appropriate) should be informed that there is a possibility of further tooth damage/dislodgement by the mouth gag applied by the surgeon.
Cardiovascular	Rarely, chronic airway obstruction (e.g., OSA) with hypoxemia may → pulmonary HTN and right heart failure. Patients with Down syndrome may have concomitant CHD. Tests: As indicated from H&P
Hematologic	✓ for recent aspirin use and Hx of excessive bleeding following minor trauma or tooth extraction.
Laboratory	CBC; other tests as indicated from H&P
Premedication	Sedative premedication is routine but should be avoided in OSA patients and patients with Sx of upper airway obstruction. Some surgeons request preop administration of an antisialagogue to achieve a dry surgical field.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Balanced GETA is most commonly used in adults. Adequate muscle relaxation is an essential surgical requirement: it facilitates surgical exposure and prevents patient from swallowing. Continuous control and protection of the airway is another major objective, along with smooth emergence from anesthesia and prevention of early postop laryngospasm. Specific airway considerations include the possibility of difficult mask

ventilation and/or difficult intubation 2° large tonsils and ↓ pharyngeal space, as well as the possibility of intraop obstruction of the ETT with the mouth gag placed by the surgeon.

Induction	An intravenous induction with propofol (2 mg/kg) and fentanyl (2–3 mcg/kg) is suitable for the majority of patients. If difficulty with mask ventilation is encountered, an oral airway is preferred to a nasal airway, 2° the possibility of trauma to the hypertrophied adenoid tissue and the resultant brisk bleeding. Patients with peritonsillar abscess may have trismus, which usually resolves after induction of anesthesia and administration of a muscle relaxant. Great care should be exercised with introduction and manipulation of laryngoscope blade to avoid bleeding 2° inadvertent trauma to the enlarged tonsils and to prevent soiling of the airway in patients with a peritonsillar abscess. The lightwand technique of tracheal intubation should be avoided in these patients. The use of a small (6.0 mm ID) reinforced ETT is preferred over an oral RAE ETT because of better resistance to kinking/obstruction by the mouth gag. The use of an FLMA can safely protect the airway during adenotonsillectomy and reduce the incidence of postextubation complications in adults and children. Its use for this procedure, however, has not become popular in the United States.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Opening of a mouth gag and the tonsillectomy itself constitute powerful noxious stimuli; therefore, an adequate depth of anesthesia must be maintained. It is essential to verify the ETT position before and after Boyle-Davis or McIvor mouth gag application (e.g., chest movement, bilateral and equal breath

sounds, and normal PIP) since the ETT may be obstructed, dislodged, kinked, or inadvertently advanced into a mainstem bronchus, or the patient's trachea can be prematurely extubated when the mouth gag is repositioned/removed. Moderate ↓ BP is desirable but not obligatory; and judicious use of vasoactive drugs (see Introduction) is preferred to deep levels of anesthesia 2° short duration of the procedure.

Low-dose remifentanyl infusion (see Introduction) as part of either balanced or TIVA technique will provide superior hemodynamic stability and facilitate smooth emergence from anesthesia. For a balanced inhalational technique, desflurane and sevoflurane are the preferred agents for these short surgical procedures. Isoflurane may be favored in pediatric patients 2° ↓ agitation on emergence.

The mouth gag may be repositioned several times during the procedure and is removed only at the end of the case. Vigorous patient movement while the mouth gag is hooked on the Mayo stand may cause C-spine injury; thus, neuromuscular blockade should usually be maintained until the end of the surgery.

Prevention of PONV is of major importance. If a throat pack is not used by the surgeon, the patient's stomach should be suctioned at the end of the case, and further gastric emptying may be facilitated by administration of metoclopramide (1–2 mg/kg up to 20 mg iv). Administration of 5-HT₃ blockers (e.g., ondansetron 0.1 mg/kg up to 4 mg iv), with or without steroids (e.g., dexamethasone 4–12 mg iv), will usually provide adequate antiemetic prophylaxis.

Emergence

Removal of a throat pack (if used by the surgeon) should be verified before extubation. Care must be exercised while suctioning the oropharynx to avoid bleeding. The use of the rubber-tipped vs. Yankauer tip suction catheter may be preferred. Extubation should be smooth and performed after the patient is

able to follow commands because this usually signifies return of protective airway reflexes. If extubation is to be performed at deeper levels of anesthesia (e.g., in small children), the patient should be placed in the lateral decubitus, head-down position ("tonsillar position") to protect the airway from soiling 2° postop bleeding or aspiration of gastric contents. The patient should remain in this position until fully awake. This method of extubation is time and labor consuming, requires adequate depth of anesthesia to prevent postextubation laryngospasm, and does not prevent postextubation upper airway obstruction. Careful attention is required to protect patient's pressure points and to prevent the patient from rolling supine. Furthermore, with the head-down position, venous pressure in the surgical wound is increased with the possibility of provoking postop bleeding. This method of extubation cannot be advocated for routine use in adults. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements	IV: 18 ga × 1 (adult) 20 ga × 1 (child) BSS @ 2–4 mL/kg/h (adult)	Blood loss is typically minimal. Adequate hydration should continue in the immediate postop period. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	See Anesthetic Considerations for Panendoscopy.
Complications	ETT obstruction dislodgement Bleeding	ETTs of different sizes should be readily available.

Blood loss sometimes may be difficult to assess 2° drainage into stomach, especially if a throat pack is not used. Careful observation of suction canisters and close communication with the surgeon are essential.

▼ POSTOPERATIVE

Complications

Retention of throat pack

Manifested by immediate symptoms of upper airway obstruction. Under direct laryngoscopy, remove pack with Magill forceps.

Laryngospasm

A relatively common complication, particularly in the pediatric population. Rx: 100% O₂ via mask ventilation with jaw thrust and CPAP. Rapid-sequence induction (RSI) and direct laryngoscopy/intubation for persistent laryngospasm.

Bleeding tonsil

Bleeding usually occurs at a slow pace; large volumes of blood may be swallowed, and hypovolemia may occur before any bleeding is detected. Frequent swallowing should alert to the possibility of ongoing

	hemorrhage. ET intubation (RSI) may prove difficult due to poor visualization secondary to bleeding (effective suction must always be available) and upper airway edema. Postobstructive (negative pressure) pulmonary edema (POPE) POPE is a rare complication; however, it may occur in patients with preexisting symptoms of severe upper airway obstruction. Although most cases resolve spontaneously, some patients may require reintubation and postop ventilatory support.
Multimodal pain management	Adult tonsillectomy is a painful procedure However, most patients are discharged home unless unable to sustain po intake. Fentanyl is the preferred PACU drug, but the addition of morphine or hydromorphone may be necessary in some patients. Ibuprofen works as well as acetaminophen/opiate combinations. Ice water or lidocaine-containing mouth rinse can provide temporary pain relief.
Tests	Hct If blood loss is suspected.
ERAS	PONV protocol (see Encourage early enteral

Suggested Readings

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3. de Almeida JR, Li R, Magnuson JS, et al: Oncologic outcomes after transoral robotic surgery: a multi-institutional study. *JAMA Otolaryngol Head Neck Surg* 2015; 141(12):1043–51.
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GLOSSECTOMY

SURGICAL CONSIDERATIONS

Description

Partial glossectomy is done for many tongue malignancies. During this procedure, nasal intubation may be helpful but is not mandatory. Complete muscle relaxation is necessary. Surgical exposure is improved with a bite block placed laterally and a lip retractor. The lesion is resected with electrocautery. Reconstruction is either advancing the tongue on itself or to the floor of the mouth, or a radial free flap may be indicated. Depending on the size and location of the tumor and whether a free flap is planned, a **tracheostomy** may be indicated; or nasal intubation alone may suffice for a few days. A short course of steroids may help reduce tongue edema. An NG tube is often used for postop feeding.

Variant procedure or approaches

Glossectomy can be done with a **neck dissection** (in which case the neck dissection may be in continuity or done via a separate sterile field) or as part of a composite resection (which includes a **partial marginal or segmental mandibulectomy** and may include an osteocutaneous free flap reconstruction and hence a tracheotomy). Rarely, a **total laryngectomy** is also done.

Usual preop diagnosis

Neoplastic disease of the tongue or adjacent structures (e.g., alveolus, floor of mouth) with involvement of the tongue

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Partial	If with composite resection
Position	Supine	⇐
Incision	Intraoral	⇐ + suprahyoid approach/neck approach
Special instrumentation	Mouth gags per surgeon	⇐
Unique considerations	Hypotensive anesthesia	⇐
	Nasal intubation	⇐
	Possible tracheostomy	⇐ (required)
	Steroids, if tracheostomy not performed	
Antibiotics	Cefazolin 2 g iv with, metronidazole 500 mg	⇐
Surgical time	1–3 h	3–8 h
Closing considerations	Local or free flap. Nasal ETT 24–48 h possibly if minimal tongue edema	Flap repair required Tracheostomy postop
EBL	50–150 mL	100–300 mL

Postop care	Intubated 24–48 h	Tracheostomy care
Mortality	<1%	⇐
Morbidity	Bleeding Infection Aspiration	Aspiration Bleeding Infection
Pain score	1–2	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adults
Male:Female	3:1
Incidence	Uncommon
Etiology	Neoplasia
Associated conditions	Nonspecific

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

See Preoperative Considerations for Neck Dissection.

Airway	Thorough airway assessment is mandatory. Nasal intubation may or may not be required, depending on location of tumor (side vs. base of tongue [BOT]) and surgeon's preference (inquire preop).
Premedication	Standard. Preop administration of antisialagogue (e.g., glycopyrrolate 0.2 mg iv) may improve operating conditions—check with the surgeon.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Complete muscle relaxation is an essential surgical requirement. For partial glossectomy, smooth extubation is desirable but not mandatory unless skin graft was used for closure (graft hematomas are the primary cause of skin graft failure). Intraop infiltration with a local anesthetic effectively supplements intraop and postop analgesia.

Induction	With a normal airway and submucosal location of tumor on the side of the tongue, standard induction is appropriate. With mobile, fungating BOT tumors, awake flexible scope intubation may be a safer approach. Similar to the considerations for laryngectomy (see Anesthetic Considerations for Laryngectomy), direct visualization of the passage of the ETT through the VC (DL or VL) is essential, to reduce the risk of tumor fragmentation with seeding of the lower airway or bleeding. Among the video laryngoscopes, the Pentax Airway Scope™ (AWS) is beneficial for use in BOT pathology, as the blade lodges under the epiglottis. Large immobile tumors may make ET intubation by conventional means extremely difficult or impossible. Awake nasal FOI may be the technique of choice in these cases. Transoral robotic surgery (TORS) is increasingly used for many BOT operations, as well as for procedures including supraglottic laryngectomy. See discussion below.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	During a partial glossectomy, closure is quick, and short-acting inhalational agents (desflurane, sevoflurane) may be favored; TIVA may also be employed. ↓ BP (see Introduction), although desirable, is not mandatory for partial glossectomy. If tracheostomy is performed as part of the planned procedure, the usual considerations apply (see

	Anesthetic Considerations for Laryngectomy, and Tracheostomy).	
Emergence	With prolonged procedures, especially involving BOT tumors, it is prudent to assess the degree of upper airway edema prior to extubation (it may be impossible to reintubate the patient if postextubation upper airway obstruction occurs). Rapid recovery with full return of protective airway reflexes is essential in patients after partial glossectomy. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Complications	Vagal reflexes: ↓ HR, ↓ BP, mediated by surgical manipulation at the base of the tongue	Rare. Rx: notify surgeon; deepen anesthetic; iv atropine (0.4 mg).

For further discussion, see Intraoperative Considerations for Neck Dissection.

▼ POSTOPERATIVE

Complications	Postop airway obstruction	May occur 2° ↑ airway edema; emergent tracheostomy may be required. Prophylactic intraop steroids are beneficial.
Multimodal pain management	PCA Parenteral opiates	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and

For further discussion, see Neck Dissection.

Suggested Readings

1. Feldman MA, Patel A: Anesthesia for eye, ear, nose, and throat surgery. In: Miller RD, ed. Miller's Anesthesia, 7th edition. Elsevier, Philadelphia: 2010, 2357–88.
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ANESTHETIC CONSIDERATIONS FOR TRANSORAL ROBOTIC SURGERY

TORS is becoming the technique of choice for treating a wide range of primary oropharyngeal and laryngeal cancers and for performing multilevel surgery for OSA, including base of the tongue (BOT) reduction and uvulopalatopharyngoplasty (UPPP). With TORS, surgery proceeds from the inside out, starting with the mucosal surface and extending outward, eliminating the need for skin incisions or transmandibular access, and also decreasing the need for tracheostomy. A highly magnified view of the surgical pathology, increased depth perception, and elimination of surgical tremor lead to shorter hospital stays and better functional outcomes affecting long-term speech and swallowing.

The essential anesthetic requirements for precision TORS surgery include the ability to safely share the patient's airway with the surgeon; enhanced situational awareness; a clear, quiet surgical field; and the absence of patient movement to ensure optimal surgical conditions to allow preservation of function. As usual, the anesthetic technique should include a stable plane of anesthesia, nonstimulating emergence, rapid return of consciousness and protective airway reflexes, and adequate PONV prophylaxis.

PREOPERATIVE

For preop considerations, see Preoperative Considerations for Neck Dissection and Glossectomy.

Airway

Thorough airway assessment is mandatory (see Introduction) and should include assessment of the patency of the nasal passages for nasotracheal intubation.

Premedication	Standard. Preop administration of an antisialagogue (e.g., glycopyrrolate 0.2 mg iv) may improve operating conditions—check with the surgeon.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of superficial cervical plexus nerve blocks for postop pain management. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Complete muscle relaxation and clear surgical field are the essential surgical requirements.

Induction and airway management	IV induction is used in the majority of patients. If the airway is compromised, strategies for atraumatic tracheal intubation should be devised and discussed with the surgeon (see Introduction and Anesthetic Considerations for Laryngectomy). A nasal intubation with a small-diameter (6.0 mm ID) microlaryngeal tracheal (MLT) tube is preferred, to improve the view and accommodate surgical instrumentation. The MLT tube is sutured in place through the nasal septum by the surgeon, and a 120° curved metal ETT connector is used to direct the anesthesia circuit over the patient's head and provide unrestricted intraoral access for the surgeon. For TORS-assisted laser surgery, patients may be intubated orally or nasally with a laser ETT. With nasal intubation, the anesthesiologist must ensure sufficient depth of ETT placement to prevent the accidental withdrawal of the ETT into the larynx during surgical manipulations and/or extending
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	patient's head intraop. A sudden air leak may immediately compromise the field during precision surgery and also create an airway fire hazard during the use of laser or electrocautery. The unrecognized ETT displacement out of the trachea and into the glottis may prompt the anesthesiologist to repeatedly add more air to ETT cuff, which overtime may lead to compression of the anterior branch of the recurrent laryngeal nerve and VC injury. Special attention should be directed to adequately protecting the patient's eyes with special safety goggles to prevent accidental injury due to heavy surgical instrumentation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Increased situational awareness is required at all times, as TORS is performed with OR lights switched off. All the lines, anesthesia circuit, and patient's position on the OR table must be secured and checked. Any intraop adjustment of equipment or the OR table by the anesthesiologist should be closely coordinated with the surgeon. BIS-type-guided TIVA with propofol and opioid represents the author's anesthetic technique of choice. The full synergistic effect of propofol with remifentanil allows quick titration of the anesthetic to the desired clinical effect. Intermittent iv fentanyl or sufentanil boluses, as well as IV administration of acetaminophen, are warranted for supplementing propofol/remifentanil TIVA, as postop pain after TORS is usually moderate. Large doses of iv fentanyl/sufentanil are usually unnecessary.
Emergence	Rapid recovery with full return of protective airway reflexes is essential. The majority of patients are

	extubated routinely after adequate reversal of NMB. A staged extubation using an airway exchange catheter (AEC) is rarely required but may be necessary following prolonged procedures, especially involving BOT, posterior airway, or laryngeal work. If postop airway patency is deemed problematic, patient should be left intubated. Postop tracheostomy is rarely indicated. Discuss with surgeon local wound infiltration or nerve block (e.g., superficial cervical plexus block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Complications	Vagal reflexes: ↓ HR, ↓ BP, mediated by surgical manipulation at the base of the tongue	Rare. Rx: notify surgeon; deepen anesthetic; iv atropine (0.4 mg).
Blood and fluid requirements	IV: 18 ga × 1. BSS @ 3–4 cc/kg/h.	Blood loss minimal. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors. ± art line.	A-line monitoring may be indicated by patient's medical history.
Positioning	Head may be elevated 20–30° Table 180° ✓ and pad pressure points ✓ and protect the eyes with goggles	

POSTOPERATIVE

Complications	Postop airway obstruction	May occur 2° ↑ airway edema. Prophylactic intraop steroids are beneficial.
Multimodal pain	PCA/parenteral opiates	Emphasize multimodal

management	Multimodal analgesia	approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

MAXILLECTOMY AND ORBIT EXENTERATION

SURGICAL CONSIDERATIONS

Description

Partial maxillectomy and/or **orbital exenteration** (OE) may be required for some malignancies. In an **OE**, the contents of the orbit are removed, including the eyeball and its attached extraocular muscles posterior toward the conus. This can be done via an incision that is made around the upper and lower eyelashes, occasionally with an extension that includes a limited incision for an external ethmoidectomy; a lateral rhinotomy incision is rarely needed. If the eyelid skin is not involved by tumor it is preserved other than the eyelashes and lid margins. Bradycardia may occur with dissection near the optic nerve.

A **maxillectomy** may be done with the orbit exenteration or independently. If the superior maxilla is removed as part of the OE, with palate preserved, the incision is the same as for just the OE as access to the superior maxilla is provided by the exenteration. If only a partial maxillectomy is to be done preserved, whether the hemipalate is to be preserved or not, the resection can often be done fully through an intraoral incision. For example, if the hard palate is to be preserved, the approach is similar to that of a Caldwell-Luc. If the hard palate is to be removed (as for a tumor involving the alveolar ridge and extending into the maxillary sinus), then the transoral resection of the alveolar ridge usually provides adequate access for the complete resection. Rarely, a transfacial partial Weber-Ferguson incision is needed, or endoscopic equipment may be used to supplement access. Osteotomies are generally made with power equipment, though osteotomes and rongeurs may at times be adequate.

Reconstruction of palate defects are generally done with an obturator with or without a split-thickness skin graft placed intraoperative on exposed soft tissue. A free flap is selectively an alternative, such as in a younger patient or when too little hard palate and too few teeth remain to stabilize a large obturator. There are a number of options to reconstruct orbit defects depending on whether skin is also resected. These options include the pericranial-galeal flap discussed for anterior cranial base surgery, a free myogenous flap such as an anterolateral thigh (ALT) flap or a radial free flap.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Maxillectomy	Orbital Resection
Position	Head raised 30°, back raised (reduces bleeding)	⇐
Incision	Intraoral	Eyelash incisions
Special instrumentation	Paranasal sinus set; headlight	⇐
Unique considerations	Nasal pledgets	⇐; see reconstruction
Antibiotics	Cefazolin 2 g iv with metronidazole 500 mg; or clindamycin 600 mg	Cefazolin 2 g
Surgical time	2–4 h	1.5–3 h
EBL	100–300 mL	25–100 mL
Postop care	Same as endoscopic surgery	Reconstruction dependent
Mortality	Rare	Very rare
Morbidity	Bleeding: 5% Transient diplopia: 5% Infection: 5%	⇐ ⇐

Pain score	2–3 if no oral entry; 4–6 if palate	2–4
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■ PATIENT POPULATION CHARACTERISTICS		
Age range	50–70	40–80
Male:Female	3:1	2:1
Incidence	Common	⇐
Etiology	Malignant tumors, selected benign tumors	Malignancies, rare benign processes
Associated conditions	Tobacco, alcohol, rare others	⇐

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Typically, these patients are older males who may have concomitant cardiopulmonary disease. Their nutritional status is not usually affected.

Airway	For patients presenting for maxillectomy, careful inspection is warranted, although it is unlikely that the disease process will → airway compromise. With large tumors, space available for laryngoscope blade manipulation may be somewhat decreased. XRT typically is done postop.
Cardiovascular	Preop evaluation should focus on the patient's suitability for controlled ↓ BP: may be contraindicated in patients with advanced cerebrovascular or cardiovascular disease. Tests: ECG; other tests as indicated from H&P
Laboratory	Tests as indicated from H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with

sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Bifrontal craniotomy may be required if the disease process extends into the paranasal sinuses, dura, and anterior skull base. These procedures can be lengthy and very stimulating; the use of potent opioids (see Introduction) is beneficial. The essential surgical requirement is ↓ bleeding into the field. Intraop hemorrhage may occasionally be brisk and substantial (in excess of 500 cc) during maxillectomy. Maintaining adequate circulating volume is crucial. Promotion of rapid awakening with full return of protective airway reflexes presents additional challenges to the anesthesiologist.

Induction	An opioid-based technique, as outlined previously (see Introduction). Any iv induction agent can be safely used; ↓ dose in elderly and patients with significant cardiac disease. Oral tracheal intubation is performed with a small (6.0 mm ID) reinforced ETT to facilitate surgical manipulation inside patient's mouth. With large tumors, especially on the right side, care is necessary when performing DL to avoid bleeding. ETT should be secured over the chin. Placement of a small-diameter esophageal temperature probe will not interfere with the surgical field.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Controlled hypotension is often requested. A combination of opioid (e.g., remifentanyl) or sufentanil infusion with a potent, quickly titratable inhalational agent (e.g., desflurane), supplemented with iv β -blockers, if required, is usually sufficient. After the specimen has been removed, surgical stimulation decreases, and the inhalational anesthetic should be significantly reduced to promote rapid awakening.

Emergence	The patients are routinely extubated, even after extended procedures. The stomach should be suctioned at the end of the case (throat pack is removed only by the surgeon to facilitate placement of a dental prosthesis). Stomach emptying should be facilitated by administration of iv metoclopramide (10–20 mg). Nasal passages may be packed → obligate mouth breathing; oropharynx must be suctioned thoroughly and carefully. No dressing is applied postop. Washing blood from the patient's face is frequently accompanied by vigorous movement of the patient's head and ETT. If a sufficient dose of opioids has been used, the patient will tolerate this well without coughing/gagging/straining while breathing spontaneously. Extubation must be smooth, with full return of protective airway reflexes. Hemoptysis may occur after extubation, and the patient's mouth should be promptly covered with an O₂ mask, to avoid contamination. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16 ga × 1 BSS @ 5–7 cc/kg/h	Blood loss minimal to severe. It can be sudden and substantial. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors ± arterial line	A-line monitoring is warranted for anterior cranial base surgery or in patients with significant cardiovascular disease.
Positioning	Head elevated 20–30° Table may be turned 90–180°	

	✓ and pad pressure points	
Complications	✓AEyes	A risk of air embolism, with surgery on the paranasal sinuses, should be kept in mind.
	Dysrhythmias	Dysrhythmias may occur 2° significant surgical stimulation and breakthrough adrenergic responses.
	Hypotension ↓↓ HR	↓ BP may be sudden 2° brisk blood loss. Trigeminocardiac reflex may occur during intraoperative manipulation of the trigeminal nerve.
	Brain edema	Careful fluid management, not to promote/exacerbate brain edema associated with the bifrontal craniotomy.

POSTOPERATIVE

Complications	Occult postop bleeding	Postop bleeding may cause the patient to swallow large quantities of blood, which represents an aspiration risk and may manifest as ↓ BP.
	Pneumocephalus	Persistent pneumocephalus (if the paranasal sinuses were opened) with symptoms of increased ICP.

Multimodal pain management	PCA Parenteral opioids	Pain is moderate. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Reading

1. Codère-Maruyama T, Moore A: Anesthesia in the elderly patient undergoing otolaryngology head and neck surgery. Clin Geriatr Med 2018; 34(2):279–88.

COMPOSITE RESECTION WITH MARGINAL OR SEGMENTAL MANDIBULECTOMY (AND NECK DISSECTION)

SURGICAL CONSIDERATIONS

Description

A composite resection means that part of the mandibular alveolar ridge is resected, often along with the floor of mouth and a part of the tongue, generally for advanced squamous cell carcinoma, though occasionally for osteoradionecrosis following complications of prior irradiation or for other malignancies such as osteosarcoma. In a **marginal mandibulectomy**, bone inferior to the plane of the inferior alveolar nerve (which runs just below the teeth and provides dental innervation and cutaneous sensation to the lower lip and chin) is preserved. In a **segmental mandibulectomy**, a through-and-through segment of bone is removed such that there is a bone gap. A marginal mandibulectomy may at times be reconstructed with intraoral advancement flaps in an edentulous patient, or a radial free flap may be indicated. A segmental resection generally is best reconstructed with an osteocutaneous free flap, such as fibula free flap or for a short distance an osteocutaneous radial free flap. A composite resection generally requires a tracheostomy although an intraoral marginal mandibulectomy repaired without a free flap may not necessitate one.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Composite Resection (with MRND)	... and Free Flap
Position	Supine, head extended, and turned to opposite side	⇐ + preparation of 2nd sterile site, such as arm or leg; two surgical teams
Incision	Intraoral and neck	⇐ and donor site
Special	Headlights, suction,	⇐ donor site specific

instrumentation Unique considerations	cautery, lip retractor Armored ETT converted to, or initial, tracheostomy	Dependent on 2nd site
Antibiotics	Common is ampicillin or cefazolin 2 g iv ($\pm$ metronidazole 500 mg) or clindamycin 600 mg	$\Leftarrow$
Surgical time	3–6 h	$\Leftarrow$ add 3–4 h
EBL	150–400 mL; transfusions uncommon	$\Leftarrow$; little additional loss
Postop care	Drain management, monitor for wound breakdown	$\Leftarrow$ + monitored bed for flap surveillance
Mortality	<1%	$\Leftarrow$
Morbidity	Bleeding: <4% Infection $\pm$ fistula: <10% DVT/PE: <1% Medical: MI < 1% Pneumonia: <3%	$\Leftarrow$ Flap failure; return to OR < 5% Donor site morbidity
Pain score	4–8	$\Leftarrow$

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–70, occasionally older and younger	$\Leftarrow$
Male:Female	3:1	$\Leftarrow$
Incidence	Common	$\Leftarrow$
Etiology	SCC and other malignancy	$\Leftarrow$
Associated conditions	Smoking and alcohol, COPD, atherosclerotic heart disease	$\Leftarrow$

NECK DISSECTION

Description

The goal of a neck dissection is a lymphadenectomy either to remove known cervical metastases (an N_+ neck) or as a staging procedure in a patient apparently without clinically or radiologically involved nodes (an N_0 neck) but at significant risk nevertheless. The most common indication by far is in the management of squamous cell carcinoma (SCC) of the head and neck. For the purpose of discussing a neck dissection the neck is divided into five levels. Level 1 is the tissue that is inferior to the mandible, anterior to the posterior belly of the digastric muscle, and superior to the hyoid bone, including the submental triangle between the left and right anterior belly of the digastric muscles. The most prominent structure in level 1 is the submandibular gland. Level 2 through 4 includes the tissue beneath and anterior to the sternocleidomastoid muscle (SCM): level 2 is superior to the hyoid bone, level 3 is between the hyoid and cricoid bone, and level 4 is inferior to the cricoid. Level 5 contains the tissue between the posterior edge of the SCM and the anterior edge of the trapezius muscle.

There are several different kinds of neck dissection, depending on which levels are removed:

1. **Radical neck dissection (RND) or modified radical neck dissection (MRND):** this involves the removal of levels 1–5. Except in markedly advanced disease, seldom is an RND (which removes the internal jugular vein, SCM, and accessory [XI] nerve necessary. An MRND preserves one or more of those three structures, generally at least the XI nerve. An MRND (or RND) is generally done only when there is N_+ disease. Similar surgery may also be necessary to remove extensive tumors other than SCC. An extended neck dissection involves removal of additional tissue, such as muscles deep to the superficial layer of the deep cervical fascia. MRND is a common additional procedure done for an N_+ neck at the time of removal of a primary SCC of the head and neck.
2. **Selective neck dissection (SND)** removes less than all five levels, such as for N_0 disease when adjacent surgery is planned or in an effort to ascertain there is no histologically demonstrable disease, thereby avoiding postop irradiation to the primary site and neck. The two most common types of SND are as follows:
 - a. **Supraomohyoid neck dissection (SOHND):** This removes levels 1–3, generally in a patient with an oral cavity SCC with a depth of >3.5 – 5 mm at risk for nodal involvement despite a clinically N_0 neck.
 - b. **Lateral neck dissection:** Removes levels 2–4, generally in N_0 patient with glottic or supraglottic carcinoma in whom, again, it is hoped that postop RT may be avoided. This may also be done in selected oropharyngeal carcinomas as part of

resection of the primary, again to further stage the malignancy and tailor subsequent intervention.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	MRND (or RND)	SOHND
Position	Supine, head extended and turned to opposite side; table 90° or 180°	⇐
Incision	Common incisions include a reverse hockey (submandibular resting skin tension line [RSTL] toward mastoid tip then extended inferiorly along anterior border of trapezius muscle); utility/hockey; superior and inferior neck RSTL (McFee)	~7-cm RSTL incision ~3 cm inferior to the mandible, beginning anterior to the SCM extending parallel to the hyoid as anteriorly as needed
Special instrumentation	Major head and neck set; headlights	⇐
Unique considerations	Avoid paralysis until after marginal mandibular nerve and XI identified. If bilateral, laryngeal edema may occur; dissection near carotid body and vagus may produce transient bradycardia	⇐
Antibiotics	Cefazolin 2 g iv (with	⇐

	metronidazole 500 mg if oral or pharyngeal entry planned) or clindamycin 600 mg	
Surgical time	2–4 h	1.5 h
EBL	50–150 mL; transfusions rarely indicated	⇐
Postop care	Drain management	⇐
Mortality	<1%	⇐
Morbidity	Unanticipated neuropathy (e.g., XII, X, VII marginal branch, recurrent laryngeal nerve, and phrenic nerve) <1% Chyle leak <2% if level 4 dissected Hematoma <4% Infection <5% DVT/PE ~1% XI weakness expected Shoulder pain ~20%	⇐
Pain score	4–8	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–80, occasionally older and younger	⇐
Male:Female	3:1	⇐
Incidence	Common	⇐
Etiology	1° site SCC; smoking and alcohol	⇐
Associated conditions	COPD, atherosclerotic heart disease	⇐

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Many of these patients are older males with a prior of tobacco and ETOH use and have a high incidence of associated cardiopulmonary diseases. Nutritional status of the patients may be poor and weight loss should be stabilized before surgery. Neck dissections are rarely associated with significant blood loss.

Airway	Airway management may be difficult 2° limited head and neck mobility, ↓ mouth opening, distorted airway, ↓ pharyngeal space and fixation of tissues 2° tumor expansion, radiation fibrosis or previous surgeries. Backup plans for airway management are essential, and a review of preop findings on indirect laryngoscopy and/or CT scan with the surgeon may be helpful in planning a strategy for intubation. If a difficult intubation is foreseen, awake FOI may be the technique of choice (see Anesthetic Considerations for Laryngectomy). Emergent tracheostomy may technically be very difficult 2° advanced disease, neck edema, previous surgeries and distorted anatomy.
Respiratory	High incidence of chronic bronchitis and COPD. Tests: CXR; preop ABG in patients with advanced COPD, CO₂ retention, and O₂ dependence. Flow-volume loops may be helpful in patients with symptoms of partial airway obstruction.
Cardiovascular	Careful assessment of cardiac risk factors and functional status. Documenting asymptomatic neck bruits or existing carotid artery stenosis, as well as any Sx of compromised cerebral circulation, is important (see Emergence, below). HTN must be controlled preop especially in patients presenting for radical neck dissection and flap reconstruction. Uncontrolled HTN in these patients carries additional risk of exaggerated hemodynamic responses postop

	2° surgical denervation of the carotid sinus (discussed below). Tests: ECG. Consider carotid ultrasound, cardiac stress testing, and imaging studies if indicated from H&P.
Neurologic	See Preoperative Considerations for Laryngoscopy/Bronchoscopy/Esophagoscopy. Tests: As indicated from H&P
Hematologic	In cases of malignancy or chronic disease, anemia or coagulopathies may be present. Tests: CBC, coagulation studies
Laboratory	LFTs, electrolytes, albumin, BUN, and creatinine; other tests as indicated from H&P
Premedication	Standard premedication. Avoid premedication in patients with symptoms of partial airway obstruction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

The anesthetic considerations for **composite resection and neck dissection** are similar. The use of deep and superficial cervical plexus blocks, as well as cervical epidural anesthesia, either alone or in combination with light GETA, has been described in the literature, but used rarely. GETA is virtually universally employed. The specific surgical requirements for the anesthetic technique are outlined above (see Introduction). Straining or coughing/gagging on the ETT during emergence from anesthesia is particularly undesirable in neck surgery 2° easily provoked bleeding and swelling at the surgical site. Smooth emergence from anesthesia is essential. Use of a FLMA in major cancer neck surgery cannot be recommended.

Induction	Choice of induction will depend on the degree of the airway compromise and anatomical location of any
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obstructing lesion. The airway may need to be secured asleep, with the patient breathing spontaneously, awake (FOI), or a tracheostomy may need to be performed preinduction. In patients with uncompromised airways, potent opioids (see Introduction) can be safely used for induction. The induction dose of a hypnotic agent frequently must be reduced because of the patients' age, preexisting medical conditions, and hypoalbuminemia.

ERAS

Verify preincision antibiotics. Assess volume status (see [Appendix K](#)). Maintain normovolemia and normothermia.

Maintenance

Maintaining moderate ↓ BP (see Introduction) and aggressive treatment of ↑ BP are necessary. In patients with flap reconstruction, ↑ BP may overcome vasospasm, dislodging a hemostatic blood clot or poorly tied ligature → formation of a flap hematoma. Muscle relaxation is usually avoided to facilitate identification of nerves.

All inspired gases should be humidified to minimize ↓ T° and propensity for mucous plugging in this patient population. However, the use of low gas flows is preferred to use of an in-circuit humidifier. Venous air embolism (VAE) during radical neck surgery occurs very rarely; maintaining adequate intravascular volume and PPV will further ↓ the incidence of this complication. If a large collection of air bubbles is observed by the surgeon in the internal jugular vein, the anesthesiologist should be notified immediately; these bubbles can be safely aspirated with a fine needle by the surgeon. Vagal reflexes from the carotid sinus may cause ↓ HR and ↓ BP. Transient prolongation of the QT interval during right radical neck dissection has been described and may potentially progress to ventricular arrhythmias and even cardiac arrest. Tracheostomy can be performed as part of the radical neck dissection with implications for anesthesia management (see

	Anesthetic Considerations for Tracheostomy).	
Emergence	If airway compromise is present preop, but tracheostomy was not performed, consider staged extubation over an airway exchange catheter to facilitate possible reintubation. The majority of patients with microvascular flap reconstruction will have a tracheostomy placed and could be awakened or continued on overnight ventilatory support in the ICU. Awakening the patient in the OR may help to exclude possible intraop embolic stroke before their transfer to the PACU or ICU. Administration of iv labetalol before awakening may prevent rebound hypertension. Up to 10% of patients may develop a sustained hypertensive response postop, probably 2° denervation of the carotid sinus; Rx aggressively 2° ↑ risk of stroke. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16(18) ga × 1 BSS @ 5–7 mL/kg/h Fluid warmer ± humidifier	Blood loss is typically gradual. Sudden blood loss resulting from injury to the internal jugular vein or carotid artery can be surgically controlled. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Control of blood loss	Surgical hemostasis Deliberate ↓ BP	Continue aggressive Rx of ↑ BP in the PACU.
Monitoring	Standard monitors ± Arterial line	Art line may be indicated for patients with preexisting severe cardiac and pulmonary disease, chronic renal insufficiency (CRI), symptoms of

	± CVP line Foley catheter	cerebrovascular insufficiency, location of tumor near the carotid artery, or in patients presenting for lengthy procedures, including microvascular flap reconstruction. CVP monitoring may be warranted in severely malnourished patients, patients with severe COPD, patients with CRI, and patients presenting for microvascular flap reconstruction.
Positioning	Usually supine, head elevated 30° Table turned 180° ✓ and pad pressure points ✓ eyes	
Complications	Vagal reflexes: ↓ HR and ↓ BP ↑ QT interval (with right radical neck dissection) Dysrhythmias	2° carotid sinus stimulation; Rx: stop surgery; lidocaine infiltration of the carotid sinus by the surgeon; iv atropine Probably caused by interruption of cervical sympathetic outflow to heart Due to either the carotid sinus stimulation or the ↑ QT interval
VAE	↓ ETCO₂ ↑ ETN₂ ↓ BP	With large veins open in the neck, VAE is possible and may account for

↑ ST segment
"Mill-wheel" murmur
Dysrhythmias

unexplained ↓ BP and/or dysrhythmia. Rx: Notify surgeon (compress open neck veins); flood field with NS; left lateral decubitus/head-down position; aspirate CVP; FiO₂ = 1.0; circulatory support (fluid, pressors, as required).

▼ POSTOPERATIVE

Complications	↑ BP and ↑ HR	2° carotid sinus denervation; Rx: aggressive pharmacologic intervention
	Nerve injury	Facial nerve injury → facial droop.
	Diaphragmatic paralysis	Recurrent laryngeal nerve injury can result in VC dysfunction. Phrenic nerve injury → respiratory problems 2° diaphragmatic paralysis
	Pneumothorax	Pneumothorax may occur with low neck dissection. Dx and Rx: see Postop Complications after Laryngoscopy/Bronchoscopy/Esophagoscopy
	Agitation	Agitation → ↑ PaCO ₂ , ↓ PaO ₂ , ↑ HR. Rx: ✓ restrictive neck dressings; evacuate any hematoma; reestablish airway with ETT; FiO ₂ = 1.0; ventilatory support, if required.
Multimodal pain management	PCA Parenteral opiates	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
Tests	CXR	For position of tracheostomy tube, evidence of pneumothorax and CVP line placement
	ECG	For diagnosis of rhythm disturbances
ERAS	PONV protocol (see Appendix	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

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LYMPH NODE BIOPSY

SURGICAL CONSIDERATIONS

Description

An open cervical node biopsy is indicated when a fine needle aspiration biopsy (FNA) has proven inadequate or is felt likely to be inadequate (as in determining a type of lymphoma). This can be done under local anesthesia, with or without monitoring, or under general anesthesia. A node is chosen based on its likelihood to answer the clinical question and minimization of operative risk to adjacent structures, especially, for example, the accessory nerve that runs in level 5 (posterior triangle) from about the midportion of the posterior border of the SCM toward the posterior clavicle and may be quite superficial.

Operative and anesthesia considerations are similar to that of a SOHND, except that the

incision should be only a few cm, and its location is variable though still in a RSTI (resting skin tension line), pain should be low, a small drain or no drain is used, and the patient should be able to go home the day of surgery. Often, the surgeon will request a frozen section to be sure there is diagnostic tissue available.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine, head extended, and turned to opposite side
Incision	Small (often requiring only local anesthesia)
Antibiotics	Cefazolin 2 g iv
Surgical time	10 min
EBL	Minimal
Postop care	Drain management
Mortality	Very rare
Morbidity	Rare injury to adjacent structures (e.g., accessory nerve)
Pain score	1–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Wide range
Male:Female	1:1
Incidence	Common
Etiology	In adults, 80% benign, pleomorphic adenoma (benign mixed tumor) 75% of these 20% malignant, low-grade mucoepidermoid carcinoma, adenoid cystic carcinoma, other Chronic sialadenitis/stone rare In children, a greater % malignancy; 60% benign, 35% malignant. Chronic sialadenitis/stone <5%

Associated conditions	None
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Suggested Reading

1. Mandel JE: Laryngeal mask airways in ear, nose, and throat procedures. *Anesthesiol Clin* 2010; 28(3):469–83.

PAROTIDECTOMY: SUPERFICIAL, TOTAL, OR RADICAL

SURGICAL CONSIDERATIONS

Description

A **superficial parotidectomy** (more accurately a lateral or a **supraneural parotidectomy**) removes tissue of the parotid lateral to the facial nerve, dissecting and protecting facial nerve branches ([Fig. 3.8](#)). It usually is done for a tumor but rarely is needed for infectious disorders. The integrity of the facial nerve is preserved during a parotidectomy, other than branches involved by malignancy. The nerve may be identified antegrade by locating it proximal to the parotid plexus (pes anserinus) ([Fig. 3.9](#)) or in a retrograde fashion that first identifies a distal branch (commonly the marginal mandibular branch or a buccal branch) and following it posteriorly as needed to resect the tumor. A parotidectomy may be combined with a neck dissection or with a lateral temporal bone resection when the tumor extends into the ear canal or middle ear or invades the facial nerve at or proximal to the stylomastoid foramen.

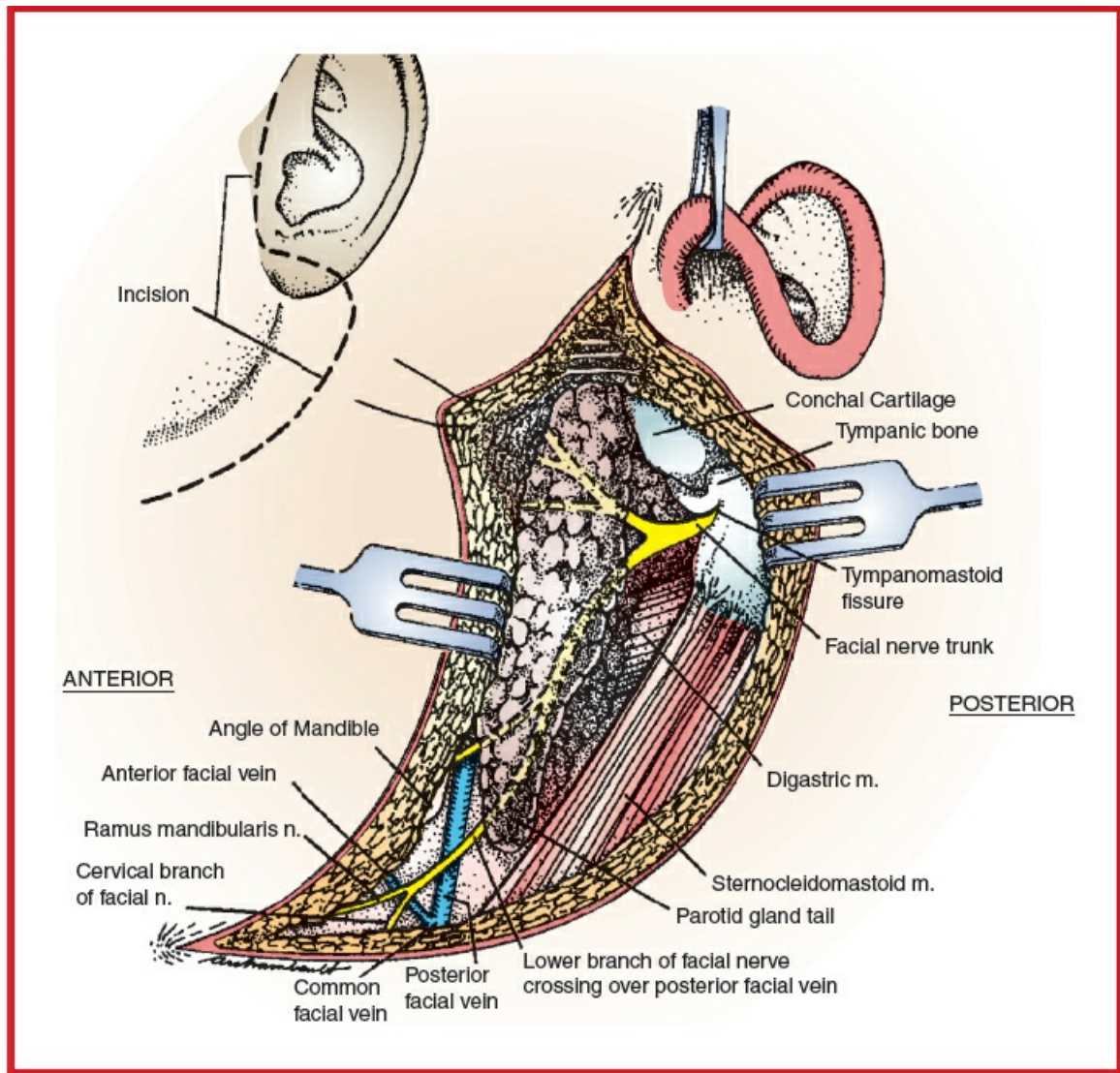


Figure 3.8 Inferior approach to facial nerve. The lower branch of the facial nerve is found immediately external to the posterior facial vein as it exits the lower pole of the parotid gland. The lower branch may divide into the ramus mandibularis and cervical branches before or after crossing the posterior facial vein. The lower branch of the facial nerve is dissected proximally to the facial nerve trunk. The posterior facial vein should not be confused with the external jugular vein, as the facial vein runs deep to the sternocleidomastoid muscle, whereas the external jugular vein lies superficial to this muscle. Elevation of the tail of the parotid gland greatly facilitates this dissection, which must be accomplished in a plane between the posterior facial vein and the parotid gland. (Reproduced with permission from Montgomery WW: Surgery of the Upper Respiratory System, 3rd edition. Williams & Wilkins, Baltimore: 1996.)

A **total parotidectomy** is simply a more extensive procedure and may be needed, for example, for a tumor that extends between branches of the facial nerve into deeper planes or for a tumor with facial nerve paralysis. If the facial nerve is resected (**radical parotidectomy**), a nerve cable graft is usually used in reconstruction. **Microsurgical techniques** are then used to graft the resected nerve. The donor nerve may be the greater auricular nerve, or if no suitable nerve is available in the neck, then the sural nerve. The mastoid may have to be drilled to get a healthy proximal end of the facial nerve.

Usual preop diagnosis

Benign or malignant tumor; occasionally infectious

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

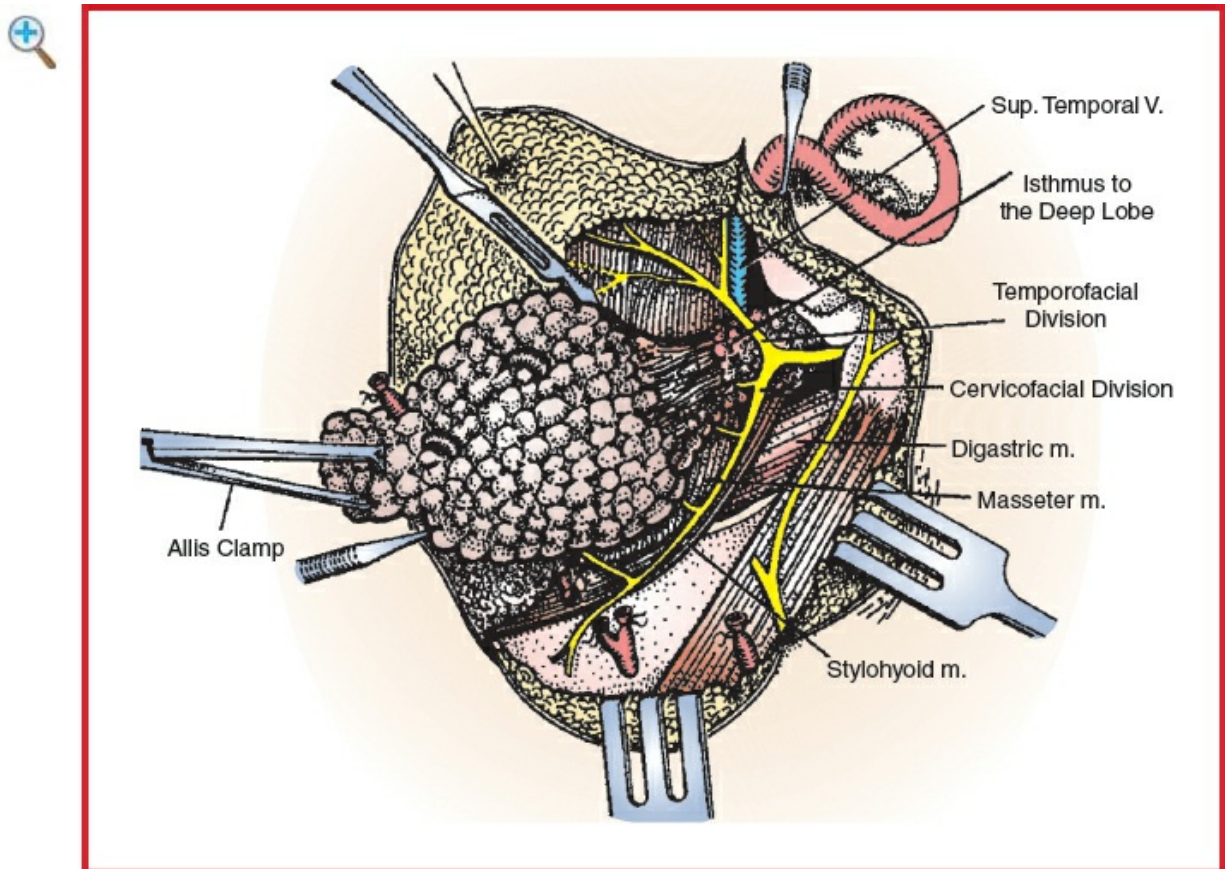


Figure 3.9 The parotid gland has been dissected from the trunk, divisions, and branches of the facial nerve. The anterior projection of the gland has been dissected free, and the parotid duct is being ligated. This dissection is conducted, for the most part, with a hemostat clamp. First, tunnels are created lateral to the branches; and the fascia between tunnels is incised as the gland is dissected forward. The facial nerve stimulator is a useful adjunct to this dissection. (Reproduced with permission from Montgomery WW: Surgery of the Upper Respiratory System, 3rd edition. Williams & Wilkins, Baltimore: 1996.)

■ SUMMARY OF PROCEDURES

	Superficial	Total/Radical
Position	Supine; head turned slightly to opposite side	⇐

Incision	Infra-auricular, extending into the neck, often with a preauricular component; many variations, such as a modified facelift incision	Postauricular incision extending inferiorly, with the extent dependent on whether a neck dissection is planned
Special instrumentation	Facial nerve stimulator, facial nerve monitor	⇒ drill for mastoidectomy; microscope/microsurgical instruments and 9-0 Nylon for nerve reanastomosis
Unique considerations	Initial muscle relaxation is not indicated 2° facial nerve identification. Tape oral ETT to the opposite side of the mandible	⇒
Antibiotics	Cefazolin 2 g iv	⇒
Surgical time	1.5–4 h	4–8 h
EBL	25–200 mL	25–400 mL for total parotidectomy, neck dissection, and modified temporal bone resection. Sudden, large blood losses do not occur; transfusion, rare
Mortality	Very rare	⇒
Morbidity	Dysesthesia or anesthesia of the greater auricular nerve: 100% (almost all will recover within 1 yr)	Possible facial nerve loss: With graft, function returns slowly over 12–18 mo in about 60%. Gold (or platinum) weight may be done at same time
	Facial nerve weakness (temporary): 0–50%	⇒
	Frey syndrome: 100%, but <5% will have clinically	⇒

	noticeable gustatory sweating	
	Bleeding: 4%	⇐
	Infection: 4%	⇐
	Permanent facial nerve paralysis: <1%	⇐
Pain score	2–4	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Wide range: infants to elderly
Male:Female	1:1
Incidence	Common
Etiology	In adults, 80% benign, pleomorphic adenoma (benign mixed tumor) 75% of these. 20% malignant, low-grade mucoepidermoid carcinoma and adenoid cystic carcinoma most common. Acinic cell carcinoma, high-grade mucoepidermoid carcinoma, adenocarcinoma, and salivary duct carcinoma less common Chronic sialadenitis/stone rare In children, a greater % malignancy; 60% benign, 35% malignant. Chronic sialadenitis/stone <5%
Associated conditions	None

SUBMANDIBULAR GLAND EXCISION

■ SURGICAL CONSIDERATIONS

Description

Removal of the submandibular gland is done for either chronic sialoadenitis due to ductal strictures and/or stones or benign or malignant tumors of the submandibular gland. (General anatomy of the area is shown in [Fig. 3.10](#).) The patient is supine with

neck mildly extended and head turned slightly to opposite side. A skin crease incision is made below the mandible and skin flaps elevated. The marginal mandibular nerve, which lies just superficial to the facial vein, is identified (or may be avoided altogether) by identifying the facial vein and dissecting deep to its plane and elevating the ligated vein superiorly (displacing the nerve superiorly out of harm's way). Dissection of the submandibular gland continues, avoiding the lingual nerve and ligating the submandibular duct.

Variant procedure or approaches

A **neck dissection** may also be done if there are involved nodes and considered for selected high-grade malignancies such as salivary duct carcinoma and high-grade mucoepidermoid carcinoma even if N₀.

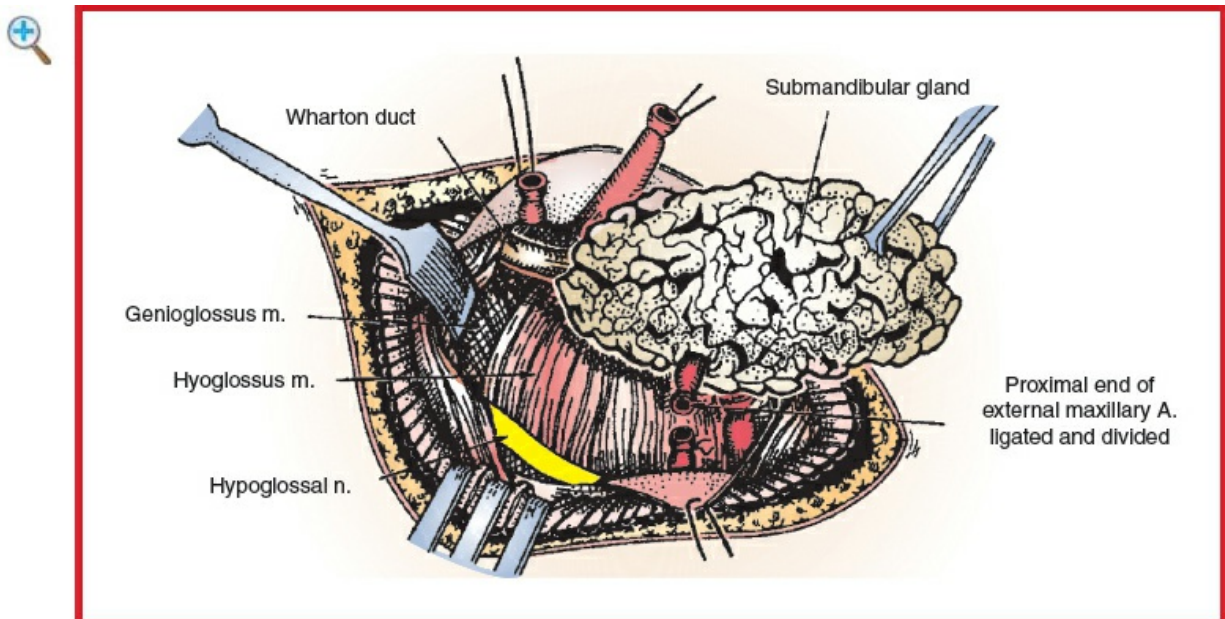


Figure 3.10 Exposure of the submandibular gland. (Reproduced with permission from Montgomery WW: Surgery of the Upper Respiratory System, 3rd edition. Williams & Wilkins, Baltimore: 1996.)

Usual preop diagnosis

Chronic sialoadenitis; stones; benign or malignant tumors

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Upper neck skin crease
Special instrumentation	Occasionally, facial nerve stimulator
Unique considerations	If surgeon plans to use facial nerve stimulator, muscle relaxation is contraindicated
Antibiotics	Usually not indicated
Surgical time	1–2 h
EBL	25 mL
Postop care	PACU → room or home
Mortality	Minimal
Morbidity	Marginal mandibular nerve paresis or paralysis: 20% Bleeding: 4% Infection: 4% Lingual dysesthesia: 1% XIIth nerve paresis or paralysis: ≤1%
Pain score	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Unlimited
Male:Female	1:1
Incidence	Rare
Etiology	Chronic infection Neoplasia: ~60% of tumors are benign with the remaining being low- and high-grade malignancies
Associated conditions	Nonspecific

ANESTHETIC CONSIDERATIONS

(Procedures covered: parotidectomy and submandibular gland excision.)

PREOPERATIVE

The majority of these patients are >40 yr, and any coexisting medical conditions should be fully evaluated. Diseases of the parotid gland have been associated with ETOH abuse and with autoimmune disease (e.g., Mikulicz-Sjögren syndrome); therefore, Sx of these conditions should be sought.

Respiratory	Airway may be affected by impaired mouth opening or parotid gland enlargement. Patients with involvement of the masseter muscle may present with trismus. Tests: As indicated from H&P
Neurologic	Surgical approach to the parotid gland will place the facial nerve in jeopardy; any preop facial nerve deficits should be documented. Also, see Preoperative Considerations for Laryngoscopy/Bronchoscopy/Esophagoscopy. Tests: As indicated from H&P
Hematologic	Submandibular and parotid malignancies may be associated with chronic debilitation and anemia. Tests: CBC
Laboratory	Other tests as indicated from H&P; liver panel and coagulation studies in patients with Hx of ETOH abuse
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of superficial cervical plexus nerve blocks for postop pain management. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA is most commonly used. Use of an FLMA also has been advocated, but does not represent the author's preference due to the possibility of distorted neck anatomy caused by the inflated FLMA cuff. The FLMA also may be displaced 2° surgical intrusion into the submandibular space and occasional intraoral surgical manipulation. These procedures involve meticulous surgical dissection and are characterized by alternating levels of stimulation: relatively long, stable levels of stimulation, with sudden adrenergic responses when surgery progresses into deeper fascial planes. (For essential surgical requirements, see Introduction.) Emergence should be rapid enough to allow immediate patient cooperation for assessment of facial nerve function by the surgeon. Muscle relaxation with nondepolarizing NMB may or may not be contraindicated depending on the surgeon's preference and type of surgery performed (inquire preop). Patients after **submandibular gland resection** may be discharged home the same day, while parotidectomy patients are admitted to the ward.

Induction	Opioid-based techniques (see Introduction) are beneficial. Standard iv induction with propofol (1–2 mg/kg) is typically employed. ETT should be carefully secured on the nonoperative side. Nasal intubation may be requested by the surgeon (inquire preop) for selected cases, to allow full manipulation of the patient's jaw.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Moderate ↓ BP (see Introduction) will greatly improve operating conditions and speed up the surgery. The use of a remifentanyl infusion supplemented with sufentanil or fentanyl will promote ↓ BP. Anticipate ↑ anesthetic requirements in patients with Hx of ETOH abuse. NMB is typically avoided to allow for intraop monitoring of the facial nerve (parotidectomy) or marginal mandibular nerve (submandibular gland excision).
Emergence	Smooth emergence is extremely desirable, and absence of reaction to the ETT is obligatory if facial nerve repair has been performed. Discuss with

	surgeon local wound infiltration or nerve block (e.g., superficial cervical plexus block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	EBL typically ~200 mL IV: 18 ga × 1 BSS @ 3–5 mL/kg/h	3rd spacing is usually minimal. See Goal-Directed Fluid Strategy, Appendix K, p. K-4 .
Monitoring	Standard monitors	
Positioning	Table usually is rotated 90–180° away from the anesthesiologist. ✓ pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	Facial paralysis 2° surgical trauma	Notify surgeons.
Multimodal pain management	Parenteral opiates PCA	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Reading

1. Mandel JE: Laryngeal mask airways in ear, nose, and throat procedures. *Anesthesiol Clin* 2010; 28(3):469–83.

ENDOSCOPIC SINUS SURGERY

SURGICAL CONSIDERATIONS

Peter H. Hwang and Sarmela T. Sunder

Description

Sinus surgery is performed most often for the management of chronic sinus disease, nasal polyps, or recurrent sinus infections or for neoplastic conditions that may cause obstruction of the sinus cavities. This surgery includes either external procedures, through the skin or oral cavity, or endoscopic approaches, through the nostrils. The endoscopic approach—the preferred technique for most surgeons—is called **endoscopic sinus surgery (ESS)** or **functional ESS (FESS)**. Other surgical procedures—including septoplasty, turbinate reduction, and, occasionally, sleep apnea surgery involving the tonsils, palate, and oropharynx—may be combined with FESS.

Endoscopic sinus surgery

ESS or FESS is a minimally invasive technique that utilizes telescopes intranasally to visualize and access the paranasal sinuses and anterior skull base. The primary indication for ESS is chronic sinusitis that has failed to clear with aggressive medical therapy. The goal of ESS is to open the sinuses through a transnasal minimally invasive approach. Inflamed bone and tissue are cleared from the outflow tracts of the sinuses under direct endoscopic illumination and magnification. By surgically enhancing the drainage and ventilation of the sinuses, normal mucociliary clearance can be restored. Depending on the extent of disease, surgery may involve the maxillary, ethmoid, sphenoid, and/or frontal sinuses. If all four sinuses are being treated, a maxillary antrostomy is typically performed first, followed by an ethmoidectomy. A sphenoidotomy and frontal sinusotomy usually follow thereafter.

Providing openings (drainage, aeration) to the sinus cavities and obtaining tissue for pathologic evaluation are the main aims of ESS. Patients are orally intubated, and a throat pack can be placed in the pharynx to decrease the amount of blood swallowed. Oral gastric tubes are recommended if more than minimal bleeding is expected. Following induction of anesthesia, local medications are applied to the nasal cavity, including injection of 1% lidocaine with 1:100,000 epinephrine and/or topical application of 4% cocaine. It is important that these local medications be noted in the anesthesia record, as complications (HTN, dysrhythmias, seizures) from local application have been reported. Patients with a history of illicit cocaine use should be warned of potential drug interactions, and the intraop use of vasoactive drugs should be

limited to avoid unexpected cardiovascular events.

At the end of the procedure, packing may be placed. It is imperative that any pharyngeal or throat pack be removed before the end of the procedure to avoid airway destruction.

Sinus surgical procedures may use **image guidance or surgical navigation tools**. These computer-assisted techniques involve a preop CT or MRI scan to create a navigational map for the surgeon to localize disease, normal tissues, and the boundaries of the surgical dissection. Surrounding the sinuses are several vital structures, including the periorbital tissues (e.g., orbital muscles, lacrimal apparatus, and the optic nerve), the sphenopalatine artery, ethmoid arteries, the carotid artery, as well as the skull base.

Variant procedure or approaches

For **external approaches** to sinus surgery, see External Sinus Surgery.

Usual preop diagnosis

Nasal polyps; chronic sinusitis; recurrent acute sinus infections, benign and malignant tumors, inverted papilloma, and management of previous surgical complications (e.g., repair of cerebrospinal fluid [CSF] leak, lacrimal duct injury, and persistent or remnant ethmoid cells or scar tissue formation)

Preoperative management

Patients should receive a decongestant nasal spray, such as oxymetazoline, preop to reduce nasal edema. This is generally given 30 min. before the start of the case, with repeat administration every 10–15 min. Preop antibiotics or steroids will vary depending on surgeon preference. Typically, an antibiotic with gram-positive and some gram-negative coverage (such as cephalexin) and an intravenous steroid such as dexamethasone are administered.

Intraoperative management

ESS is usually performed under GA, using either endotracheal intubation or an LMA. The LMA offers the potential advantages of reducing the entry of blood into the stomach and causing less bucking or coughing during emergence (which may provoke unnecessary bleeding from the surgical site). The tube should be taped securely to the left side or midline of the patient's mouth. The OR table is routinely turned 90° or 180°, and the patient is positioned with the head slightly elevated. The surgeon sits or stands at the patient's right and operates from a video monitor positioned at the patient's head.

The eyes are taped but should remain visible in the surgical field throughout the procedure so that the status of the orbits may be noted by the surgeon. The

presence of ecchymosis or proptosis suggests a transgression of the thin bony orbital wall and requires immediate attention. Throughout the procedure, the surgeon may palpate the orbit externally while endoscopically evaluating for dehiscences of the orbital wall, which if present will reveal transmitted movement of orbital fat or periorbita.

In more complex FESS or endoscopic skull base surgery, the surgeon may use a stereotactic navigation system to map out the surgical field according to a preoperatively acquired CT or MRI scan. During preparation of the patient, a localizing headset is placed over the patient's forehead, and a computer workstation tower is positioned adjacent to the head of the patient. After a registration process, a navigating probe communicates with the workstation via electromagnetic or infrared technology to provide triplanar localization of the probe tip relative to critical anatomic structures in and adjacent to the sinuses.

Hemostasis is critical in ESS, where the small lens of the endoscope (4 mm) can be easily obscured by blood. When possible, a hypotensive anesthesia regimen can reduce bleeding and thereby optimize the surgeon's view of the operative field. It may be reasonable to maintain blood pressure at ~80% of the preop level.

To improve hemostasis, the surgeon may place pledgets soaked in a vasoconstrictive agent such as cocaine, oxymetazoline, or epinephrine in the nasal cavity before starting the case. A transoral or transnasal sphenopalatine ganglion block may be performed using 1% lidocaine with 1:100,000 epinephrine, which will provide vasoconstriction of the sphenopalatine artery. The lateral nasal wall and middle turbinate are also infiltrated with lidocaine/epinephrine. Additional vasoconstrictor-soaked pledgets may be placed in the operative field during the procedure to control oozing.

Complications of endoscopic sinus surgery

Along the boundaries of the paranasal sinuses are important structures that may be injured in the course of the procedure.

Maxillary Sinus: Just anterior to the maxillary sinus ostium is the nasolacrimal duct. Injury to this structure, manifesting as excessive tearing, will not usually be evident until postop.

Ethmoid Sinus: Two critical boundaries of the ethmoid sinus are the medial orbital wall laterally and the skull base superiorly. Transgression of the orbit may result in medial rectus injury, prolapse of orbital fat, or orbital hemorrhage. If orbital hemorrhage progresses to the point of marked proptosis, vision may be threatened,

and an emergent decompression of the orbit may be required via lateral canthotomy and cantholysis.

The superior extent of the ethmoid dissection is the cribriform plate and the anterior skull base. A breach through the ethmoid roof may result in a cerebrospinal fluid leak. CSF leaks require immediate repair using autogenous tissue grafts and require an inpatient recovery period. Vigorous bag-mask ventilation should be avoided in patients with known skull base defects because of the risk of creating a pneumocephalus.

Sphenoid Sinus: The sphenoid sinus, the posteriormost of the paranasal sinuses, lies adjacent to critical structures of the parasellar region. The carotid artery and optic nerve run along the superolateral wall of the sphenoid sinus. Any significant bleeding on entry into or while working in the sphenoid sinus is concerning for potential carotid injury. Carotid artery injury requires emergent tamponade, possible fluid resuscitation, and angiography with embolization.

Frontal Sinus: Endoscopic approaches to the frontal sinus are especially challenging because of the narrowness of the region. Violation of the cribriform plate medially or the orbital wall laterally may lead to complications of CSF leak or eye injury, respectively.

Postop Management: In the PACU, patients s/p ESS require continued monitoring for possible delayed manifestation of complications. Maintaining normotensive hemodynamics is extremely helpful in controlling postop bleeding. Changing nasal dressings for oozing every 20–30 min is expected, but brisker bleeding should be evaluated by the surgeon for possible arterial sources requiring intervention.

Orbital injury can also present in the postop period with significant pain, diminished visual acuity, diplopia, proptosis, or ecchymosis. Any complaints of eye pain or visual changes need to be addressed urgently. The surgeon must be alerted immediately, and an ophthalmology consult should be considered.

Altered mental status, focal neurologic exam, or severe headache may indicate a possible intracranial complication. Clear rhinorrhea, especially unilateral, should be closely evaluated for the possibility of CSF leak.

Other applications of endoscopes in sinonasal surgery

Although the most common indication for ESS is medically refractory chronic sinusitis endoscopic techniques can be applied to a wide range of sinonasal and skull base

pathology. OR setup and patient positioning are identical to that for ESS (except as noted).

Turbinate reduction

The inferior turbinate is the largest of the nasal turbinates (inferior, middle, superior) and when hypertrophied can cause nasal obstruction. Various methods exist for inferior turbinate reduction and include radiofrequency ablation, cauterization, submucosal resection, partial resection, and outfracture. Both turbinate cauterization and RF ablation involve inserting a probe into the soft tissue of the turbinate and producing a coagulative lesion, which ultimately contracts to reduce the turbinate soft tissue. Similarly, submucosal reduction involves making an incision through the mucosa and mechanically reducing the underlying soft tissue. Partial resection of the turbinate involves cutting through the bone, soft tissue, and mucosa of the anterior one-third of the turbinate. At the conclusion, the cut edges are cauterized. The main adverse event from the majority of procedures on the turbinate is bleeding, which is usually controlled with cautery or nasal packing.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine; table turned 90° or 180°
Incision	Mucosal, intranasal
Special instrumentation	4-mm nasal endoscopes, camera monitor, computer-guided navigation system
Premedication	Cefazolin 2 g iv, dexamethasone 8–12 mg iv
Surgical time	1–3 h
EBL	50–300 mL; transfusions rare
Postop care	PACU monitoring for surgical complications; discharge home from PACU
Mortality	Rare
Morbidity	1% major complication rate (see complications of ESS above)

Pain score	2–3
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■ PATIENT POPULATION CHARACTERISTICS

Age range	15–80
Male:Female	1:1.5
Incidence	Common
Etiology	Medically refractory chronic sinusitis
Associated conditions	Asthma, allergic rhinitis, migraine

Septoplasty

Septal deviation may arise congenitally or during periods of rapid craniofacial growth or from nasal trauma. Septoplasty is indicated when the septal deviation causes obstructed nasal breathing. Through an incision in the nasal vestibule, a mucoperichondrial septal flap is elevated to reveal deflections in the underlying septal cartilage and bone. Selective excision of bone and cartilage allows the septum to return to a midline position. Although the procedure may be performed using direct visualization with a headlight, nasal endoscopy provides superb visualization and allows for precise surgical maneuvering.

Control of epistaxis

Control of epistaxis can usually be performed in the clinic; however, the OR provides a controlled environment suitable for high-risk patients and patients with profuse bleeding. Under endoscopic visualization, bleeding sites can be readily identified and controlled with cautery or laser. A more definitive procedure to control posterior epistaxis is sphenopalatine artery ligation. The artery is identified at the posteriormost aspect of the middle turbinate through a mucoperiosteal incision. The artery is then cauterized or ligated with surgical clips.

Orbital decompression

Endoscopic orbital decompression is performed for Graves ophthalmopathy to preserve diminishing vision and to improve cosmesis associated with exophthalmos. An ethmoidectomy is performed first, followed by removal of the medial orbital wall. After adequate exposure is achieved, the orbital periosteum is incised, causing the orbital fat to prolapse into the ethmoid cavity, thereby decompressing the orbit contents. Decompression may be extended to the floor of the orbit if medial decompression alone

is insufficient.

Repair of CSF leak

Discontinuity of the anterior skull base, occurring traumatically or spontaneously, may result in CSF rhinorrhea. The defect may be a small crack in the skull base or a larger defect with prolapse of meninges or brain tissue through the defect. The majority of anterior skull base CSF leaks can be closed effectively with endoscopic approaches. Meningoencephaloceles that protrude through a bony skull base defect are reduced with bipolar cautery. After endoscopic visualization of the bony defect, a layered repair is performed using bone or cartilage, followed by a free mucosal graft.

A lumbar subarachnoid drain may be placed prior to starting the case. Fluorescein can be placed through the drain to aid in identifying the leak site during surgery. If fluorescein is used, it must be carefully measured and administered to avoid the neurotoxic effects of overdose (accepted dose is 0.1 cc of 10% fluorescein mixed in 10 cc of nonbacteriostatic saline, administered slowly over 5 min). Postop, patients remain at bedrest for 1–3 d, typically as an inpatient.

Pituitary surgery

The application of endoscopic technique to transnasal transsphenoidal approaches to the sella provides an excellent panoramic view unavailable with microscopic approaches. Pituitary adenomas, craniopharyngiomas, and other sellar masses are suitable for endoscopic resection. The operating team consists of a neurosurgeon and an otolaryngologist working in concert. The approach is performed by the otolaryngologist, while the tumor resection is performed by the neurosurgeon under endoscopic guidance by the otolaryngologist (“four hands technique”). After performing intraoral greater palatine blocks and topically decongesting the nasal mucosa, the surgeon performing the approach creates bilateral sphenoidotomies. The posterior portion of the septum is then resected, allowing for the subsequent removal of the remainder of the sphenoid rostrum. This provides wide exposure of the sella and parasellar anatomy. The sella is then entered, and tumor resection is performed endoscopically.

The dissection of the sella risks injury to the adjacent optic nerve, cavernous sinus, and carotid artery. The anesthesiologist should be prepared to hemodynamically resuscitate the patient in the rare event of a carotid injury. Additional monitoring such as an arterial line may be indicated at the discretion of the anesthesiologist.

A CSF leak may occur if the arachnoid is breached during the tumor resection. This is repaired easily with abdominal fat, fibrin glue, and an absorbable reconstruction plate. In the event of a CSF leak, a lumbar subarachnoid drain may be placed.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The majority of these procedures are performed on an outpatient basis. Patients with extensive polypoid disease may be at risk for substantial (>400 mL) blood loss. Many of these patients will have received a short preop course of steroid therapy to decrease tissue swelling and reduce bleeding intraop. Routine intraop administration of dexamethasone (8–12 mg iv) to minimize postop edema makes additional stress dose of glucocorticoids unnecessary.

Respiratory	Many patients with nasal polyposis have a high incidence of reactive airway disease (up to 26%) as well as a hypersensitivity to aspirin, which can precipitate bronchospasm (triad patients). In these patients, NSAIDs, including ketorolac, should be avoided. In general, ketorolac administration for postop pain relief is not recommended in these patients because it may promote microvascular bleeding. Tests: CXR, as indicated by H&P
Cardiovascular	See Anesthetic Considerations for Nasal Surgery.
Premedication	See Anesthetic Considerations for Nasal Surgery.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

FESS can be performed under local anesthesia with sedation (MAC) or GA. The essential surgical requirements are similar to those for other types of nasal surgery (see Anesthetic Considerations for Nasal Surgery), although more meticulous control of intraop BP is crucial. Local anesthesia offers the advantage of reduced blood loss, but GA is performed more frequently and may be associated with ↓ incidence of surgical complications. Patients with extensive sinus disease, history of multiple previous nasal

and sinus surgeries, patients feeling uneasy about being awake, and those with reactive airway disease or labile HTN may not be candidates for MAC. Use of an FLMA instead of an ETT offers significant advantages (see Introduction), including decreasing airway reactivity and improving mucociliary clearance. For tracheal intubation, the use of wire-reinforced or oral RAE ETTs is preferred to facilitate surgical access.

Maintaining intraop controlled hypotension (see Introduction) is essential for improving surgical visibility and operating conditions. Although MAP may largely affect the arterial inflow, other factors including the condition of the vascular network, local mechanisms regulating functional nasal capillary density, and local venous pressure also play a significant role in regulating overall nasal capillary perfusion. As with other types of nasal surgery, these procedures are typically characterized by minimal immediate postop pain 2° submucosal infiltrative local anesthesia used by the surgeon.

Induction	See Anesthetic Considerations for Nasal Surgery. The patient's eyes should be protected in a manner that allows the surgeon to quickly diagnose proptosis in case of inadvertent orbital wall penetration and intraorbital hemorrhage.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	See Anesthetic Considerations for Nasal Surgery. TIVA, with or without nitrous oxide, promotes superior hemodynamic stability and minimizes direct peripheral vasodilation. When additional ↓ BP is required esmolol is preferred over SNP. Breakthrough adrenergic responses occur more frequently with FESS compared to other types of nasal surgery because many reflexogenic areas (e.g., skull base) cannot be blocked by infiltrative local anesthesia. These responses can be effectively controlled by vasoactive agents or additional boluses of remifentanyl (0.5–1.0 mcg/kg iv). The anesthesiologist should frequently observe the video monitor for signs of possible complications—prolapse of periorbital fat (orbital injury) and “washout” of blood in the surgical field (CSF leak).

Emergence	See Anesthetic Considerations for Nasal Surgery. If a dural injury has occurred, a smooth extubation is mandatory. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 20 ga × 1 (unilateral FESS) 20–18 ga × 1 (bilateral FESS) BSS @ 3 cc/kg/h Blood loss generally minimal	Blood loss is usually minimal but can be substantial in high-risk patients (as described above). See Goal-Directed Fluid Management, Appendix K, p. K-4. Meticulous surgical technique and the use of injectable vasoconstrictors are essential in reducing intraop bleeding. An iv is usually placed at the side where the surgeon is positioned (patient's right upper extremity).
Monitoring	Standard monitors BP cuff left arm	Prevent BP cuff from inflating against the surgeon's hands during delicate endoscopic manipulation.
Positioning	Table turned 90–180° Head elevated 15–30° ✓ and pad pressure points ✓ eyes	Patient's arms usually tucked and carefully padded.
Complications	Dysrhythmias Tachycardia	Usually related to use of vasoconstrictor agents; and may present a problem in patients with

CAD. Short-acting β -blocker (esmolol 0.5–1.0 mg/kg iv boluses) should be available to blunt these hemodynamic effects.

▼ POSTOPERATIVE

Complications	Occult bleeding.	Occult postop bleeding may cause the patient to swallow large quantities of blood ($\uparrow$ aspiration risk), and repacking the nostrils may be required.
Multimodal pain management	Fentanyl 25–50 mcg iv prn is usually sufficient in PACU.	Occasionally, pain after FESS may be significant. If external approaches to the sinuses were utilized, higher doses of opioids may be necessary. Emphasize multimodal approach combining nonopioid (e.g., ibuprofen) and bolus/PCA opioid analgesics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1).	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

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EXTERNAL SINUS SURGERY

SURGICAL CONSIDERATIONS

Description

Transoral or transfacial approaches to the paranasal sinuses ([Fig. 3.11](#)) include a transoral sublabial approach to the maxillary sinus and pterygomaxillary space (**Caldwell-Luc**), rarely done external ethmoidectomy (often with medial orbitotomy and/or medial maxillectomy), and osteoplastic approaches to the frontal sinus via a bicoronal incision just behind the hairline. The usual indication for such approaches is resection of malignant and occasionally benign locally extensive tumors such as angiofibroma, especially when for technical reasons (e.g., angle of access or extension into the orbit) it is felt that an endoscopic approach is not indicated. The continued

improvement in endoscopic equipment along with increasing surgical experience makes endoscopic resection increasingly more common and appropriate.

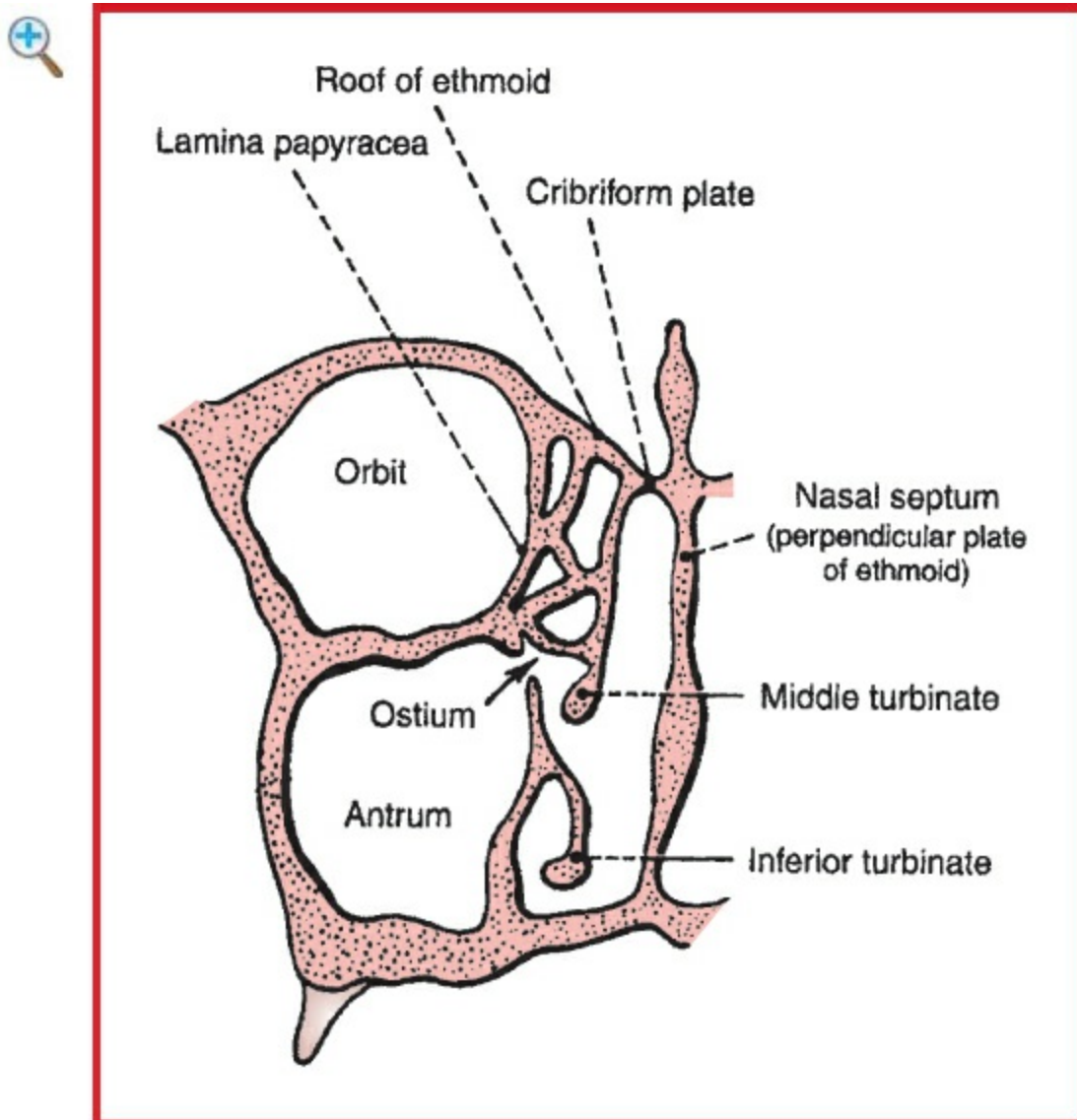


Figure 3.11 Diagrammatic representation of the relationships of the orbit, maxillary antrum, ethmoid labyrinth, and nasal cavity. Note that the medial wall of the ethmoid labyrinth is an upper extension of the attachment of the middle turbinate. The attachment of the upper extension separates the roof of the ethmoid and the cribriform plate. (Reproduced with permission from Montgomery WW: *Surgery of the Upper Respiratory System*, 3rd edition. Williams & Wilkins, Baltimore: 1996.)

Anterior cranial base surgery

When tumors transgress the anterior cranial base at the cribriform plate and ethmoid roof, an anterior cranial base resection is needed. If the extent of the anticipated dural resection is limited and the lateral extent amenable, then the resection can be accomplished endoscopically; otherwise, an open approach is done. A central facial incision today is almost never required as adequate exposure may be obtained via a

bicoronal incision and bifrontal craniotomy, with endoscopic assist if needed. The CSF leak is addressed by a combination of dural substitution (autologous tissue, such as temporalis fascia, bovine pericardium, or manufactured substitutes) and a pericranial-galeal flap. A lumbar subarachnoid drain is sometimes used in this setting.

Variant approaches

Approaches to the middle cranial base include an orbitozygomatic approach to the infratemporal fossa. The zygomatic arch and part of the zygomatic body is temporarily removed, the temporalis major muscle is transposed inferiorly. The lateral orbital bone with or without the orbital rim may be temporarily removed. A middle fossa craniotomy or craniectomy is usually done. This provides access to the foramen ovale, foramen rotundum, and carotid artery at the skull base, as well as the lateral orbit and superior infratemporal fossa and pterygomaxillary fossa to the pterygoid plates, and, as needed, lateral sphenoid sinus and maxillary sinus. Following tumor resection, part of the temporalis muscle may be used to assist in CSF leak management.

Usual preoperative diagnosis

Malignant and some benign tumors

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	External Sinus Surgery	Anterior Cranial Base Surgery
Position	Head up 30°, back raised (reduces bleeding)	⇐ also reduces intracranial pressure
Incision	Access to the ethmoid and maxilla is via a 3–4-cm incision midway between the medial canthal ligament and the midline, in the shadow of the nasal bone; a lateral rhinotomy incision is not needed	A bicoronal incision posterior to the hairline masks the scar. A supplemental facial incision is rarely needed

Special instrumentation	Paranasal sinus set;	⇐ + craniotomy set
Unique considerations	headlight Nasal packing	⇐ see frontal craniotomy
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	1.5–4 h	5–8 h
EBL	25–300 mL	150–750 mL
Postop care	Same as endoscopic surgery	Same as craniotomy
Mortality	Very rare	Rare
Morbidity	Bleeding: <5%	Anosmia if bifrontal craniotomy CSF leak: <2% CVA: <1% Bone flap loss: <1% Meningitis: <1% Infection: <1%
Pain score	4–6	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Children–seniors
Male:Female	1:1
Incidence	Common
Etiology	Malignant tumors, selected benign tumors, occasionally infectious
Associated conditions	Dependent on specific tumor; also, see Endoscopic Sinus Surgery

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Endoscopic Sinus Surgery.

NASAL SURGERY (RHINOPLASTY, SEPTOPLASTY, SEPTORHINOPLASTY)

SURGICAL CONSIDERATIONS

Sam P. Most

Description

Nasal surgery is performed for cosmetic and functional restoration of the airway. Functional restoration is usually performed for either congenital or posttraumatic deviations of the septum. The nasal cavity is first cocaine-soaked with 4% cocaine-soaked pledgets placed in each nostril for 5–10 min. **Septoplasty** (reconstruction of the nasal septum) usually can be carried out under sedation with local anesthesia, using 1% lidocaine with 1:100,000 epinephrine. **Rhinoplasty-septorhinoplasty** usually is carried out under local anesthesia, but if GA is used, a mouth pack is inserted. Local infiltration with 1% lidocaine with 1:100,000 epinephrine is used to ensure vasoconstriction and to minimize bleeding. Intranasal incisions are made and septal problems corrected. Generally, an anterior hemitransfixion incision is made down to the cartilage, and a submucoperichondrial flap is elevated the length of the septum. A similar flap may be elevated on the contralateral side. Bony deformities are resected with an osteotome, whereas cartilaginous deformities are either resected or weakened by morselizing, either in situ or after removal, and then replaced. The incision is closed with interrupted absorbable sutures. In **rhinoplasty**, tip remodeling, hump reduction, and bony osteotomies are performed to remodel the nasal contour. Surgery on the inferior turbinates in the form of intramural cautery, resection of turbinate bone, resection of turbinate mucosa, or, in some cases, complete turbinectomy may be required to produce a satisfactory airway. After the surgery is complete, both nasal cavities are packed and external splints may be used for rhinoplasty and septorhinoplasty cases. For the plastic surgeon's perspective, see [Chapter 11.1](#).

Usual preop diagnosis

Nasal deformity or deviation; deviated septum

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Head up 30° to reduce bleeding. Table may be turned 90–180°
Incision	Intranasal, usually; extended only in open septorhinoplasty
Unique considerations	Nose initially cocaineized; use of 1% lidocaine with 1:100,000 epinephrine to ↓ bleeding
Antibiotics	Cefazolin 2 g iv; routinely used as long as nasal packs are in place
Surgical time	1–2.5 h
Closing considerations	Nose often packed postop, necessitating oral airway after extubation
EBL	50–100 mL (excessive blood loss rare)
Postop care	PACU
Mortality	Minimal
Morbidity	Septal perforation: 5% Bleeding: 4% Infection: 4%
Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Young teens–young adults
Male:Female	1:1
Incidence	Common
Etiology	Congenital/traumatic septal and/or nasal deviation

FACIAL PLASTIC SURGERY

Overview

Facial plastic surgery encompasses both reconstructive and aesthetic surgery of the face

and neck. Typically, these surgeons are trained in otolaryngology—head and neck surgery/facial plastic surgery, general plastic surgery (see [Chapter 11.1](#)), or both. The procedures to be covered in this section are rhinoplasty, rhytidectomy (facelift), and blepharoplasty (eyelid rejuvenation).

SURGICAL CONSIDERATIONS

RHINOPLASTY

Description

Rhinoplasty has traditionally been considered an aesthetic procedure. However, when combined with a septoplasty (rhinoseptoplasty, septorhinoplasty), it is both a functional and aesthetic operation. In some cases, a purely functional operation (functional rhinoplasty) is performed to improve the ability to breathe through the nose. The approach and anesthetic requirements for these can be considered together.

Technique

A septoplasty involves a unilateral intranasal incision along the anterior septum with minor variations in the placement of the incision in the anteroposterior plane. The approaches to rhinoplasty can be intranasal, extranasal, or both. Frequently, the external (or “open”) approach to rhinoplasty is employed. The intranasal approach involves incisions along the septum anteriorly and some combination of incisions in the nasal vestibule. The external approach combines these approaches with a midcolumellar incision. In both functional and aesthetic rhinoplasty, the maneuvers include cartilage remodeling or resection and possibly osteotomies to manipulate the bony pyramid of the nose.

Usual preop diagnosis

For aesthetic patients, the typical preop diagnosis is nasal deformity. However, many patients also complain of nasal obstruction, and some rhinoplasty patients have purely functional concerns. Patients with significant nasal obstruction may have some component of sleep-disordered breathing or even OSA. Because of this, patients with nasal obstruction are routinely asked about snoring and daytime somnolence (if they have not already had a diagnostic sleep study).

Preop and intraop preparation

Preoperatively, patients receive a nasal decongestant spray (0.05% oxymetazoline spray, 3 doses 10 min apart) to reduce intraop bleeding. Preop corticosteroid (4–8 mg dexamethasone iv) and antibiotics (cefazolin, 2 g iv) are administered. Intraop, we prefer LMA for protection of the airway and esophageal inlet, as some bleeding in the

nasopharynx is expected. If an ETT is used, a throat pack can be placed to reduce blood entrance into the esophagus/stomach. The tube should be secured to the midline of the lower lip to avoid distortion of the nose/nasal base.

Postop care

At the end of the procedure, prior to extubation, the surgeon will place an external splint of some type on the nose. Any mask placed on the patient should be done in such a fashion as to not place pressure on the nose. We prefer the transport oxygen mask be cut off such that no contact occurs. Any Valsalva can result in epistaxis and/or bleeding under the septal or nasal skin flaps. Thus, any maneuvers to reduce bucking or coughing during extubation are in order.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Intranasal + columella (external rhinoplasty)
Special instrumentation	Headlight
Unique considerations	Nasal obstruction postop
Antibiotics	Often administered preop: cefazolin 2 g iv
Other preoperative meds	Nasal decongestant spray, dexamethasone 4–8 mg iv
Surgical time	1–3 h
EBL	25–50 mL
Mortality	Minimal
Morbidity	Epistaxis, septal perforation, unsatisfactory aesthetic result
Pain score	1–4, typically mild

■ PATIENT POPULATION CHARACTERISTICS

Age range	15+ yr
Male:Female	1:1
Incidence	Common
Etiology	Nasal obstruction, desire for change in nasal form

RHYTIDECTOMY (FACELIFT)

Description

Rhytidectomy (facelift) is almost always an aesthetic procedure. In the common vernacular, “facelift” is often used to describe total surgical facial rejuvenation (including eyelids and forehead). Strictly speaking, rhytidectomy involves aesthetic improvement of the lower half of the face and upper neck. For this reason, some have called it a “neck lift” or “lower facelift” to avoid confusion. Perhaps the latter term is more appropriate, as a separate “neck lift” procedure does exist. In any event, the anesthetic requirements for these are similar.

Technique

Rhytidectomy incisions are bilateral, extending pre- and postauricularly with minor variations. In addition, the surgeon may place a small (0.5–2 cm) incision submentally for access to submental fat and/or platysma muscle. As dissection may be near branches of the facial nerve, **avoidance of muscle relaxants** is required.

Usual preop diagnosis

For aesthetic patients, the typical preop diagnosis is simply desire for reduction in nasolabial and labiomandibular lines, jowling, and neck skin laxity. Patients are typically over the age of 40 and usually in good health. Smoking is a contraindication due to the risk of poor wound healing and/or skin flap necrosis.

Preop and intraop preparation

Preop patients receive corticosteroid (4–8 mg dexamethasone iv) and antibiotics (cefazolin, 2 g iv). Intraop, we prefer LMA for protection of the airway and esophageal inlet, though an ETT can be used. The tube should be secured to the **midline of the lower lip** to avoid distortion of the lower face/neck. The tube is prepped into the field and may be manipulated side to side during the procedure as the submental incision is accessed. In addition, the head may be turned about 30–45° to the side opposite the surgeon. Typically, dissection near the branches of the facial nerve is undertaken, and

visual monitoring for any twitches is performed by the assistant; thus, **avoidance of muscle relaxants is imperative.**

Postop care

At the end of the procedure, prior to patient extubation, the surgeon will place a circumferential external pressure dressing of some kind on the face. Typically, cotton balls soaked in antibiotic ointment are placed in the ears. The patient may have difficulty hearing due to this dressing. Drains may have been placed on each side of the neck. Any Valsalva can result in bleeding under the skin flaps. Hematoma is one of the most common periop complications and requires evacuation. Thus, any maneuvers to reduce bucking or coughing during extubation are in order.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Pre- or postauricular and submental
Special instrumentation	Headlight
Unique considerations	Facial nerve is monitored (visually); avoid paralytics
Antibiotics	Often administered preop: cefazolin 2 g iv
Other preop meds	Dexamethasone: 4–8 mg iv
Surgical time	3 h
EBL	25–50 mL
Mortality	Minimal
Morbidity	Hematoma, facial paralysis, unsatisfactory aesthetic result
Pain score	1–4, typically mild

■ PATIENT POPULATION CHARACTERISTICS

Age range	40+
Male:Female	1:8
Incidence	Common
Etiology	Desire for change in facial aesthetics
Associated conditions	Other facial aesthetic considerations

BLEPHAROPLASTY

(For the plastic surgeon's perspective, see [Chapter 11.1](#))

Description

Blepharoplasty (eyelid rejuvenation surgery) is, in the overwhelming majority of cases, an anesthetic procedure. In some cases, functional upper lid blepharoplasty may be performed to reduce visual field obstruction and is typically covered by insurance. These procedures are often performed under local anesthesia with sedation. However, as they are often combined with other procedures (brow lift, facelift), the patients are often under general anesthesia.

Technique

Upper blepharoplasty incisions are placed on curvilinear fashion along the upper eyelid creases with minor variations. The amount of skin to be excised is marked by the surgeon, and local anesthetic with epinephrine is administered. The evolution of upper eyelid blepharoplasty has led to a more conservative approach with less muscle and fat excision being performed. As the procedure becomes more skin only, many are being performed under local anesthesia as an isolated procedure.

Lower blepharoplasty is generally performed transcutaneously with a subciliary incision or transconjunctivally via the inner portion of the lower eyelid. In the former approach, a small amount of lower lid skin and muscle is typically excised. With both approaches, the surgeon removes or repositions a small amount of the intraorbital fat pockets for the desired aesthetic result.

Usual preop diagnosis

For aesthetic patients, the typical preop diagnosis is simply desire for reduction of upper lid skin excess, which may obstruct their vision (see above). In the lower lid, the most common complaint is “puffiness” of the lower eyelid, which is often due to

pseudoherniation of orbital fat and descent of the midface fat with age. Patients are typically over the age of 30 and usually in good health.

Preop and intraop preparation

Preop patients receive corticosteroid (4–8 mg dexamethasone iv) and antibiotics (cefazolin, 2 g iv). Intraop, we prefer LMA for protection of the airway and esophageal inlet. The tube should be secured to the midline of the lower lip to avoid distortion of the lower face/neck.

Postop care

At the end of the procedure, prior to patient extubation, the surgeon will confirm satisfaction with symmetry and aesthetic result. No dressings are used. Any Valsalva can result in bleeding under the skin flaps. More important, intraorbital pressure must be avoided as it may result in a serious complication, placing the patient's vision at risk. Thus, any maneuvers to reduce bucking or coughing during extubation are in order.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Upper lid crease, lower lid via transconjunctival or transcutaneous approaches
Special instrumentation	None
Unique considerations	None
Antibiotics	Often administered preop: cefazolin 2 g iv
Other preop meds	Dexamethasone: 4–8 mg iv
Surgical time	1.25 h (upper or lower only); 2 h (both upper and lower)
EBL	<5 mL

Mortality	Minimal
Morbidity	Asymmetry, dry eye, ectropion, corneal abrasion, blindness (rare)
Pain score	1–4, typically mild

■ PATIENT POPULATION CHARACTERISTICS

Age range	30+
Male:Female	1:8
Incidence	Common
Etiology	Desire for change in facial aesthetics; in some cases, upper blepharoplasty is performed for functional reasons
Associated conditions	Other facial aesthetic considerations

ANESTHETIC CONSIDERATIONS FOR FACIAL COSMETIC SURGERY (NASAL SURGERY, RHYTIDECTOMY, AND BLEPHAROPLASTY)

PREOPERATIVE

These cases typically are performed on an outpatient basis. Most of the patients are young and otherwise healthy, but some patients presenting for rhinoplasty/septoplasty and turbinate reduction surgery may have OSA with the corresponding implications for intraop management. Older patients may present for major reconstructive nasal surgery following resections of a basal cell carcinoma of the nose. Patients presenting for closed reduction of nasal fractures may have suffered concomitant closed head trauma. If the nose injury is recent, blood may have been swallowed, and the patient should be considered “full stomach.”

Patients with diabetes mellitus (DM) and those with rheumatologic diseases may be particularly prone to bruising, infection, and delayed healing. The use of nonsteroidal anti-inflammatory medications (NSAIDs), aspirin, and certain herbal supplements, such as ginkgo or echinacea, and certain vitamins (e.g., vitamin E) should be stopped at least 2 wk prior to surgery to avoid the risk of excessive microvascular bleeding.

Respiratory	Some patients may present with a Hx of reactive airway disease; their functional status should be checked and optimized preop. Increased airway reactivity may provoke excessive postop coughing and straining, increasing the risk of bleeding. Tests: As indicated from H&P
Cardiovascular	Special attention should be directed to the presence of HTN, which may predispose the patients to hematoma formation. HTN should be well controlled preop. In patients with preexisting cardiac disease, cocaine for topical anesthesia or control of bleeding may be contraindicated. Tests: ECG in older population; others as indicated from H&P
Neurologic	Psychologic disturbances are common in patients presenting for rhinologic surgery.
Premedication	Standard premedication. Sedation should be avoided or used with caution in patients with OSA.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

The anesthetic plan should provide for patient immobility, smooth emergence from anesthesia, a rapid return of consciousness, prevention of PONV, and fast-tracking patients for discharge. A wide variety of office-based cosmetic procedures, including rhinoplasty, rhytidectomy, and blepharoplasty, can be performed under either local anesthesia with iv sedation or GA. The choice is frequently dictated by the patient's desires and the surgeon's level of comfort. MAC removes the triggers for malignant hyperthermia (MH) GA-induced emergence phenomena, reduces the incidence of PONV, and facilitates patient discharge; however, administration of a successful MAC without immediate access to the patient's airway is frequently more challenging than conducting GA. With MAC, FiO_2 must not exceed 0.3 to prevent intraop fire.

Induction	Standard induction with propofol (1–2 mg/kg) is an ideal induction agent because of its short duration and intrinsic antiemetic effect. For GETA, the use of an oral RAE or reinforced ETT will facilitate surgical access. Spontaneous or controlled ventilation can be used, and FLMA is strongly preferred over tracheal intubation (see Introduction). Both ETT and FLMA are usually taped midline.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	All GA techniques (balanced, inhalational, or TIVA) can be used safely, although TIVA is preferred. Propofol infusion (80–150 mcg/kg/min) may offer the advantage of ↓ BP without compensatory ↑ HR and ↓ PONV. Remifentanyl infusion (0.1–0.25 mcg/kg/min) will improve hemodynamic stability and facilitate rapid and smooth emergence. Maintaining moderate degree of ↓ BP is essential.
Emergence	Aggressive prevention of PONV, usually with iv 5-HT₃ antagonist (e.g., ondansetron 4–8 mg) in combination with iv dexamethasone (8–10 mg) is highly effective and may lead to improved patient outcome. Dexamethasone is administered routinely during facial cosmetic surgery. Multimodal PONV prophylaxis is warranted in high-risk patients. Avoid ↑ BP, bucking, and coughing on emergence (FLMA is superior to ETT in this regard) to prevent microvascular bleeding and hematoma formation, especially after rhinoplasty and rhytidectomy. Emergence from anesthesia after nasal surgery presents significant challenges to the anesthesiologist. Nasal passages may be packed at the end of surgery, making the patients obligate mouth breathers; the oropharynx should be suctioned carefully. Avoid pressure on the patient's nose with the face mask. Full return of airway

	reflexes should occur before extubation. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Blood loss usually minimal IV: 20 ga × 1 BSS @ 2–3 mL/kg/h	Blood loss is controlled with surgical hemostasis and topical applications of vasoconstrictor (epinephrine) but also may include cocaine. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors	
Positioning	Head elevated 15–30° Table turned 90–180° ✓ and pad pressure points ✓ eyes	Patient's arms usually tucked in and should be carefully padded. The eyes are usually protected by the surgeon according to surgeon's preference.
Complications	Tachycardia, dysrhythmias Laryngospasm, negative pressure pulmonary edema	These are usually related to use of vasoconstrictor agents and may present a problem in patients with CAD. Short-acting β-blocker (esmolol, 0.5–1 mcg/kg in boluses) should be available to blunt these hemodynamic effects. Treatment is supportive: reestablishment of airway patency, oxygen supplementation, ventilatory support (CPAP

or tracheal intubation with positive pressure ventilation and PEEP), management of fluid shifts, and maintenance of normal intravascular volume.

▼ POSTOPERATIVE

Complications	Occult bleeding	Occult postop bleeding may cause the patient to swallow large quantities of blood (↑ aspiration risk); and repacking the nostrils may be required. Adequate prophylaxis of hypertensive responses postop (iv bolus doses of labetalol, 0.1–0.2 mg·kg ⁻¹) and PONV are essential.
Multimodal pain management	Fentanyl 25–50 mcg iv prn is usually sufficient.	Emphasize multimodal approach using acetaminophen/NSAIDs (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

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4. Webster AC, Morley-Forster PK, Janzen V, et al: Anesthesia for intranasal surgery: a comparison between tracheal intubation and the flexible reinforced laryngeal mask airway. *Anesth Analg* 1999; 88:421–5.
5. See references for Introduction and for Anesthesia Considerations for Endoscopic Sinus Surgery.

OTHER ANESTHETIC CONSIDERATIONS FOR FACIAL COSMETIC SURGERY

See Anesthetic Considerations for Facial Cosmetic Surgery.

OTOLOGY AND NEUROTOLOGY

SURGICAL CONSIDERATIONS

Otologic Surgery

Otologic surgery is indicated for a spectrum of disorders, including traumatic, developmental, infectious, and neoplastic disease. Procedures involving the external auditory canal, middle ear, and inner ear will be considered in this section. Surgery confined to the pinna or that requiring craniotomy is considered elsewhere.

The temporal bone is as complex as any region of human anatomy (Figs. 3.12 and 3.13). Multiple vital neurovascular structures interrelate in a small confined space. These include the organs of hearing and balance, the facial nerve, the carotid artery, the jugular vein, and the delicate mechanisms of the middle ear. The density and complexity of the anatomy dictate the delicate nature of otologic procedures and hence many of its anesthetic requirements. Most procedures are performed via a transcanal approach (through the natural meatus of the external ear), a retroauricular incision, or a combination of the two.

Many otologic procedures are undertaken to control recurrent or chronic ear infections (otomastoiditis). The simplest of these is the **myringotomy**, in which the tympanic membrane (TM) is incised through the ear canal, and the middle ear contents are aspirated. In most cases, a myringotomy tube is also placed to maintain middle ear aeration and drainage. This procedure usually takes just a few minutes and is performed under mask GA. A myringoplasty involves the repair of a persistent TM perforation and usually involves placement of autologous tissues as a patch to enable healing. When combined with repair of chronic middle ear changes, the procedure is termed a tympanoplasty.

The mastoid is an area of bone and air spaces situated behind the middle ear and external auditory canal. The removal of this bone, through a **simple mastoidectomy**, enables the removal of loculated infection and diseased tissue and provides access to additional anatomy. This is often combined with tympanoplasty (**tympanomastoidectomy**) to address the changes from chronic infection. This is the standard surgical approach to resect a **cholesteatoma**, which is a keratin-containing epithelial cyst originating from the TM that has grown into the middle ear and mastoid cavity. When the posterior bony wall of the external auditory canal is left in place to keep the mastoid cavity anatomically distinct from the external auditory canal, the procedure is called a **canal wall-up tympanomastoidectomy**. In contrast, for more aggressive disease, the partition between the external canal and the mastoid cavity can be removed, thereby exteriorizing the mastoid cavity in what is termed a **canal wall-down tympanomastoidectomy**. During this procedure, the external auditory canal meatus should also be enlarged to accommodate the resulting mastoid bowl. The

complete removal of mastoid and middle ear contents is sometimes referred to as a **radical mastoidectomy**. Tympanomastoid surgery is usually performed under GA and may take several hours to complete.

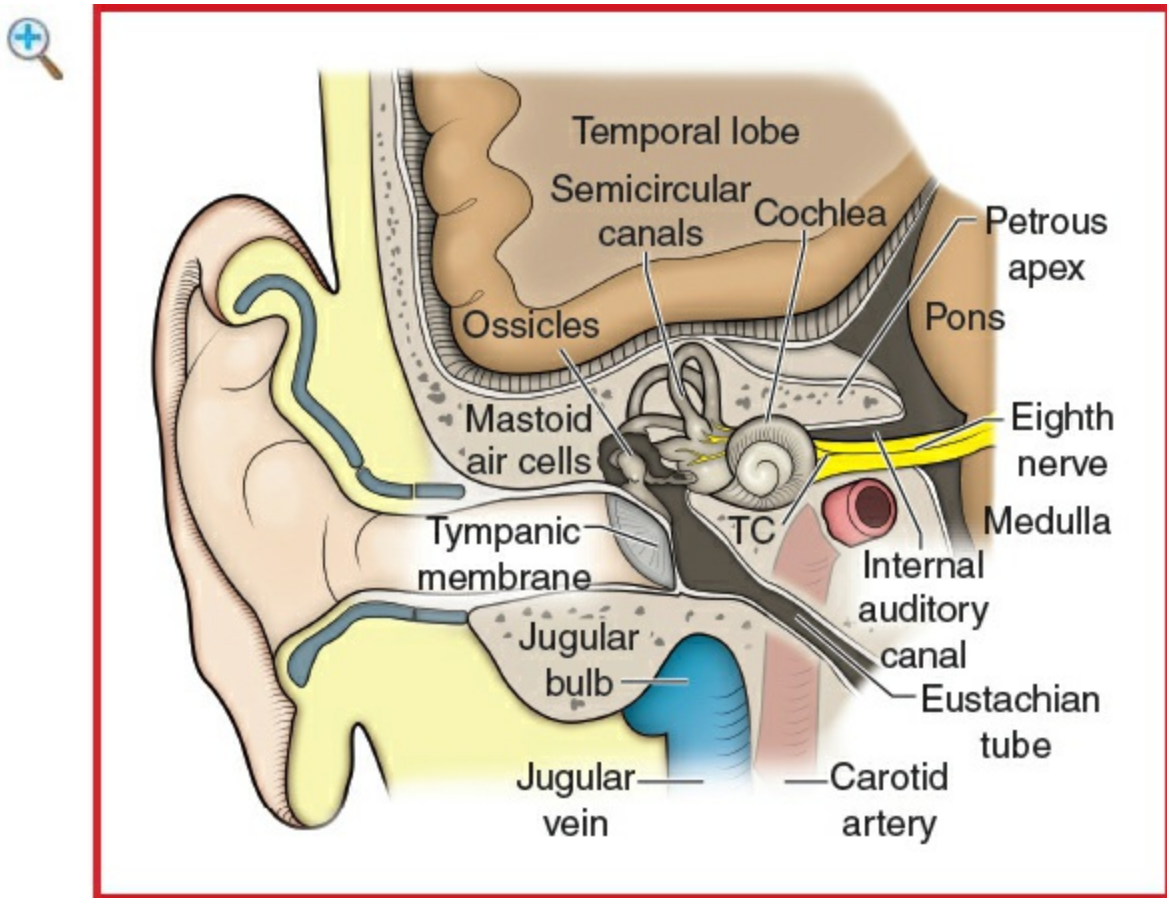


Figure 3.12 Coronal diagram of the right ear.

Surgery is often performed to improve the mechanical conduction of sound vibrations through the middle ear into the inner ear (“conductive hearing loss”). If the cause of the mechanical dysfunction is not clear, the surgeon may perform an **exploratory tympanotomy**, during which the TM is elevated and the middle ear structures are inspected and palpated to reveal deficits in their function. **Ossicular chain reconstruction** involves rebuilding or replacing the bones of hearing (malleus, incus, and stapes) with either the patient’s own tissues or with alloplastic prostheses. Otosclerosis is a disorder characterized by the progressive fixation of the stapes to the surrounding bone, resulting in hearing loss. One option for affected patients is to perform a **stapedectomy** (or stapedotomy) in which the stapes is replaced by a prosthesis to restore more normal continuity of sound conduction. This procedure can be performed under either local or general anesthesia. It involves extreme accuracy for success and can be one of the most challenging cases for the otologic surgeon.

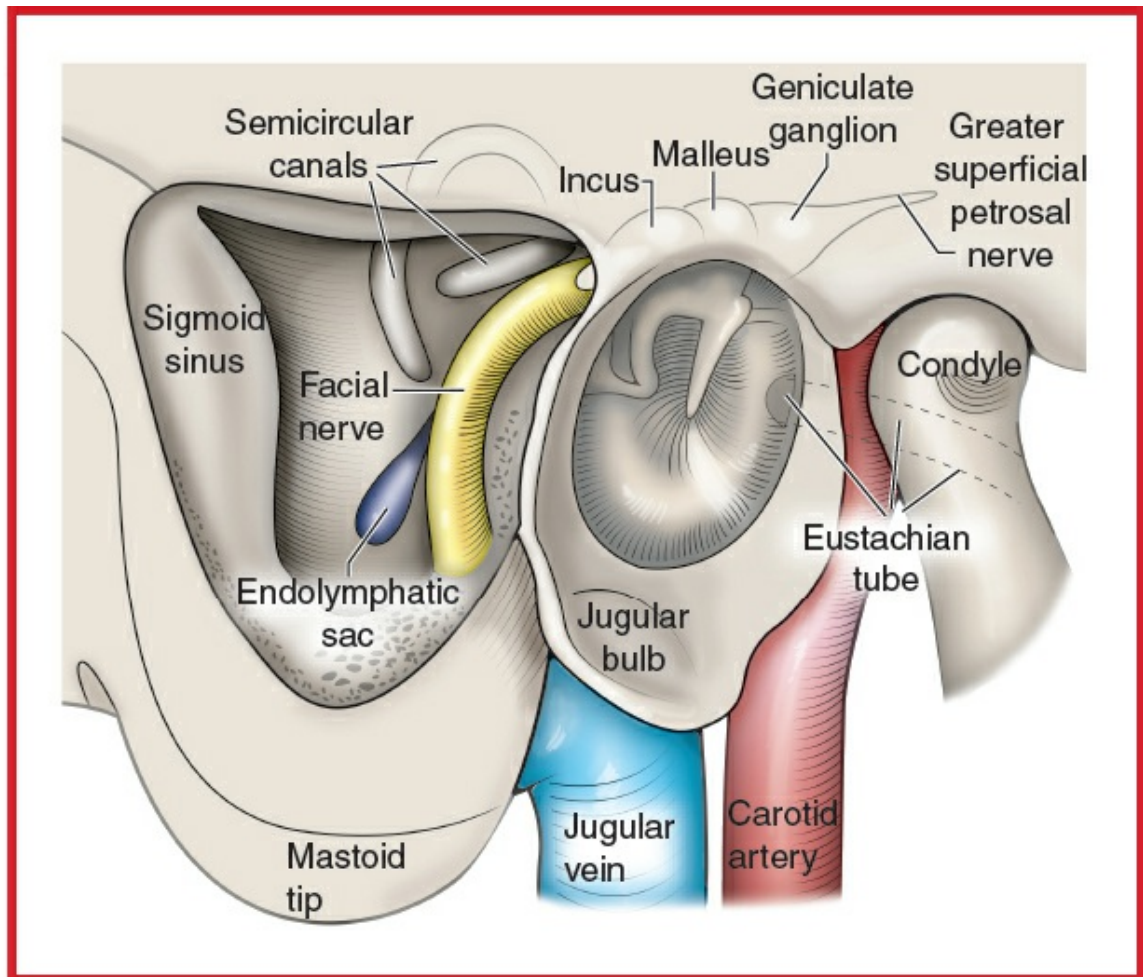


Figure 3.13 Lateral diagram of the right ear.

Cochlear implantation is an option to address severe or profound inner ear hearing loss (“sensorineural hearing loss”). During implantation, a retroauricular incision and mastoidectomy are performed to provide a pathway for device placement into the inner ear. A receiver-stimulator is placed under the skin behind the ear, and a flexible electrode array is threaded into the turns of the cochlea to later be used to stimulate the cochlear nerve directly and thereby bypass the dysfunctional inner ear. The devices are activated usually several weeks following placement, at which time hearing function is expected to increase.

A number of procedures address disorders specifically confined to the external auditory canal. Patients may develop extra bone growths (exostoses) in the ear canal that obstruct the canal. **Exostoses resection** can be performed through a retroauricular incision or via a transcanal approach if the disease is limited. Patients may lack an ear canal (canal atresia) as the result of a congenital disorder, from chronic inflammation, or posttraumatic scarring. In such patients, a **canalplasty**, involving removal obstructing skin, bone, and scar, can be performed. Similarly, patients may require the resection of

a number of neoplastic lesions of the ear canal, most commonly of cutaneous origin. Such resections may require additional resection of adjacent tissues affected by the neoplastic process. A skin graft may be used to reconstruct the external auditory canal. This may be harvested from the retroauricular region, the inner aspect of the arm, or from the hip or thigh.

The majority of otologic surgery requires the use of a surgical microscope. The presence of the microscope has a number of implications for anesthesia. Often, the OR table is turned 180°. Of critical importance is that the patient remains entirely motionless. Any adjustment of equipment or position can have profound consequences during microdissection. Therefore, the surgeon must be made aware of any plans that may result in patient motion—even those that would otherwise seem insignificant. During microsurgery, even small amounts of bleeding can have dramatic implications; thus, maintenance of a stable low blood pressure is useful.

One common theme to otologic procedures is the need to identify and preserve the **facial nerve**. The 7th cranial nerve winds its way through the temporal bone from its origin in the brainstem to the stylomastoid foramen. It passes through the middle ear and is at risk during almost all otologic procedures. The use of an EMG-based neural monitor is now commonplace to assess neural function. The monitor will not function if paralytic agents are in use, and their untimely use can prevent the surgeon from receiving critical warnings regarding nerve activity. As a standard rule, always check with the surgeon prior to the administration of any paralytic agents during otologic procedures.

Although the internal carotid artery and jugular bulb pass through the middle ear, significant vascular injury is quite rare. Most if not all bleeding can be readily controlled with packing in the surgical field. The use of absorbable gelatin sponges (e.g., Gelfoam™) is routine to hold reconstructive materials in place and also can help with hemostasis. Venous injury, such as laceration of the sigmoid sinus during mastoidectomy, can usually be controlled using bone wax or with other hemostatic materials (e.g., Surgicel™) held with pressure against the site of bleeding. In the case of a large venous injury, a **venous air embolism** can potentially result, and timely communication between the surgical and anesthesia teams can facilitate its identification and treatment. More severe arterial injury may require additional neuroradiologic or neurosurgical intervention. Even in these rare instances, hemodynamically significant blood loss can usually be avoided with prompt and secure packing.

The temporal bone abuts the dura of the middle fossa above and the posterior fossa behind. Rarely, transgression of the dura can result either as the result of pathology or

from dissection of adjacent tissues. When this occurs, the surgeon can usually close the leak using autologous tissues. The surgeon may request administration of a Valsalva maneuver to check the integrity of the repair.

Usual preop diagnoses

Acute or chronic otitis media, cholesteatoma, hearing loss (conductive or sensorineural), otosclerosis, aural atresia (acquired or congenital), tympanic membrane perforation, temporal bone fracture, temporal bone neoplasm

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Postauricular or transcanal
Special instrumentation	Binocular surgical microscope, otologic microinstruments, otologic drill, laser
Unique considerations	Complex microanatomy, no paralytics given, need for facial nerve monitoring, possible avoidance of nitrous oxide
Antibiotics	Often administered preop
Surgical time	<1 h to >4 h
EBL	Usually minimal
Mortality	Minimal
Morbidity	Infection, hearing loss, vertigo, facial nerve dysfunction, spinal fluid leak
Pain score	1–6, typically mild

■ PATIENT POPULATION CHARACTERISTICS	
Age range	All ages
Male:Female	1:1

Incidence	Common
Etiology	Infectious, traumatic, congenital, neoplastic
Associated conditions	Hearing loss, head trauma

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for ear surgery are generally young and healthy, and most procedures are performed on an outpatient basis. Communicating with these patients may be difficult 2° ↓ hearing ability. Myringotomy and pressure equalization (PE) tube insertion are very short procedures. In contrast, ossicular and tympanic membrane reconstruction with facial nerve preservation may last several hours. Patients presenting for external ear reconstructions may have associated congenital abnormalities, which should be considered preop.

Respiratory	Patients may present with Sx of concurrent URI including cough (see Anesthetic Considerations for Tonsillectomy). Tests: As indicated from H&P
Cardiovascular	No special considerations Tests: EKG in older population.
Neurologic	Document brief neuro exam (including facial nerve function) and extent of any hearing loss
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GA, often with moderate controlled hypotension

Induction	Standard iv induction with propofol is used most
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often. If GETA is planned, a single intubating dose of an intermediate-acting nondepolarizing NMB (e.g., rocuronium) is usually inconsequential with regard to intraop monitoring of the facial nerve. In the absence of contraindications, succinylcholine can also be safely administered for tracheal intubation. Coadministration of iv remifentanyl bolus 3–4 mcg/kg (or alfentanil 40–60 mcg/kg) given over 90 sec, and propofol iv bolus 2–2.5 mg/kg, will provide excellent intubating conditions without the need for NMB in the majority of patients. PPV through the ETT or LMA are used most commonly, although spontaneous ventilation techniques can also be employed in selected patients.

ERAS

Verify preincision antibiotics. Assess volume status (see [Appendix K](#)). Maintain normovolemia and normothermia.

Maintenance

Moderate ↓ BP (see Introduction) may be requested. If an inhalational agent is used, N₂O is best avoided. Inhalational anesthesia, balanced inhalational anesthesia, TIVA, and even combined propofol-inhalational anesthetics have all been successfully used for maintenance. Sevoflurane may be preferred for outpatient surgery: it improves the quality of postop patient recovery, shortens the discharge time, is associated with reduced incidence of coughing and postop agitation compared to desflurane, and produces less somnolence and PONV compared to isoflurane.

The use of an LMA in children undergoing myringotomy and PE tube placement may be associated with ↓ incidence of intraop hypoxemia and better operating conditions as compared to the standard face mask technique. The second-generation LMA family of devices (LMA ProSeal™ or LMA Supreme) will offer the greatest advantages (see Introduction).

	As the patient's head is usually maintained in a fixed position (turned away from the surgeon), the position of these devices remains stable. A total intraoperative dose of fentanyl can be safely limited to 2–3 mcg/kg in the majority of patients, especially when remifentanyl is used intraoperatively.	
Emergence	At the end of the procedure, the patient's head is moved during dressing application, which may provoke gagging or bucking. Smooth emergence from anesthesia is mandatory. The use of the LMA eliminates laryngeal stimulation and facilitates smooth emergence (see Introduction). If postextubation laryngospasm occurs, the use of CPAP or face mask ventilation may be contraindicated, as it may unseat the tympanic membrane graft or disrupt other repairs. Postoperative pain is usually mild to moderate but may be minimal (e.g., laser stapedectomy, tympanoplasty) or severe (e.g., radical mastoidectomy). Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 18–20 ga × 1 BSS @ 3–4 mL/kg/h	Blood loss is usually minimal. An IV is usually placed at the side where the surgeon is positioned, to eliminate the possibility of triggering sudden surgical tremor during cuff inflation. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Control of bleeding	Head-up position (10–15°), per surgeon's preference Injection of epinephrine-	These special considerations are applicable to surgical procedures carried out

	containing solutions by surgeon Deliberate moderate ↓ BP	under microscopic control, where even the smallest quantity of blood may interfere with visualization.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes ✓ opposite ear	The LMA should be diligently secured midline. Surgical drapes can usually be arranged in a way that allows the anesthesiologist limited access to the patient's airway.

POSTOPERATIVE

Complications	PONV Facial nerve injury	Liberal prophylactic use of antiemetics is indicated.
Multimodal pain management	Fentanyl (outpatient) Hydromorphone (inpatient)	The use of a PCA may be appropriate in older children and adults. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

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- comparison of desflurane, isoflurane and sevoflurane. *Eur J Anaesthesiol* 2004; 21:902–6.
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NEUROTOLOGIC SKULL BASE SURGERY

SURGICAL CONSIDERATIONS

The skull base refers to the floor of the cranial cavity and can be divided into anterior, middle, and posterior fossae. It is formed by the frontal, ethmoid, sphenoid, temporal, and occipital bones. Neuro-otologists primarily deal with lesions in the posterior fossa, which is bordered by the clivus (anterior), temporal bone (lateral), and occipital bone (posterior). An axial view of the skull identifying these structures is shown in [Figure 3.14](#).

The term **skull base surgery** is a misnomer as the majority of lesions treated are located adjacent to the brainstem and not intrinsic to the skull base itself. Removal of the skull base bone allows exposure to these lesions while minimizing cerebral and cerebellar

retraction.

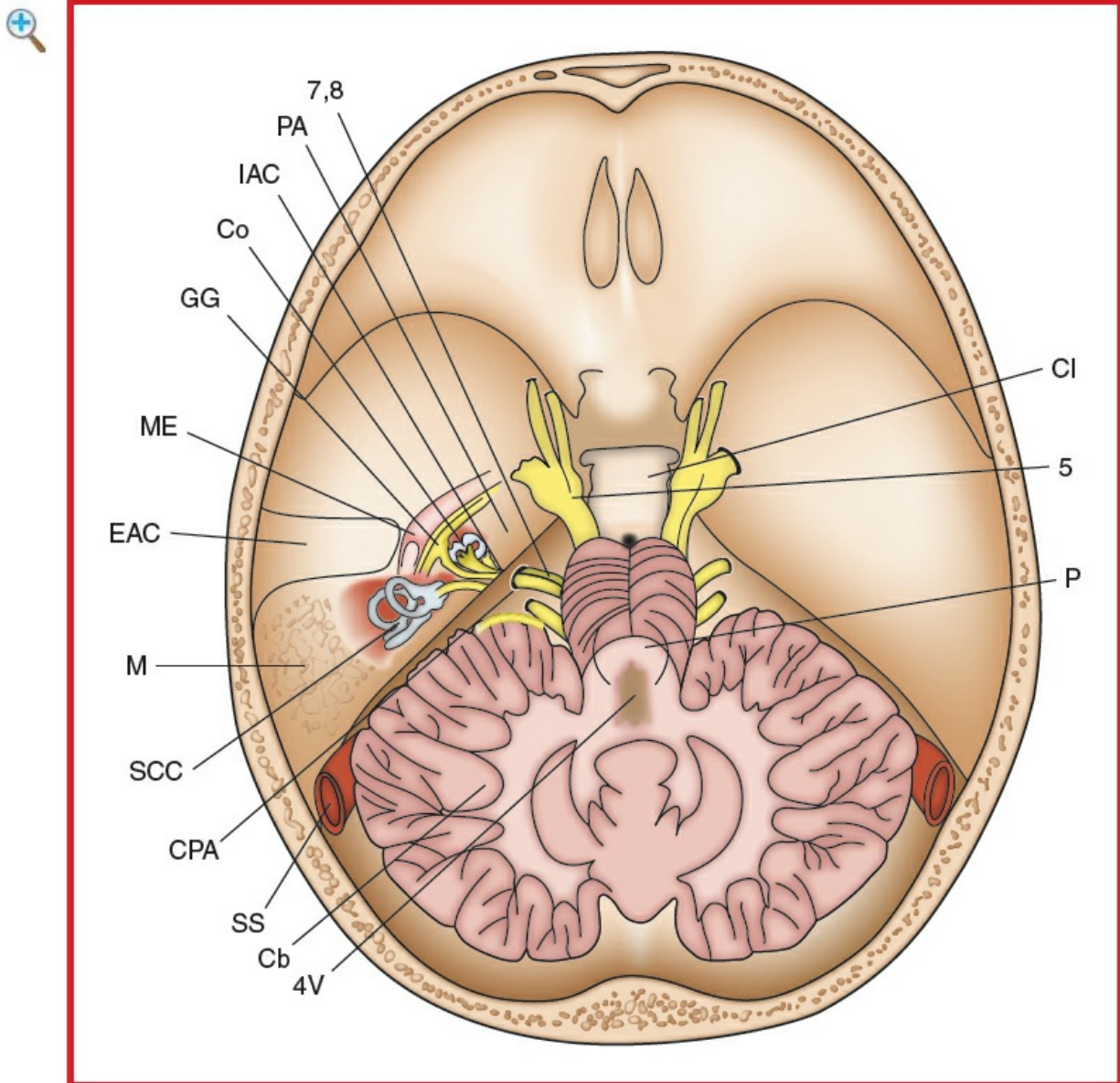


Figure 3.14 An axial view of the skull through the level of the internal auditory canal and cerebellopontine angle. 5, trigeminal nerve; 7, facial nerve; 8, audiovestibular nerve; PA, petrous apex; IAC, internal auditory canal; Co, cochlea; GG, geniculate ganglion of the facial nerve; ME, middle ear; EAC, external auditory canal; M, mastoid air cell system; SCC, semicircular canals; CPA, cerebellopontine angle; SS, sigmoid sinus; 4V, 4th ventricle; Cl, clivus; P, pons; Cb, cerebellum. (Republished with permission from Jackler RK: Atlas of Neurotology and Skull Base Surgery. Mosby, St. Louis: 1996. Copyright Dr. R. K. Jackler © 2007.)

The cerebellopontine angle (CPA) is a fluid-filled space containing the facial nerve (VII) and vestibulocochlear nerve (VIII) coursing laterally toward the internal auditory canal. A depiction of the CPA and associated cranial nerves is shown in [Figure 3.15](#). This is one of the most commonly approached areas in skull base surgery, and lesions include vestibular schwannomas (91.3%), meningiomas (3.1%), epidermoids (2.4%),

nonvestibular schwannomas (1.4%), and arachnoid cysts (0.5%).

Vestibular schwannomas (also known as **acoustic neuromas**) are removed via three surgical approaches: translabyrinthine, retrosigmoid, and middle fossa. The translabyrinthine approach uses a postauricular incision to access the temporal bone (Fig. 3.16). Retraction of the temporal lobe and cerebellum is minimized as the entire mastoid and labyrinth are removed to create access. This approach results in complete sensorineural hearing loss and is most commonly used in patients with large tumors and/or nonserviceable hearing. Patients with serviceable hearing may elect for a hearing conservation approach for their tumors (retrosigmoid or middle fossa). The retrosigmoid approach uses a more posterior craniotomy between the sigmoid and transverse sinuses. The cerebellum is retracted posterior away from the petrous face of the temporal bone (Fig. 3.15). Disadvantages include the increased incidence of postop headache and the need for rigid skull fixation (e.g., Mayfield). The middle fossa approach places the craniotomy above the ear and requires retraction of the temporal lobe (Fig. 3.17). This approach has the highest rate of hearing conservation but can only be used in smaller tumors without increasing the risk of postop facial palsy.

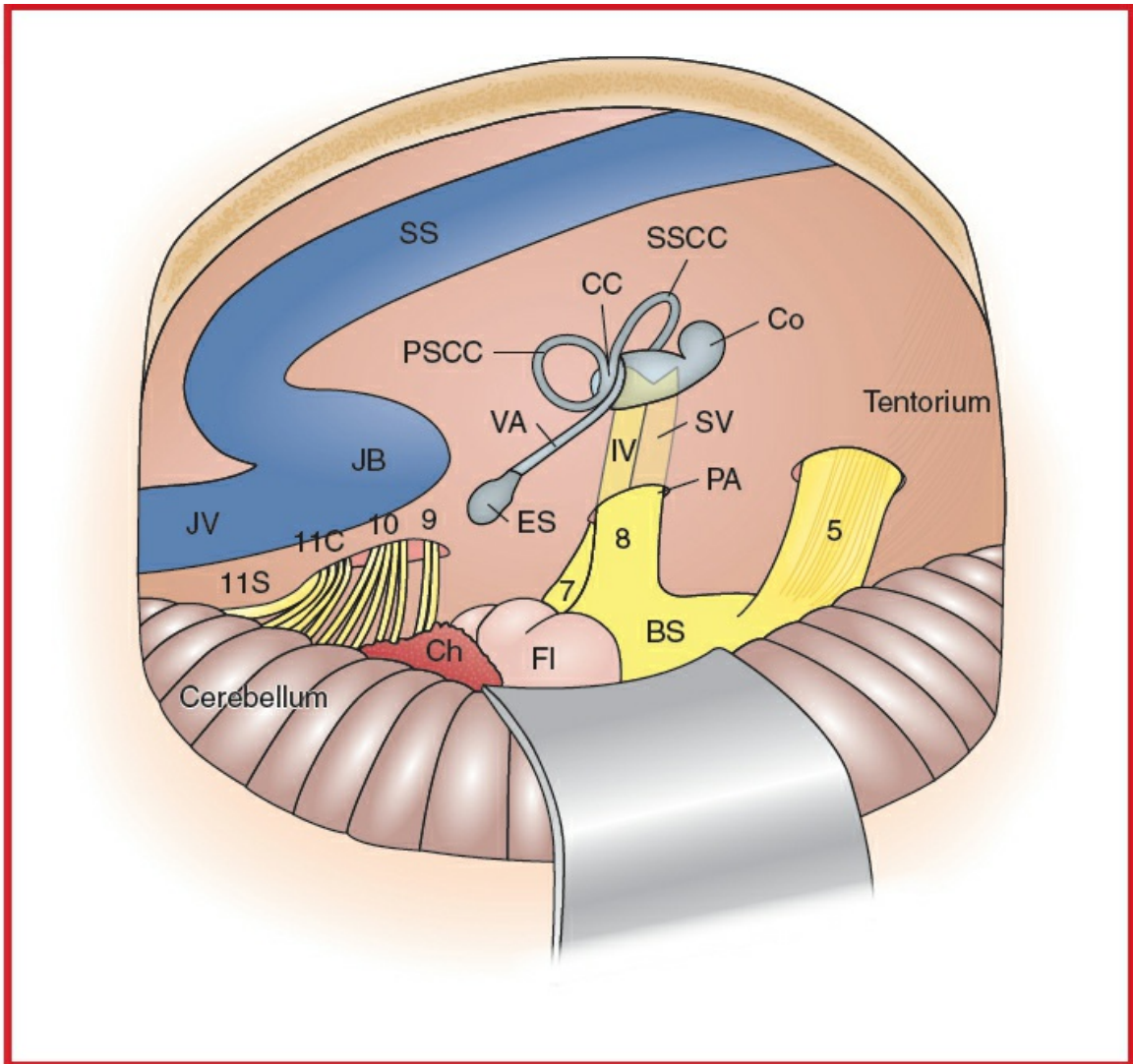


Figure 3.15 Anatomical relationships of the cerebellopontine angle shown through a retrosigmoid posterior fossa craniotomy. JV, jugular vein; JB, jugular bulb; SS, sigmoid sinus; 11s, spinal component of the accessory nerve; 11c, cranial component of the accessory nerve; 10, vagus nerve; 9, glossopharyngeal nerve; Ch, choroid plexus emanating from the lateral recess of the 4th ventricle; Fl, flocculus; BS, brainstem surface (pons); 5, trigeminal nerve; 7, facial nerve; 8, audiovestibular nerve; PA, porus acusticus; IV, inferior vestibular nerve; SV, superior vestibular nerve; ES, endolymphatic sac; VA, vestibular aqueduct; PSCC, posterior semicircular canal; CC, common crus; SSCC, superior semicircular canal; Co, cochlea. (Republished with permission from Jackler RK:Atlas of Neurotology and Skull Base Surgery. Mosby, St. Louis: 1996. Copyright Dr. R. K. Jackler © 2007.)

Another commonly approached area is the jugular foramen, which lies at the junction of the petrous apex and occipital bone. This foramen connects the intracranial compartment to the neck and contains the glossopharyngeal nerve (IX), vagus nerve (X), accessory nerve (XI), jugular bulb, and inferior petrosal sinus. The most common tumors in this area are paragangliomas. Paragangliomas are divided into two groups: (a) adrenal paragangliomas (also known as pheochromocytomas) and (b) extra-adrenal paragangliomas located in the abdomen, chest, and head and neck regions. In the jugular foramen, these tumors are often referred to as **glomus jugulare**. Less common

pathologies in this area include meningiomas and lower cranial nerve schwannomas.

Glomus jugulare tumors are highly vascular, and patients undergo preop embolization before surgery. Two to 4 U of autologous or cross-matched blood are kept available depending on the size and extent of the tumor. Large-bore ivs should be placed in case rapid transfusion is needed. In addition to a lateral craniotomy similar to the translabyrinthine approach, a limited neck dissection is performed to gain vascular control of the jugular vein and carotid artery. This transjugular craniotomy exposes the entire jugular fossa and involves resection of the involved sigmoid sinus and jugular vein ([Fig. 3.18](#)). New lower cranial nerve deficits occur in 25–50% of patients; therefore, appropriate airway and aspiration precautions must be taken. Rehabilitation often involves speech therapy and VC medialization. Tracheotomy and PEG tube placement are rarely necessary.

In all of the preceding craniotomies, the patient is turned 180° away from the anesthesiologist. This allows room for the operating microscope, scrub assistant, and surgeon to all be placed around the head of the patient (Figs. 3.19 and 3.20). The patient's head is turned to one side to allow exposure of the lateral cranium and neck. In the retrosigmoid craniotomy, a Mayfield is used for rigid skull fixation. As lateral table rotation is often used during these cases, three straps should be used to secure the patient, and the contralateral arm should be well padded in order to avoid ulnar neuropathy. Neurophysiologic monitoring (EMG, auditory brainstem responses, and SSEPs) is used throughout the case; therefore, muscle relaxants should be avoided.

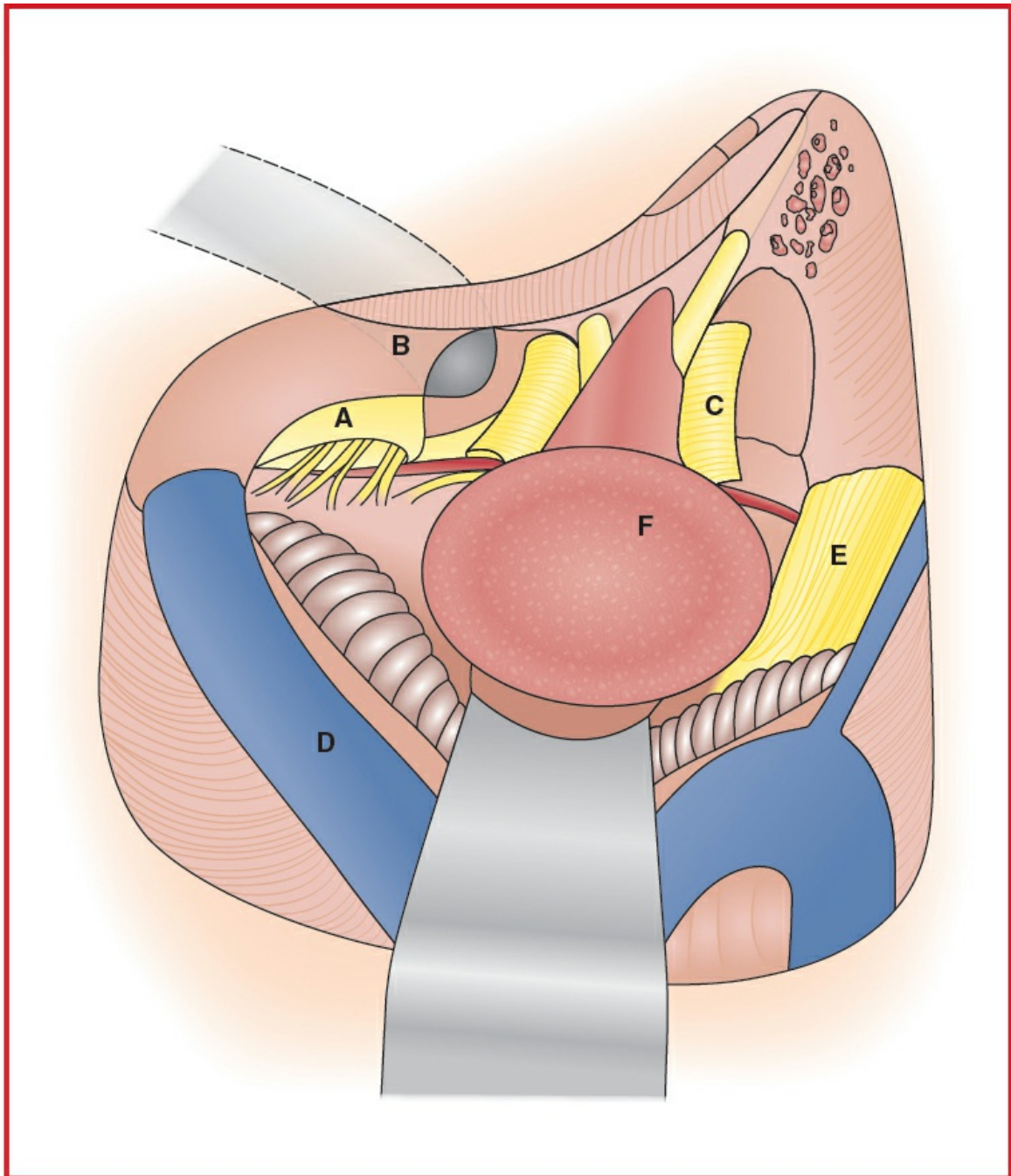


Figure 3.16 Typical left translabyrinthine posterior fossa craniotomy exposure of a medium-sized tumor. Inferiorly, the lower cranial nerves (A) are visible, and the jugular bulb (B) has been identified. Troughs have been drilled above and below the IAC, and the dura (C) has been reflected off the tumor surface. The sigmoid sinus (D) and cerebellum are gently retracted posteriorly. The trigeminal nerve (E) is located superiorly. The facial nerve (F) takes a variable and often serpentine course across the medial side of the tumor. (Republished with permission from Jackler RK: Atlas of Neurotology and Skull Base Surgery. Mosby, St. Louis: 1996. Copyright Dr. R. K. Jackler © 2007.)

Stimulation of the trigeminal or vagus nerves can cause bradycardia; therefore, the surgeon should be informed of any unexpected physiologic changes occurring with the patient. Stable low blood pressure is useful in controlling slow bleeding in the surgical field. Significant hemodynamic blood loss is rare and can usually be controlled with

secure packing. Large venous injuries may result in air embolism, and arterial injuries may require the assistance of interventional neuroradiology. Having 1 U of cross-matched blood available is prudent in most craniotomies, whereas 2–4 U may be required in glomus jugulare cases. CSF leaks are controlled using an abdominal fat graft.

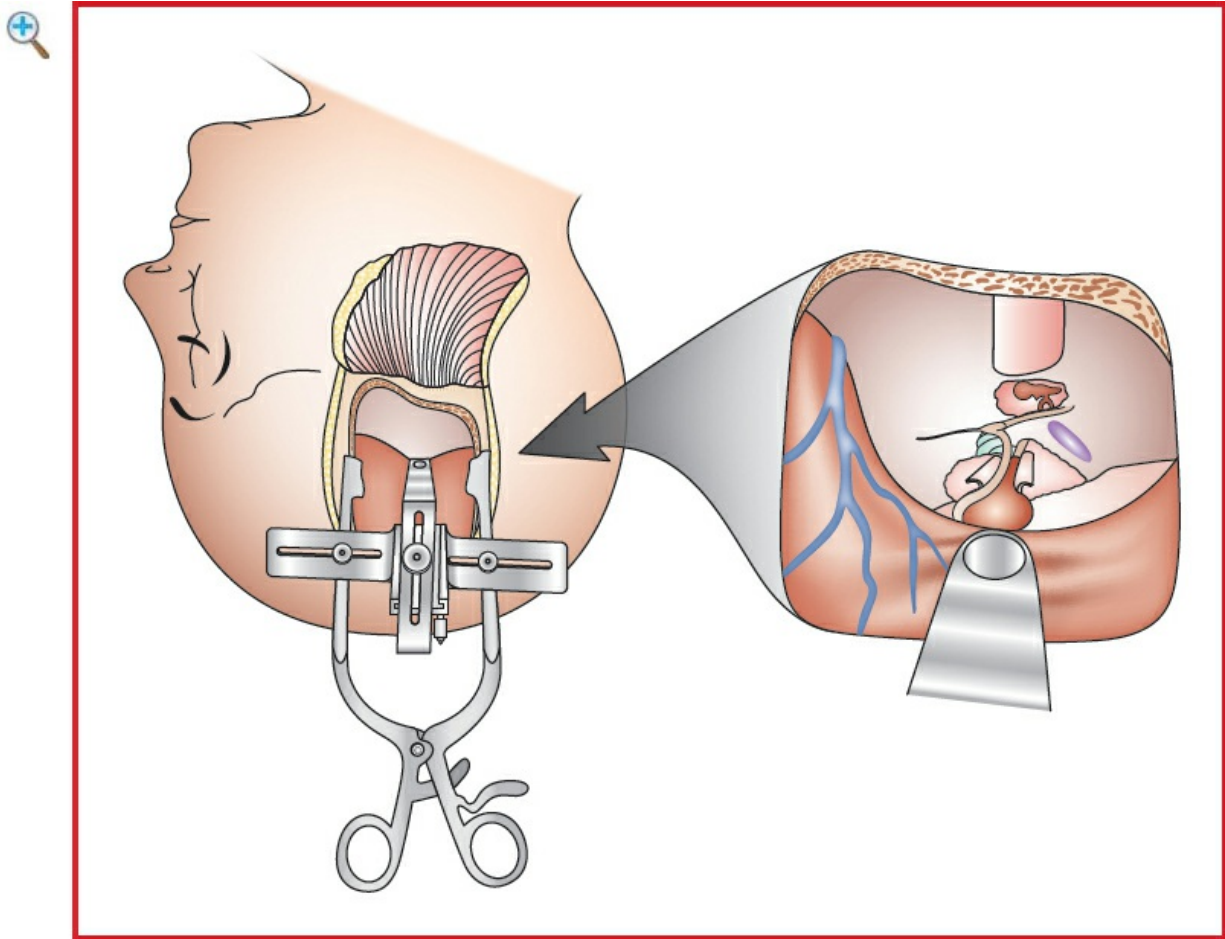


Figure 3.17 Middle fossa exposure of a small right-sided tumor. The retractor is engaged over the posterior lip of the petrous bone and retracts the temporal lobe. Bone has been removed around the IAC, and the dura has been opened to expose the tumor. The facial nerve typically courses across the superior aspect of the tumor surface. (Republished with permission from Jackler RK:Atlas of Neurotology and Skull Base Surgery. Mosby, St. Louis: 1996. Copyright Dr. R. K. Jackler © 2007.)

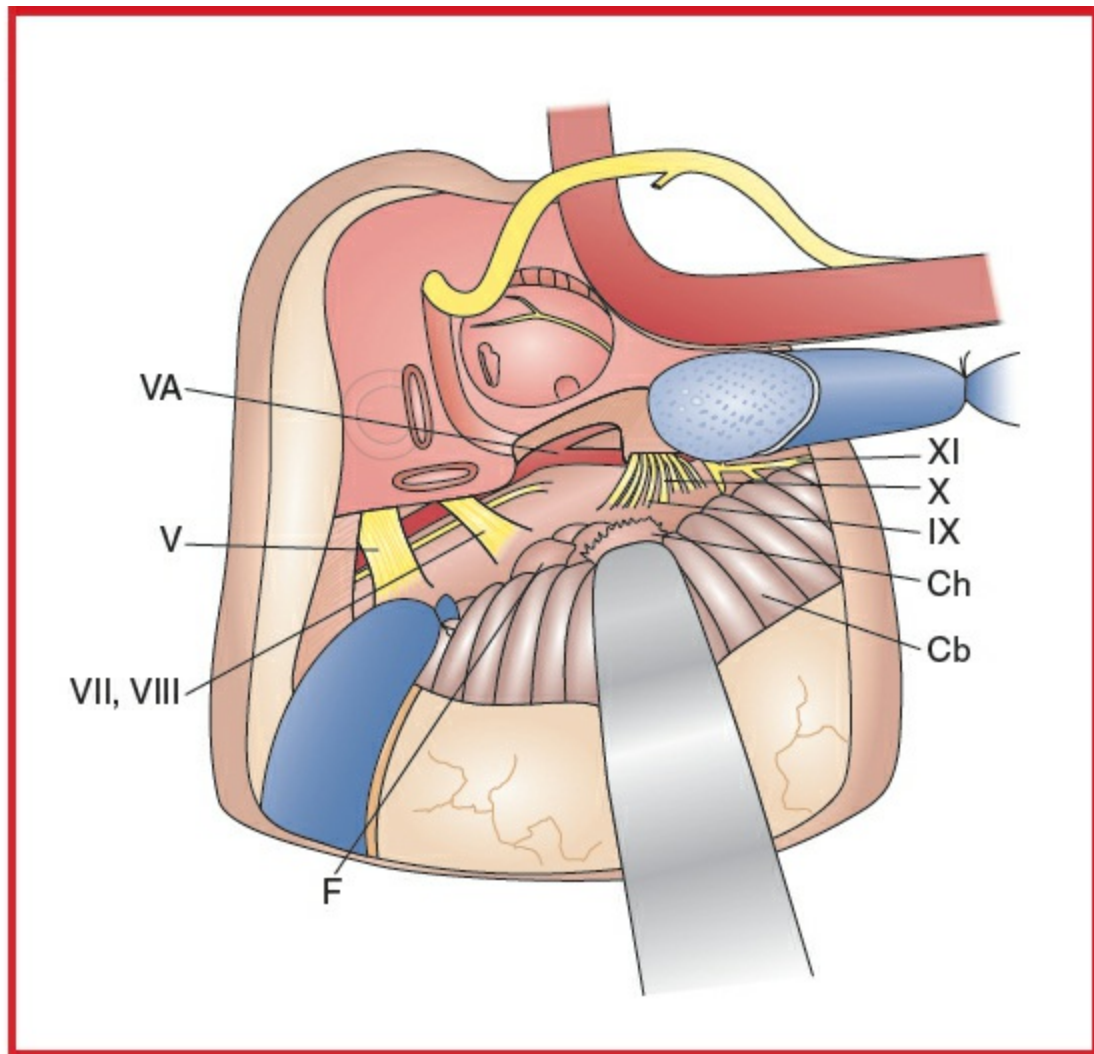


Figure 3.18 Transjugular craniotomy illustrating the degree of intracranial exposure obtained following resection of the sigmoid-jugular system and wide opening of the posterior fossa dura. Note the multiple small rootlets of the lower cranial nerves emanating from the lateral surface of the medulla. In contrast to extracranial procedures, the sigmoid sinus is ligated proximally rather than packed extraluminally. While this illustration depicts anterior rerouting of the facial nerve, this is not necessary in most intracranial jugular foramen tumors. The sigmoid sinus has been controlled with a suture ligature just distal to the transverse-sigmoid junction. VA, vertebral artery; F, flocculus; Ch, choroid; Cb, cerebellum; V, trigeminal nerve; VII, facial nerve; VIII, audiovestibular nerve; IX, glossopharyngeal nerve; X, vagus nerve; XI, accessory nerve. (Republished with permission from Jackler RK Atlas of Neurotology and Skull Base Surgery. Mosby, St. Louis: 1996. Copyright Dr. R. K. Jackler © 2007.)

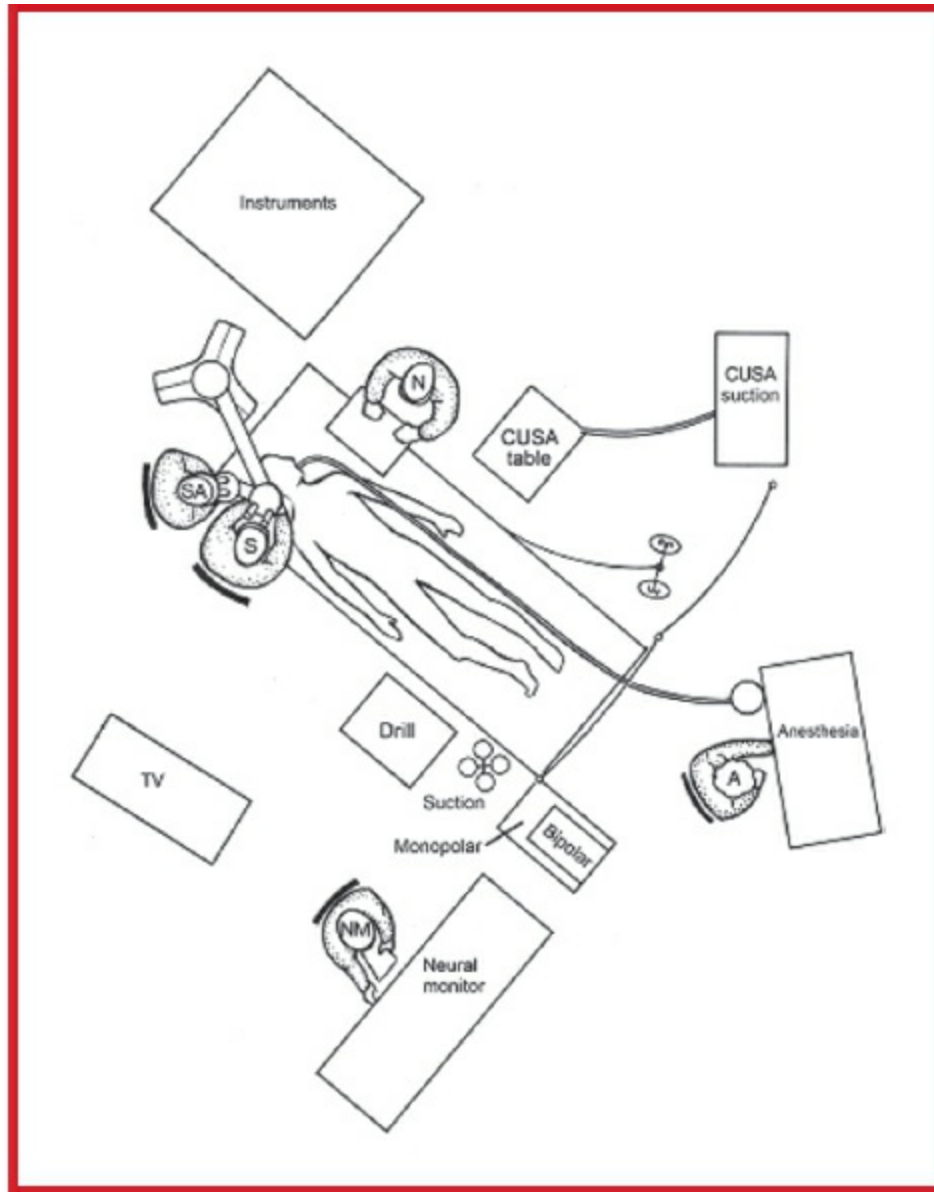
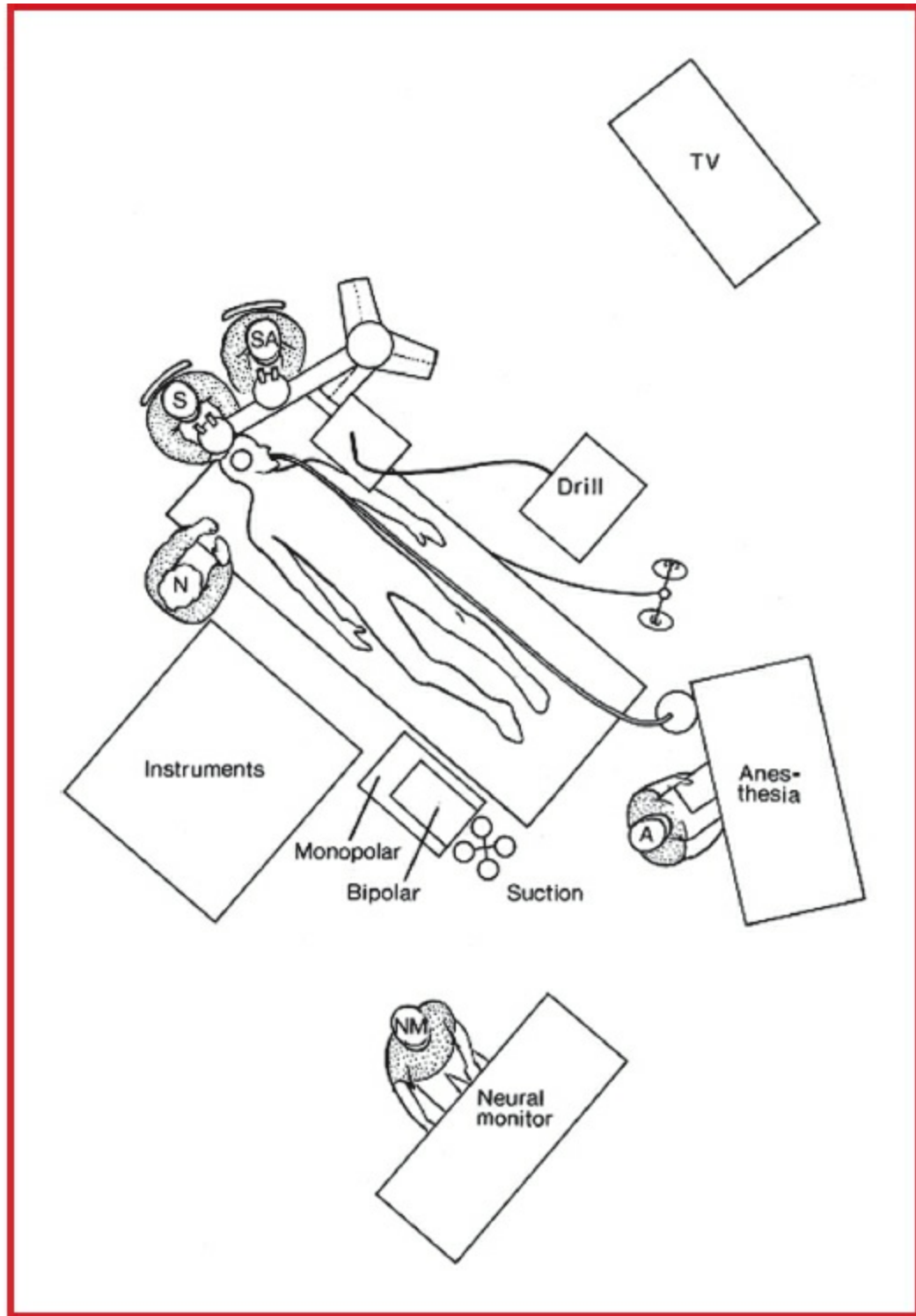


Figure 3.19 Operating room setup for posterior fossa craniotomy (translabyrinthine, retrosigmoid, and transjugular). (Republished with permission from Jackler RK:Atlas of Neurotology and Skull Base Surgery. Mosby, St. Louis: 1996. Copyright Dr. R. K. Jackler © 2007.)



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Usual preop diagnoses

Vestibular schwannoma (acoustic neuroma); meningioma; paraganglioma (glomus jugulare); lower cranial nerve schwannoma; epidermoid cyst; arachnoid cyst;

cholesterol granuloma; chondrosarcoma; chordoma

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine, turned 180° from anesthesia (Figs. 3.19 and 3.20). Three straps to secure patient and arms well padded. Mayfield rigid fixation in retrosigmoid craniotomies
Incision	Postauricular or temporal with possible extension into neck depending on approach
Special instrumentation	Microscope, neurophysiologic monitoring team, ultrasonic aspirator, neck dissection and vascular instruments (transjugular), middle fossa retractor (middle fossa), otologic drill
Antibiotics	Ceftriaxone 2 g iv or vancomycin 1 g iv for CSF penetration
Procedure time	4–12 h
Closing considerations	Minimize coughing and ↑ ICP. Abdominal fat often harvested for CSF leak closure. Head dressing may be placed
EBL	100–1000 mL depending on approach and tumor. One unit of cross-matched or autologous blood for craniotomies (2–4 U for glomus jugulare tumors) should be available
Postop care	Extubation in majority of cases. Monitoring in the ICU for the first night. Minimal sedatives to allow neurologic monitoring. Airway and aspiration precautions in patients with possible lower cranial nerve palsies
Mortality	Rare

Morbidity	Hearing loss (approach dependent) Vertigo Facial nerve dysfunction Hoarseness Aspiration CSF leak Meningitis Stroke (rare)
Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Fifth to sixth decades for most pathology Younger patients in tumors associated with neurofibromatosis (NF-2)
Male:Female	1:1 for vestibular schwannoma
Incidence	1/100,000
Etiology	Neoplastic
Associated conditions	Hearing loss (use assistive devices when possible), vertigo, lower cranial nerve dysfunction (aspiration precautions), neurofibromatosis

ANESTHETIC CONSIDERATIONS FOR NEUROTOLOGIC SURGERY

PREOPERATIVE

Large **vestibular schwannomas** (acoustic neuromas) can compress the brainstem, cause cranial nerve deficits, and occasionally produce symptoms of ↑ intracranial pressure (ICP). If ↑ ICP is present, anesthetic management should follow the same general principles as for other intracranial mass lesions.

Highly **vascular tumors** are usually embolized during preop neuroangiography. If the tumor's blood supply is from the anterior and posterior ethmoidal arteries, these vessels cannot be safely embolized unless the patient has already lost vision. ICA balloon occlusion test is performed during angiography if the ICA is involved or encased.

Large **glomus tumors** (GT) are capable of secreting catecholamines and occasionally

serotonin, presenting with symptoms of pheochromocytoma (e.g., hypertension, sweating, cardiovascular instability) or carcinoid syndrome. Urine and plasma catecholamine levels should be routinely checked preop for all extensive GT, even if the patient is asymptomatic. The same preop pharmacologic stabilization used for patients with a pheochromocytoma may be required for these patients. Dehydration and/or electrolyte abnormalities must be corrected preop. Intraop, the release of histamine and bradykinin can cause bronchospasm, tachycardia, and both hypotension and hypertension. Large GT can cause cranial nerve IX, X, XI, and XII palsies, which may predispose to airway compromise, by either obstruction or aspiration (the risk of gastroparesis is increased in these patients).

The **anterior skull base** is a common location for many intradural and extradural cranial and/or facial pathologies. The anterior cranial base surgery may involve combined transthemoidal incision and bifrontal craniotomy approaches. With that approach, the intraop surgical navigation set is frequently used, and a lumbar drain may be placed after induction of anesthesia.

Respiratory	GT patients may be at ↑ risk for aspiration (cranial nerve impairment) or airway obstruction depending on tumor location. GT may release histamine or bradykinin → bronchospasm. Tests: Imaging studies
Cardiovascular	~5% of GT secrete catecholamines → cardiovascular instability. Tests: ECG, UA catecholamine metabolites
Neurologic	Examine cranial nerves and document preop function. Check for Sx of ↑ ICP.
Laboratory	CBC; coagulation studies; other tests as indicated from H&P
Premedication	In the absence of airway compromise, standard premed is appropriate.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: The essential anesthesia considerations for neurotologic surgery are largely similar to otologic surgical procedures. The anesthesiologist must be familiar with the principles of neuroanesthesia and understand both the pathologic process involved and the planned surgical approach.

Standard premedication with short-acting benzodiazepine (e.g., iv midazolam) is warranted in the majority of cases, but should be done with caution or avoided if symptoms of $\uparrow$ ICP are present. IV antibiotics with good CSF penetration (e.g., iv ceftriaxone 1–2 g) are uniformly used. Corticosteroids (e.g., iv dexamethasone 10–12 mg) may or may not be given, depending on the size of the tumor and presence of brain edema.

The most critical aspect of the skull base surgery is identification and preservation of the cranial nerves. Intraoperative neurophysiologic monitoring (IOM) is used widely, especially when the tumors are located at the cervicomedullary junction, jugular foramen (JF), the upper clivus, when the tumors are large, and during revision surgeries. Cranial nerves VII, IX, X, XI, and XII are most commonly monitored.

These surgeries can be extremely lengthy, and meticulous attention to proper patient positioning is paramount.

Induction

Preop airway compromise is rare. Standard GETA is used in the majority of cases. Patients with GT or other lesions extending into the JF, upper neck, or skull base may require intraop identification and monitoring of the vagus nerve. In these patients, placement of specialized ETT (e.g., Xomed ETT, Medtronic Xomed Inc., Jacksonville, FL) to monitor EMG activity in the vocal cords may be requested by the surgeon.

The use of an intermediate-acting NMB (e.g., rocuronium) for tracheal intubation is inconsequential with regard to IOM. In patients with $\uparrow$ ICP, standard induction precautions should be observed. With large tumors, a lumbar drain will be placed after induction to remove CSF to reduce surgical brain retraction.

ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	The anesthetic technique should facilitate IOM. Intraop use of NMB is avoided. Remifentanyl-based techniques, allowing a safe reduction of the concentration of inhalational agent to ≤ 0.5 minimal alveolar concentration (MAC) are widely used. Administration of iv mannitol (0.5–1 g/kg), started after the scalp incision, and the use of mild hyperventilation will facilitate brain relaxation for dural entry. Patients should be kept euvolemic, and electrolyte status should be monitored during the surgery. Stimulation of the trigeminal or vagus nerves during surgery can cause $\downarrow\downarrow$ HR, $\downarrow$ BP or $\uparrow$ BP, and arrhythmias and even transient asystole, and the surgeon should be promptly informed of their occurrence.
Emergence	A smooth extubation is important because coughing and bucking on the ETT may provoke postop bleeding, increase cerebral edema, and cause tension pneumocephalus (TP). Tracheal extubation is carried out in the majority of cases. The exceptions are the patients who had undergone free flap reconstruction of the cranial base and possibly those where cranial nerve deficits existed preop. New paralysis of the lower cranial nerves with dysphasia remains the most serious complication of surgery, and delayed extubation and tracheotomy may be required in some patients. The patients are observed in the ICU overnight. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.
Blood and fluid	IV: large-bore access With the exception of GT,

requirements	BSS @ 3–4 mL/kg/h	blood loss is usually gradual and rarely requires transfusion. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors Arterial line ± CVP Foley catheter	A-line is placed after induction of anesthesia. CVP should be guided by the patient's medical history; consider if significant blood loss is anticipated. A Foley catheter is used routinely.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Massive blood loss Cranial nerve injury VAE Aspiration	Glomus tumors are highly vascular. Airway compromise may preclude extubation. Monitor for VAE (e.g., ETCO ₂).

POSTOPERATIVE

Complications	Trigeminal and vagal reflexes Cranial nerve injury Tension pneumocephalus	IV atropine Extubation considerations (cranial nerve deficits)
Multimodal pain management	Fentanyl/hydromorphone	The use of a PCA may be appropriate in older children and adults. Emphasize multimodal with peripheral local anesthetics (see

ERASPONV protocol (see
[Appendix B, p. B-1](#))Encourage early enteral
nutrition, Foley catheter
removal, and mobilization**Suggested Readings**

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RECONSTRUCTIVE SURGERY FOR SLEEP- DISORDERED BREATHING

SURGICAL CONSIDERATIONS

Robert W. Riley, Nelson B. Powell, Donald M. Sesso, and Jerome E. Hester

Description

The surgical approaches to the upper airway attempt to relieve obstruction occurring most commonly at the level of the palate, base of the tongue, or pharynx. These fall into three categories: (a) classic procedures that directly enlarge the upper airway, (b) specialized procedures that directly enlarge the upper airway, and (c) tracheotomy to bypass the pharyngeal portion of the upper airway. The surgeon performs a preop evaluation, including complete head and neck exam, fiberoptic examination of the upper airway, and cephalometric radiographs. This, together with the results of the polysomnogram, will enable the surgeon to determine what levels of the airway need to be surgically modified. Individuals with severe obstruction may require a multistage approach to treatment.

Uvulopalatopharyngoplasty (UPPP)([Fig. 3.21](#)) is a procedure that removes a rim of the soft palate, including the uvula. This shortens and tightens this tissue, thus preventing collapse during sleep. The tonsils, if present, are removed. The muscular crease of the palate is used as a landmark to prevent overly aggressive resection, which could → velopharyngeal insufficiency (VPI), an uncommon but serious complication. The wound is then closed, using interrupted absorbable sutures.



UPPP

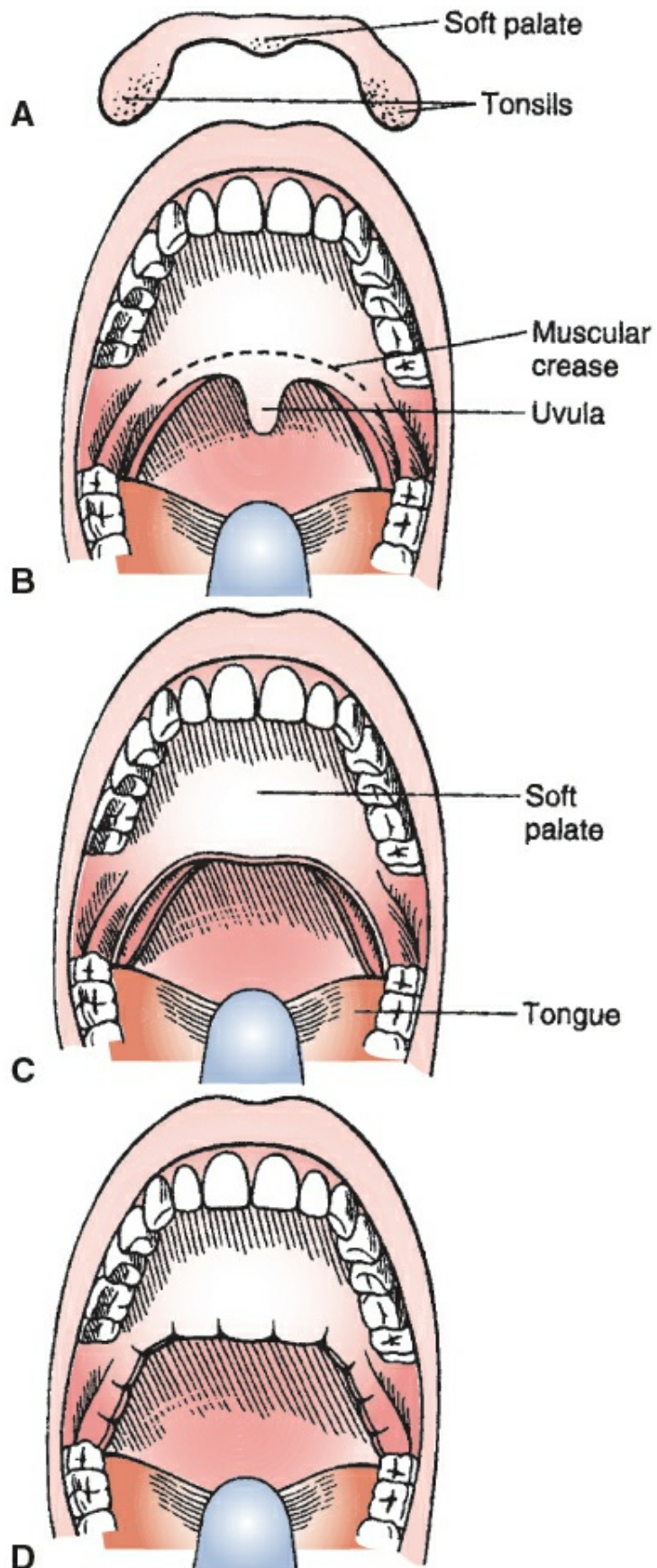


Figure 3.21 Uvulopalatopharyngoplasty (UPPP)(**A and B**) tonsils and redundant soft palate are excised; (**C**) mucosal flaps are prepared for closure; (**D**) the soft palate is closed to itself, and the anterior and posterior tonsillar pillars are sutured to each other.

Uvulopalatal flap (UPF) ([Fig. 3.22](#)) is a variation of UPPP, used for treating palatal obstruction. Rather than excising a rim of the soft palate, the mucosa of the anterior aspect of the uvula is removed, along with a corresponding area of the soft palate. The uvula is then reflected superiorly and sutured into place with absorbable suture. This procedure may be done in combination with a tonsillectomy. UPF is theoretically reversible if signs of VPI become evident. UPF also may be somewhat less painful than UPPP.

Uvulopalatopharyngoglossoplasty (UPPGP) is a rarely performed, intraoral procedure incorporating a modified UPPP with limited resection of the base of the tongue for both retropalatal collapse and retrolingual collapse.

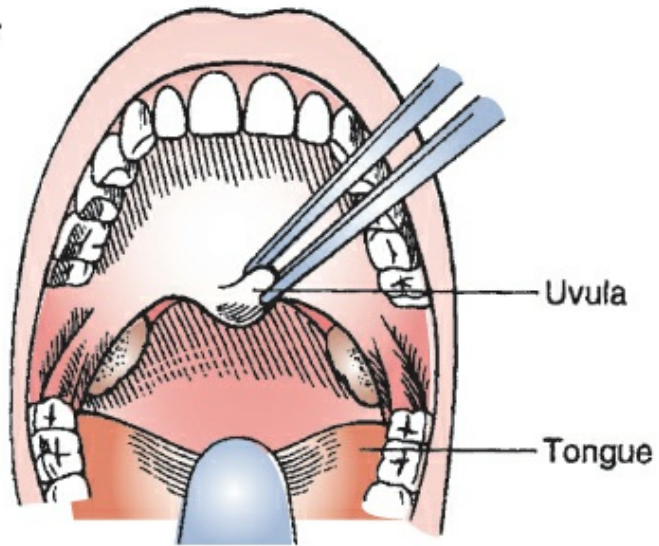
Laser midline glossectomy (LMG) is used to enlarge the retrolingual airway by excision of $\sim 2.5 \times 5$ cm of midline tongue tissue through an intraoral approach. This also may require lingual tonsillectomy, reduction of the aryepiglottic folds, and partial epiglottectomy ([Fig. 3.23](#)). LMG usually is combined with a tracheotomy for airway protection.

Lingualplasty (LP) is the same procedure as LMG, except that additional tongue tissue is extirpated posteriorly and laterally to that portion removed by LMG ([Fig. 3.23](#)). It is usually combined with a tracheotomy (see below) for airway protection.

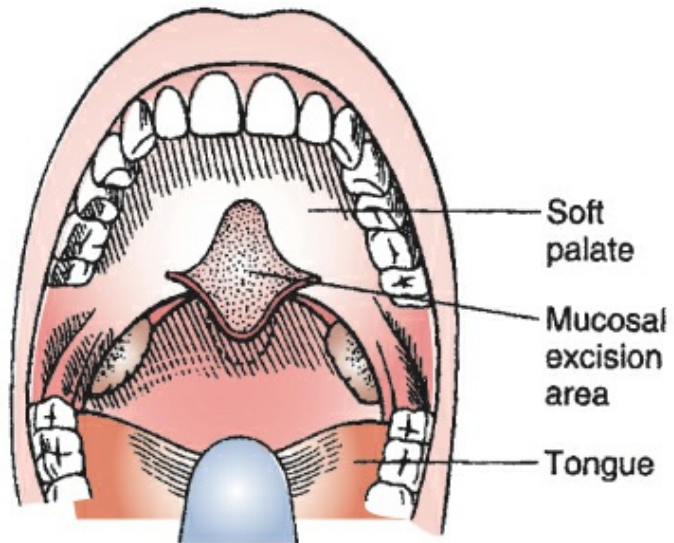


UPF

A



B



C

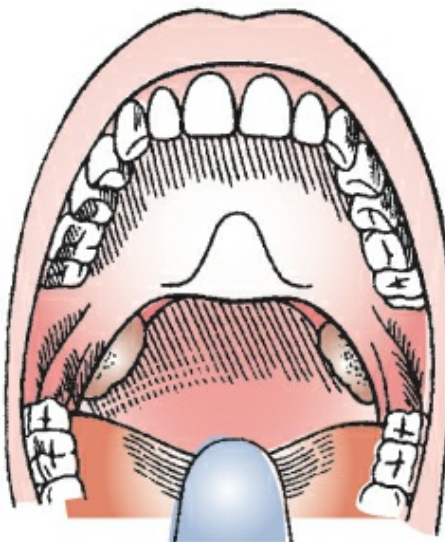


Figure 3.22 Uvulopalatal flap (UPF):(A) the uvula is reflected back to the hard palate to identify the muscular crease; (B) the mucosa on the oral side of the uvula and soft palate are removed, and part of the uvula is amputated; (C) mucosal incisions are closed with absorbable suture.

Inferior sagittal mandibular osteotomy and genioglossal advancement (MOGA) (Fig. 3.24) is an intraoral approach designed to enlarge the retrolingual area. This procedure relies on the firm attachment of the genioglossus muscle to the geniotubercle, a bony protuberance on the medial (lingual) aspect of the mandible. A mucosal incision is made intraorally, and soft tissue, including the mentalis muscle, is elevated off the mandible. Osteotomies, which include the geniotubercle on the inner cortex, are then performed. The segment is advanced and rotated to lock it in place. The outer cortex is removed, and the fragment is fixated to the inferior mandible with a titanium screw. The advancement is limited by the width of the mandible and laxity in the genioglossus muscle.

Hyoid myotomy and suspension (HM) is a retrolingual procedure that alleviates obstruction by redundant lateral pharyngeal tissue or a retrodisplaced epiglottis. A horizontal cervical incision above the hyoid bone is performed, and the dissection is carried down to the suprahyoid musculature. The midline hyoid bone is isolated and then advanced over the thyroid ala. It is then immobilized with two medial and two lateral permanent sutures (Fig. 3.25). The wound is closed, a drain placed, and a pressure dressing applied.

Maxillomandibular osteotomy and advancement (MMO) prevents retropalatal collapse through stenting of the superior pharyngeal muscles and widening of the nasopharyngeal inlet. It also minimizes retrolingual obstruction by placing the genioglossus muscle under tension, providing more room in the oral cavity for soft tissues, and stenting the lateral pharyngeal wall. An outer-table cranial bone graft usually is performed, along with arch bar placement (or orthodontic banding in an outpatient setting) prior to the osteotomies. A LeFort I maxillary osteotomy and bilateral sagittal-split mandibular osteotomy are performed. The skeletal arches are advanced forward ~10 mm and secured with the aid of a methylmethacrylate dental splint (Fig. 3.26). Immobilization with wires, plates, and screws follows, along with wound closure, intermaxillary fixation, and pressure dressing application. This procedure usually is performed if previous upper airway procedures have not completely relieved the sleep-related obstruction.

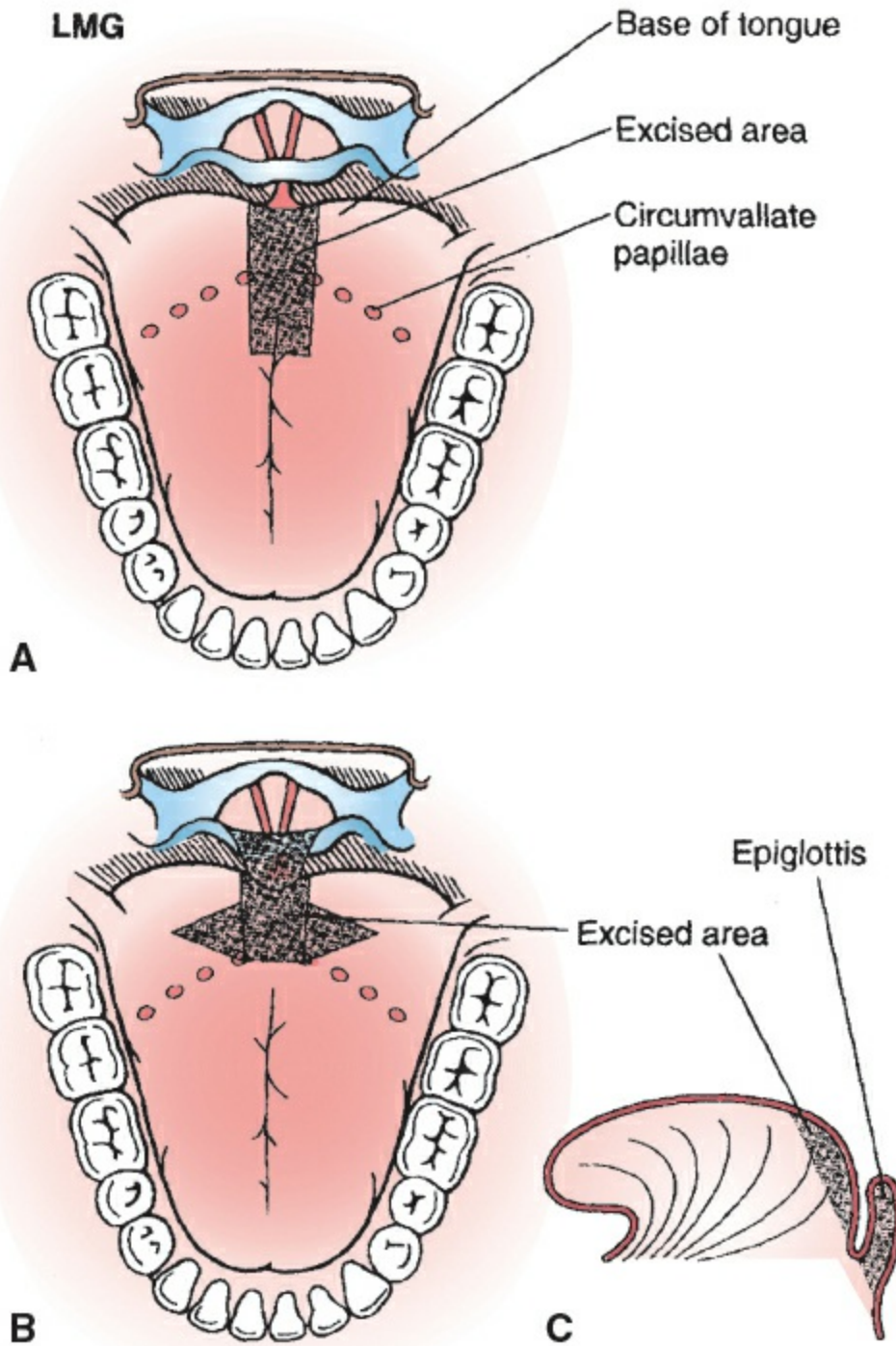


Figure 3.23 LMG/lingualplasty technique. **A:** Excision of a midline segment of the base of the tongue is performed with a laser or electrocautery; this excision occasionally is carried lateral. **B:** The remaining tongue muscle edges are reapproximated with absorbable suture. **C:** Lateral view of the tongue.

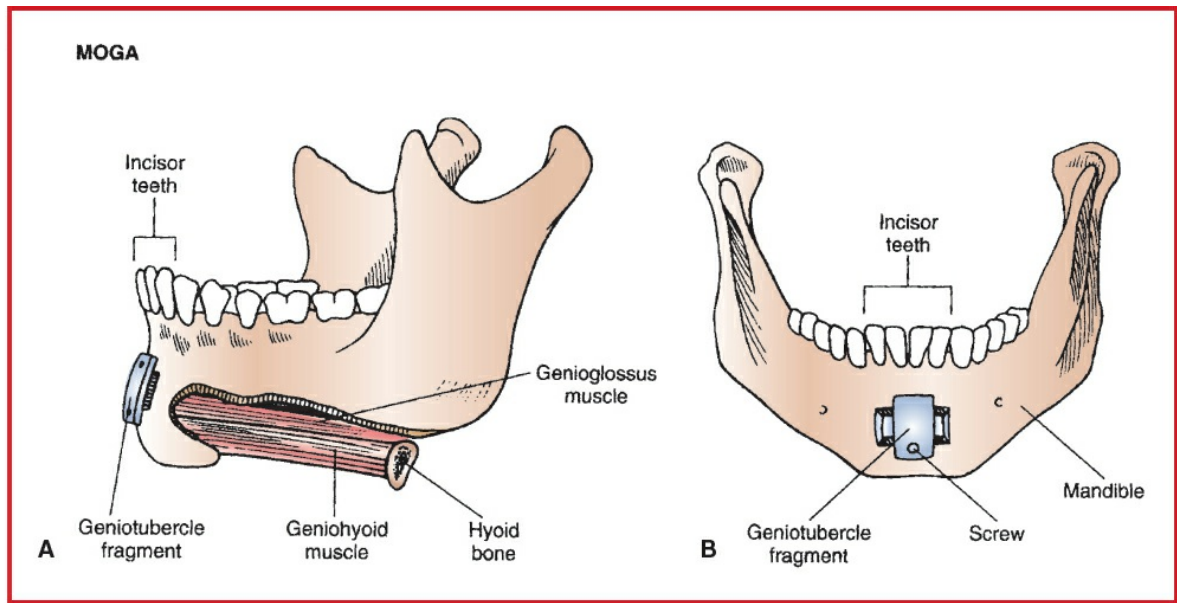


Figure 3.24 The mandibular osteotomy and genioglossus advancement (MOGA) technique. **A:** Lateral view. **B:** Anterior view. A rectangular anterior mandibular osteotomy below the incisor teeth is advanced, rotated, and immobilized.

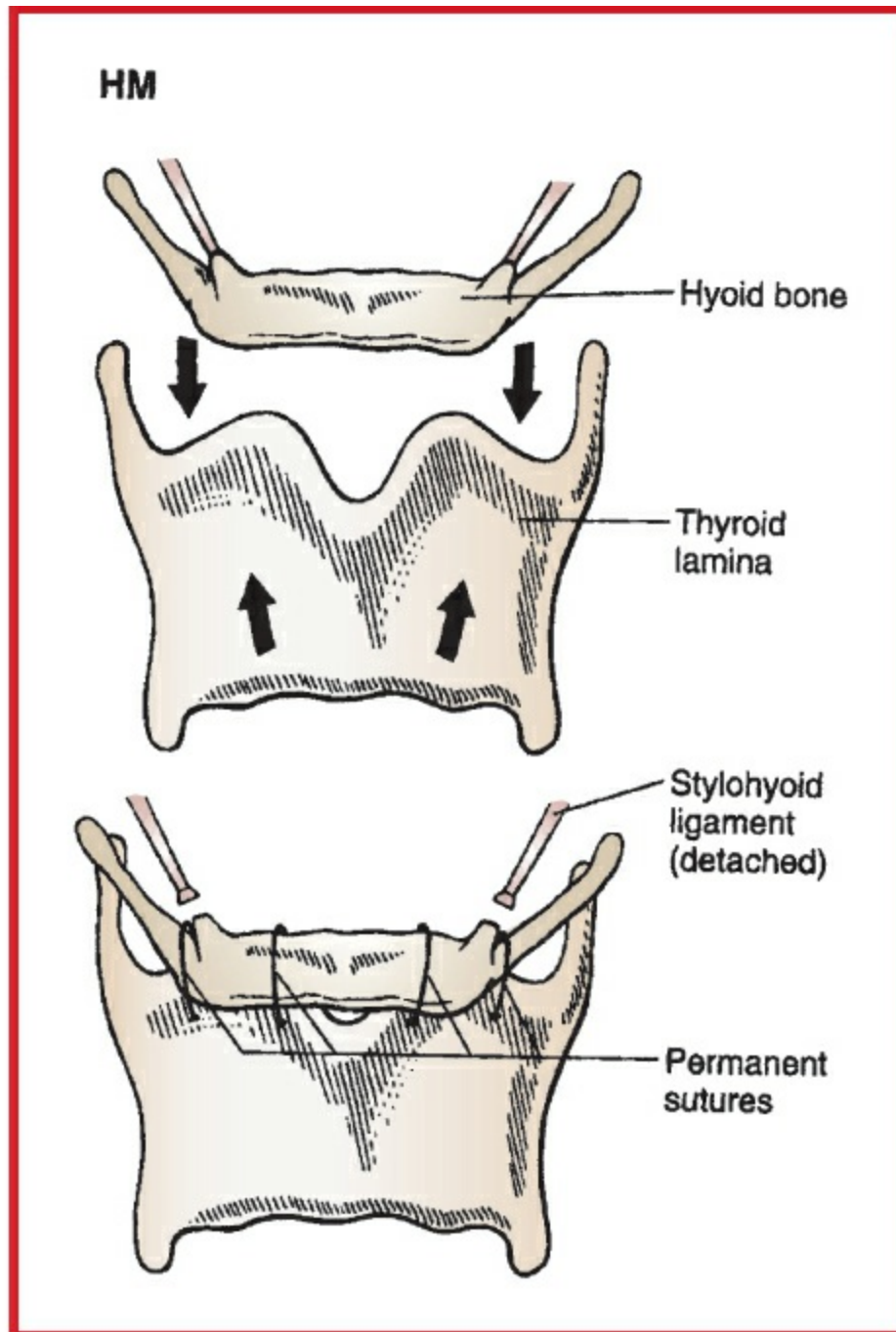


Figure 3.25 The hyoid myotomy (HM) and suspension technique. The hyoid is advanced over the thyroid lamina and immobilized.

Tracheotomy is a procedure performed to bypass pharyngeal obstruction. It is usually reserved for those individuals with severe OSA and may be used to secure the airway before other procedures. Preop evaluation, including fiberoptic examination, will help identify those individuals whose airways are so compromised that the tracheotomy should be done with the patient awake and under local infiltration anesthetic. A horizontal cervical incision is performed midway between the manubrium and the cricoid cartilage. Dissection is carried out in the midline down to the trachea, frequently

transecting the thyroid gland; then, an opening in the superior trachea allows placement of a tracheotomy tube.

Radiofrequency (RF) probes use heat to cause reduction and tightening of tissue. This may be used to enable the airway at the level of the nose (by reduction of the turbinates), the palate, or the base of the tongue. Whereas the nasal and palatal procedures are usually performed as outpatient under local anesthesia, the initial treatment of the base of the tongue commonly is performed in the OR, either alone or in conjunction with other airway procedures. The area of the tongue just anterior to the circumvallate papillae is infiltrated with local anesthetic. A needle-like RF probe is then used to heat the tissue. There is usually little or no immediate edema, although the surgeon may admit the patient overnight for airway observation.

Usual preop diagnosis

Sleep-disordered breathing, including upper airway resistance syndrome and OSA syndrome

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

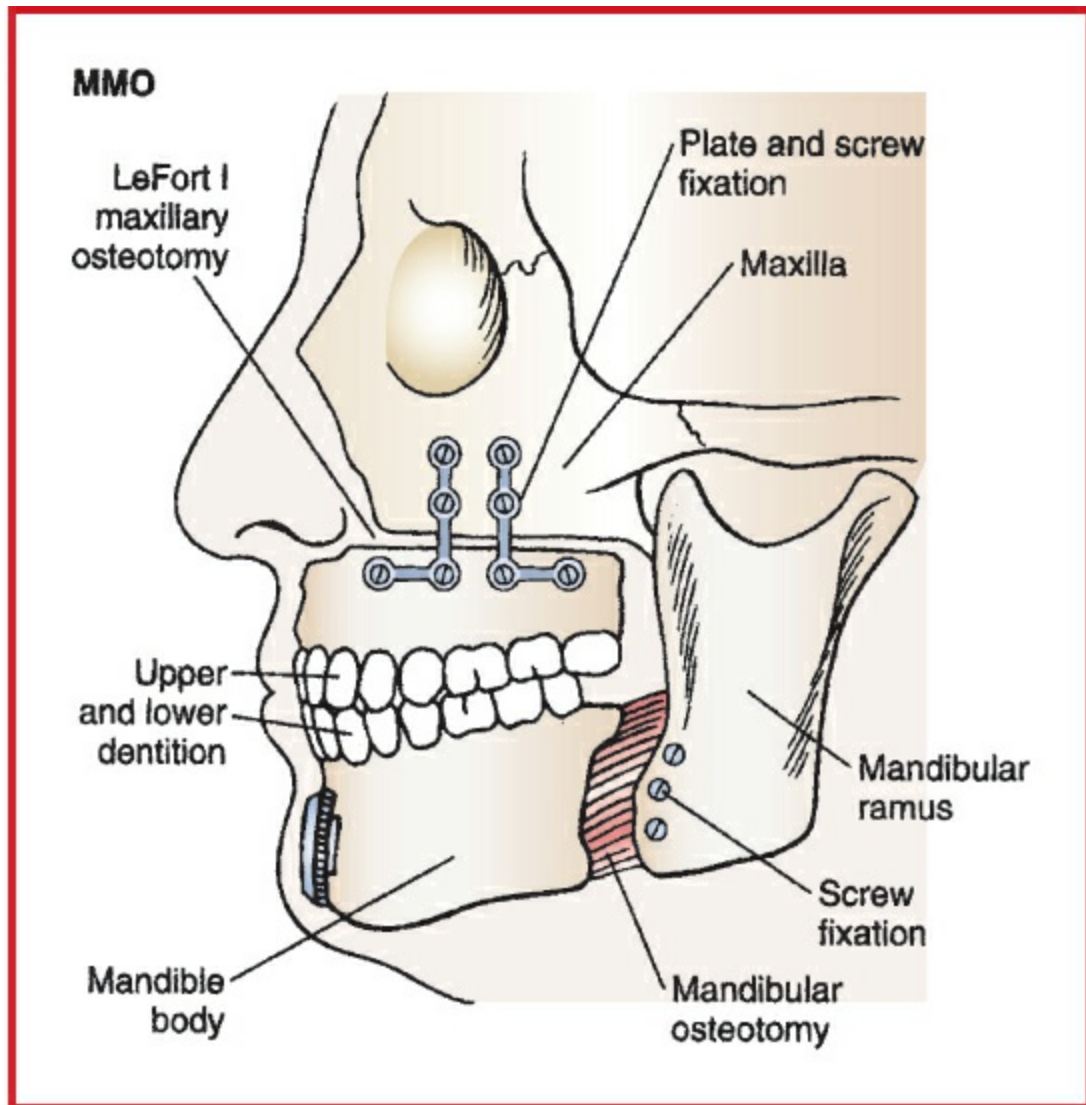


Figure 3.26 Maxillomandibular osteotomy (MMO) and advancement surgery technique.

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Intraoral (except HM—horizontal cervical above hyoid bone)
Special instrumentation	For osteotomy cases (MOGA, MMO), sagittal and reciprocating saws, drills
Unique considerations	When using a CO ₂ laser (LMG, LP), a laser-safe ETT or, more commonly, a tracheotomy tube is required. MMO requires a nasal intubation. Tracheotomy may be performed with patient sitting with iv sedation and/or local infiltration. Dexamethasone 8 mg iv

Antibiotics	Cefazolin 2 g iv
Surgical time	UPPP, UPF: 20–60 min UPPGP, LMG, LP: 1–3 h MOGA, HM: 30–60 min MMO: 3–5 h
EBL	UPPP, UPF, MOGA, HM: 0–100 mL UPPGP, LMG, LP: 50–250 mL MMO: 100–500 mL
Postop care	UPPP, UPF: PACU → ward Multiple procedures, MMO, labile HTN: ICU → ward
Mortality	Rare
Morbidity	Paresthesias: 10% HTN: 5% Wound dehiscence: 5% Bleeding: 1–2% Infection: 1–2% Hematoma/seroma: 1–2% Upper airway obstruction: 1%
Pain score	UPPP, UPPGP, LMG, LP: 8–10 UPF, MOGA, HM, MMO: 6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Children, adolescents, adults
Male:Female	3–10:1
Incidence	4–10% of adult males; 2–4% of adult females
Etiology	Upper airway collapse by frank obstruction or muscular relaxation
Associated conditions	Systemic and pulmonary HTN; CAD; cardiac arrhythmias; GERD; depression; obesity; polycythemia

ANESTHETIC CONSIDERATIONS

Vladimir Nekhendzy and Tiffany Cheng

 PREOPERATIVE

Sleep-disordered breathing is extremely common, affecting nearly 20% of adult population, with 7–8% suffering from moderate-to-severe OSA. It is estimated that over 20% of adult surgical population suffer from OSA, where it remains undiagnosed in the majority (70%) of patients. OSA is characterized by periods of intermittent upper airway occlusion during sleep, leading to repeated episodes of hypoxemia/hypercapnia, sympathetic activation, and sleep fragmentation. OSA adversely affects a variety of organ systems, with widespread implications for the surgical patient. The presence and severity of OSA are strongly correlated with obesity and metabolic syndrome, systemic and pulmonary HTN, CHF, arrhythmias, heart attacks, and stroke. The mainstay of treatment is continuous positive airway pressure (CPAP), which acts to stent open collapsible airways, and the CPAP treatment must be continued in the perioperative setting, especially in patients with severe OSA.

When screening for the presence of OSA during preop office visit, a physician may wish to use the highly sensitive and validated STOP-BANG questionnaire (Table 3.4). The patients with either known or presumed OSA should undergo outpatient surgery only if their comorbidities are optimized, if postop pain is going to be mild and can be predominantly managed with nonopioid analgesics, and if (for CPAP users) the patients can continue to use CPAP postop. For practical purposes, patients with severe OSA (ASA physical status III) should not routinely undergo office-based surgery under GA or sedation. Postop, the ASA recommends extended monitoring of outpatients with OSA for an additional 3 h.

Other ENT pathology (e.g., septal deviation, laryngotracheomalacia, tonsil and adenoid hypertrophy) may be present in OSA patients. Intraop, the OSA patients demonstrate enhanced sensitivity to opioids and benzodiazepines, which may lead to rapidly developing respiratory depression, upper airway collapse, and airway obstruction. Conscious sedation of these patients should be performed with extreme caution. A multimodal approach that minimizes narcotic medications (e.g., NSAIDs acetaminophen, cyclooxygenase-2–specific inhibitors) should be used perioperatively whenever feasible.

Table 3.4 The STOP-BANG Questionnaire for Screening for the Risk of Obstructive Sleep Apnea (OSA)	
Snore loudly?	BMI > 35 kg/m²?
Tired during the day?	Age > 50 yr?

Observed cessation of breathing during sleep?	Neck circumference > 40 cm?
Pressure (high blood pressure)?	Gender male?

Answering “Yes” to three or more questions confers a high risk of OSA. The higher the cumulative risk score, the greater the possibility of severe OSA.

Respiratory	OSA patients may present with difficult airway problems, related to both mask ventilation and tracheal intubation. The awake versus asleep approach to tracheal intubation should be guided by careful airway evaluation with special attention to the presence of the predictors of difficult mask ventilation (DMV) and their association with difficult intubation (see Introduction, Anesthesiologist’s Perspective, above). Awake intubation is rarely required but should be considered if the predictors of DMV coincide with factors predicting difficult intubation. Mask ventilation problems encountered in OSA patients are most commonly due to the pharyngeal airway obstructing at the level of soft palate and base of the tongue. This obstruction can usually be managed with supraglottic airway, which can also serve as a conduit for a subsequent tracheal intubation, either blind (e.g., intubating LMA) or fiberoptic guided. Discussion with the surgeon of the results of preop nasopharyngoscopy is useful. COPD/asthma often coexists and has a synergistic effect on worsening OSA. In advanced cases, chronic OSA → hypoxia/hypercarbia → pulmonary HTN → right heart failure. Tests: CXR, ABG, and PFT as indicated from H&P
Cardiovascular	Patients are at ↑ risk for systemic and pulmonary HTN (Sx: loud P₂, clubbing, ↑ JVD, cyanosis, RVH, right-axis deviation), CHF, cardiac arrhythmias (nonsustained VT, second-degree block, sinus arrest), MI, and stroke. Patients may have dyspnea on exertion and at rest, making routine assessment of cardiovascular function and reserve difficult.

	Tests: ECG; stress ECHO if abnormal ECG; others as indicated from H&P
Neurologic	A polysomnogram is typically performed preop, reporting the number of apneas (cessation or near cessation of airflow for 10 sec) and hypopneas (30% reduction in airflow over 10 sec with 3% hemoglobin desaturation). The apnea hypopnea index (AHI) is the number of apneas and hypopneas per hour of sleep. AHI of <5 is normal, 5–15 indicates mild OSA, 15–30 indicates moderate OSA, and >30 indicates severe OSA. OSA patients exhibit various degrees of daytime somnolence and may be chronically fatigued and irritable due to sleep fragmentation. Daytime sleepiness also may be associated with hypothyroidism or anemia, which should be ruled out. The impaired sleep/arousal cycle leads to autonomic hyperactivity; peripheral neuropathy and gastroparesis due to obesity may occur. An exaggerated response to sedative medications is frequently observed. Tests: As indicated by H&P
Hematologic	Chronic hypoxia → polycythemia (Sx: clubbing, cyanosis) → ↑ risk of CVA. Hypercoagulability may be present due to chronic oxidative stress that leads to associated systemic inflammation and endothelial dysfunction. Test: Hct, coagulation studies as required
Gastrointestinal and Renal	This patient population has a higher incidence of GERD and hiatal hernia; thus, full-stomach precautions may be required. Renal function may be impaired, likely due to chronic sympathetic activation.
Laboratory	Tests: As indicated from H&P. Impaired glucose control is often present. Elevated plasma bicarb levels point to coexisting obstructive respiratory disease (asthma, COPD) and/or CO₂ retention.
Premedication	Typically, sedative medications should be kept to a

minimum or avoided in this patient population. If sedative premedication is given, the anesthesiologist should remain with the patient and be prepared to manage the airway. Occasionally, OSA patients may have been prescribed stimulants (i.e., amphetamine) to combat daytime somnolence. These patients may not respond to indirect sympathomimetics (i.e., ephedrine) used to treat postinduction hypotension.

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 500–1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique: GETA.

Induction

Asleep airway management is safe in the majority of OSA patients (see Preoperative Respiratory Considerations, above). Supraglottic airways with gastric access (e.g., LMA ProSeal/LMA Supreme™) should be readily available for rescue ventilation (see above). Topical anesthesia of the upper airway may impair airway reflexes and airway muscle tone, therefore provoking airway collapse and possibly worsening OSA postop.

Orotracheal intubation is preferred, except in patients undergoing base of the tongue surgery (e.g., lingual tonsillectomy) or MMO, where nasotracheal intubation is required (see Anesthetic Considerations for TORS, [p. 204](#)). Preoxygenation with CPAP in 25° head-up position for 3–5 min will prolong safe apnea time. OSA patients with severe GERD should receive aspiration prophylaxis, and RSI should be performed, when indicated.

Standard induction with propofol and an opioid (e.g., fentanyl) is safe, followed by administration of an

	intermediate-acting neuromuscular blocker (e.g., rocuronium) for tracheal intubation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Maintenance using TIVA with propofol (80–180 mcg/kg/min) and remifentanyl (0.05–0.3 mcg/kg/min), supplemented with iv boluses of fentanyl (usually not exceeding 2–4 mcg/kg total dose), with or without N₂O, is useful for OSA surgery. It facilitates maintenance of moderate controlled hypotension and improves hemodynamic stability during even most stimulating parts of surgery. Propofol administration causes profound depression of airway reflexes, thereby facilitating smooth emergence from anesthesia. A “sandwich anesthetic” technique utilizing a combination of TIVA and a low-dose inhalational agent (e.g., sevoflurane 0.4–0.6%) can also be used to achieve the above-stated objectives. Hypertensive responses should be treated aggressively, especially during craniofacial advancement with down fracture of the pterygoid plate in MMO. Maintaining low SBP (below 100 mm Hg) and low MAP (60–70 mm Hg) is critical during this part of the procedure. Administration of “ad lib” iv boluses of remifentanyl 1–3 mcg/kg, supplemented with iv labetalol (5–10 mg increments) and hydralazine (5 mg increments), is usually effective. Esmolol and SNP infusions can be considered, but are usually not necessary. Administration of a dexmedetomidine infusion should be considered. A typical loading dose is 0.5–1.0 mcg/kg iv bolus over 10 min, followed by a continuous infusion of 0.2–1.0 mcg/kg/hr. Dexamethasone 0.1–0.15 mg/kg is recommended to decrease postop airway edema.
Emergence	The patient must remain intubated until sufficiently

awake to respond to verbal commands. Neuromuscular blockade should be fully reversed, and the return of protective airway reflexes (e.g., swallowing) should be observed. The patient's stomach should be suctioned to remove swallowed blood. Premature extubation can cause complete loss of airway (2° glossopalatal obstruction) or laryngospasm. With extensive pharyngeal and base of the tongue work, significant upper airway edema must be anticipated, and extubation over a tube changer may be appropriate in selected cases. An insertion of a nasopharyngeal airway before emergence can be recommended in patients who had extensive surgery or had severe OSA preop. Anticipate pain and ↑ BP during emergence, which will continue into first postop day. Narcotic analgesics (e.g., morphine 0.1–0.15 mg/kg) should be administered before emergence. Antihypertensive agents usually required to prevent rebound hypertension and associated postop bleeding, particularly from osteotomy sites. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements

IV: 16–18 ga × 1 BSS @ 2–3 mL/kg/h	Minimal-to-little blood loss with all except MMO. EBL with MMO is 100–500 mL. Attempt to minimize fluids to reduce severity of postop airway edema (<1000 mL of NS/LR for all except MMO; <2000 mL for MMO). See Goal-Directed Fluid Management, Appendix K, p. K-4.
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Monitoring

Standard monitors	Arterial line also
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	Arterial line (MMO and ICU admits)	suggested for postop BP control in all patients with preexisting HTN and in any patient who displays labile BP intraop. Control of BP in the postop period is an extremely high priority.
Positioning	✓ and pad pressure points ✓ eyes	Occasional requests for minimal Trendelenburg or semisitting positioning
Complications	ETT damage Airway Hemorrhage	Airway problems and respiratory depression are the most common complications (hypercapnia, pulmonary edema, reintubation, bronchospasm, and laryngospasm). In addition, OSA patients are at increased risk for CHF, MI, arrhythmia, CVA, and delirium. ETT may become dislodged or kinked during surgical manipulations.
	Dysrhythmias	Vagally mediated severe bradycardia/asystole may occur with maxillary or mandibular advancement during MMO.

POSTOPERATIVE

Complications	Airway compromise Airway obstruction	Airway compromise 2° hematoma, edema, excessive sedation, or
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		underlying disease. Consider observation in ICU for patients having multiple procedures, MMO, or labile HTN.
	HTN Aspiration	Postop HTN is extremely common and can contribute significantly to likelihood of postop airway obstruction 2° hematoma.
	Infections	The risk of wound infection is increased postop in OSA patients.
Multimodal pain management	Minimize iv narcotics	Avoid excessive postop sedation. UPPP patients, in particular, should be warned preop of anticipated significant postop discomfort. CPAP should be continued throughout the entire postop period. Consider BiPAP if noncompliant with CPAP.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. American Society of Anesthesiologists Task Force on Perioperative Management of patients with obstructive sleep apnea: Practice guidelines for the perioperative management of patients with obstructive sleep apnea: an updated report by the American Society of Anesthesiologists Task Force on Perioperative Management of patients with obstructive sleep apnea. *Anesthesiology* 2014; 120(2):268–86.
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sleep apnea. *Chest* 2010; 138:1489–98.

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7. Hillman DR, Platt PR, Eastwood PR: Anesthesia, sleep, and upper airway collapsibility. *Anesthesiol Clin* 2010; 28:443–55.
8. Isono S: Obstructive sleep apnea of obese adults: pathophysiology and perioperative airway management. *Anesthesiology* 2009; 110:908–21.
9. Joshi GP, Ankichetty SP, Gan TJ, et al: Society for ambulatory anesthesia consensus statement on preoperative selection of adult patients with obstructive sleep apnea scheduled for ambulatory surgery. *Anesth Analg* 2012; 115:1060–8.
10. Kheterpal S, Han R, Tremper KK, et al: Incidence and predictors of difficult and impossible mask ventilation. *Anesthesiology* 2006; 105:885–91.
11. Kheterpal S, Martin L, Shanks AM, et al: Prediction and outcomes of impossible mask ventilation: a review of 50,000 anesthetics. *Anesthesiology* 2009; 110:891–7.
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ANESTHESIA FOR DRUG-INDUCED SLEEP ENDOSCOPY

PREOPERATIVE

General

Patients with OSA may undergo drug-induced sleep endoscopy (DISE) to evaluate the upper airway collapsibility pattern in an attempt to stratify and personalize the required surgical intervention.

Preop considerations for DISE are typical for patients with OSA, with particular attention to upper airway exam and careful review of comorbidities to determine their ability tolerate periods of desaturations.

Airway

A thorough airway examination is essential, as no airway supporting maneuvers are usually made, until the patient desaturates significantly ($\text{SpO}_2 < 85\%$).

	These patients should be carefully assessed preop for the predictors of difficult mask ventilation or difficult intubation (see Introduction). Rescue maneuvers to relieve upper airway obstruction during DISE must be performed gently and expeditiously.
Respiratory	These patients may have ↓ respiratory reserve due to ↓ FRC and ↑ O ₂ consumption for patients with significant coexisting obesity. Evaluation for underlying pulmonary disease that may further compromise oxygen exchange should be undertaken.
Cardiovascular	Careful evaluation for underlying cardiac is essential given that the common comorbidities in patients with OSA (↑ age, male, HTN, obesity, smoking history, etc.) are also notable cardiac risk factors. Tests: ECG; other tests as indicated from H&P
Premedication	Sedative premedication and iv opioids should be avoided in patients undergoing DISE to optimize visualization of dynamic changes in the upper airway. An anticholinergic agent (e.g., glycopyrrolate 0.1–0.2 mg iv) should be considered preop for baseline HR < 50 bpm, if dexmedetomidine is used for DISE.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Most commonly, DISE using a flexible fiberoptic endoscope is performed under MAC allowing the surgeon to visualize and identify the sites of dynamic upper airway obstruction. The goal of anesthesia in DISE is to utilize sedative-hypnotics to produce loss of consciousness and any associated obstruction in patients with OSA while permitting spontaneous ventilation and avoiding prolonged periods of oxygen desaturation.

Currently, no consensus exists for the ideal sedative agent or agents to achieve adequate

sedation for DISE. The three agents studied for DISE include midazolam, propofol, and dexmedetomidine. The drawbacks of midazolam and propofol infusions include their respiratory depressant effects and dose-dependent increases in airway collapsibility (particularly with propofol), thereby increasing the possibility of a false-positive study. IV propofol has a propensity to cause anteroposterior upper airway collapse. As a result, we prefer iv dexmedetomidine as the primary DISE agent.

Induction	IV dexmedetomidine (1.5 mcg/kg iv bolus over 10 min) or propofol (in a ramp-up function) is used most often, with the goal of gradually inducing sleep and airway obstruction. Some patients may require additional boluses of 0.25 mcg/kg iv dexmedetomidine or propofol (10–20 mg iv) to achieve adequate sedation/obstruction.	
ERAS	Verify preincision antibiotics.	
Maintenance	Dexmedetomidine at 1.2–1.5 mcg/kg/hr iv infusion. Maintain good communication with otolaryngologist during procedure regarding intervention for low SpO ₂ levels. Be prepared to intervene with jaw-thrust/chin-lift maneuvers, supplemental O ₂ , or advanced airway intervention (e.g., LMA or ETT) if SpO ₂ falls below 85%. Judicious crystalloid loading intraop 10 cc/kg, to minimize incidence of postop orthostatic hypotension	
Emergence	Monitor spontaneous ventilation and evaluate degrees of upper airway obstruction following the endoscopy. Can apply supplementary O ₂ , reposition head and neck, perform jaw thrust/chin lift, and/or place oral/nasal airways to maintain SpO ₂ > 95%.	
Fluid Requirements	IV: 20 ga × 1 BSS @ 2–3 mL/kg/h	10 mL/kg for the procedure is usual.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	The procedure is usually performed on the gurney. Patients should be positioned comfortably in

		a way that will allow for comfortable light sleep while supine. Checking padding of pressure points.
Complications	Upper airway obstruction	Apply jaw-thrust/chin-lift, supplemental O ₂ , or advanced airway maneuvers if SpO ₂ falls below 85% or when procedure is completed.
	Orthostatic hypotension	Postop; particularly following the use of iv dexmedetomidine. PRN iv crystalloid bolus in PACU, as required. Assess for postop orthostatic hypotension prior to ambulation.

Suggested Readings

1. Atkins JH, Mandel JE: Drug-induced sleep endoscopy: from obscure technique to diagnostic tool for assessment of obstructive sleep apnea for surgical interventions. *Curr Opin Anaesthesiol* 2018; 31(1):120–6.
2. Atkins JH, Mandel JE, Rosanova G: Safety and efficacy of drug-induced sleep endoscopy using a probability ramp propofol infusion system in patients with severe obstructive sleep apnea. *Anesth Analg* 2014; 119(4):805–10.
3. Capasso R, Rosa T, Tsou DY, et al: Variable findings for drug-induced sleep endoscopy in obstructive sleep apnea with propofol versus dexmedetomidine. *Otolaryngol Head Neck Surg* 2016; 154(4):765–70.
4. Kuk TS, So E, Karm M, et al: Anesthetic management for simultaneous drug-induced sleep endoscopy and maxillomandibular advancement in a patient with obstructive sleep apnea. *J Dent Anesth Pain Med* 2017; 17(1):71–6.
5. Shteamer JW, Dedhia RC: Sedative choice in drug-induced sleep endoscopy: a neuropharmacology-based review. *Laryngoscope* 2017; 127(1):273–9.

Suggested Viewing

Links are available online to the following videos:

Segmental Anatomy (www.bronchoscopy.nl): <https://www.youtube.com/watch?v=ThYHLG50pH0>

Principles of Laryngoscopy (Fibreoptic): <https://www.youtube.com/watch?v=6MKKdu2bFDw>

TMJ Anatomy “The Yoint” Video: <https://www.youtube.com/watch?v=nH8KG3floBM>

Clinical Complications of Tracheostomy: <https://www.youtube.com/watch?v=UCYBGRj4Teo>

Transoral Robotic Surgery: Supraglottic Laryngectomy – Johns Hopkins Hospital
<https://www.youtube.com/watch?v=6q9q4umrBAI>

Endoscopic Septoplasty: <https://www.youtube.com/watch?v=bpSVrVL77kc&list=PL3F3EEEEA621>

Tonsillectomy: <https://www.youtube.com/watch?v=RTFUpoqhbHo>

Thyroidectomy: <https://www.youtube.com/watch?v=i1y1RVagkfs&t=5s>

Modified Radical Neck Dissection: <https://www.youtube.com/watch?v=ALcizhZln58>

Modified Radical Neck Dissection, part 1: <https://www.youtube.com/watch?v=2uilaptogdU>

Modified Radical Neck Dissection, part 2: <https://www.youtube.com/watch?v=pp6oqT2Pvr4>

Modified Radical Neck Dissection, part 3: https://www.youtube.com/watch?v=HkdiXwN_yR8

Endoscopic View of Zenker Diverticulum Repair: <https://www.youtube.com/watch?v=P79SsJxO2SA>

Endoscopic Sinus Surgery: <https://www.youtube.com/watch?v=NX9NjsvLJwE>

SECTION 4.0

DENTAL SURGERY

SURGEON

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TEMPOROMANDIBULAR JOINT ARTHROSCOPY/ARTHROTOMY/ARTHROPLASTY

SURGICAL CONSIDERATIONS

Description

Temporomandibular joint (TMJ) surgical procedures include both open and closed surgical techniques.

TMJ arthrotomy involves a preauricular, postauricular, or endaural incision to gain access to the joint compartment. It usually is performed for severe fibrous adhesion removal in the TMJ, bony or fibrous ankylosis, tumor resection, chronic dislocation, painful nonreducing disc dislocation, and severe osteoarthritis. **Open TMJ** surgery may range from discoplasty, discectomy, arthroplasty, and/or eminoplasty (reshaping of articular eminencia) to optimize the fit of the disc, condyle, and fossa to total joint replacement utilizing costochondral grafts or total joint metal implants. For the open treatment of condylar fractures, extraoral approaches (e.g., preauricular, retromandibular, and submandibular) are used. All extraoral approaches to the TMJ have the risks of facial nerve damage and the creation of visible scars. Due to those possible complications, endoscopically assisted transoral approaches for open reduction and miniplate fixation of condylar mandible fractures are being used more often. In any of these procedures, facial nerve monitoring is indicated.

TMJ arthroscopy is a minimally invasive technique that has reduced the need for open surgery of the TMJ. Arthroscopic TMJ surgery is indicated for treatment of internal derangements and intracapsular disorders. The major advantage is that it results in less periarticular tissue disruption and better preservation of vascular supply and lymphatic drainage of the joint. The procedure involves insertion of a TMJ miniscope through a preauricular puncture on the canthus-tragus line and insertion of an outflow needle. The joint compartment is continually lavaged with LR. A second cannula may be inserted. Arthroscopic TMJ procedures include lysis of adhesions and lavage, partial synovectomy, and abrasion arthroplasty. Sometimes a holmium:YAG laser is used to make intra-articular incisions anterior to displaced discs and to treat inflamed synovial tissue. Usually, at the end of the procedure, 2-mg dexamethasone is injected into the joint space. Injection of 2 mL 0.5% bupivacaine mixed with 1 mL sterile saline solution has been shown to significantly reduce postop pain. Arrhythmia, reflex bradycardia, and pulmonary edema have been reported as general complications in TMJ arthroscopy.

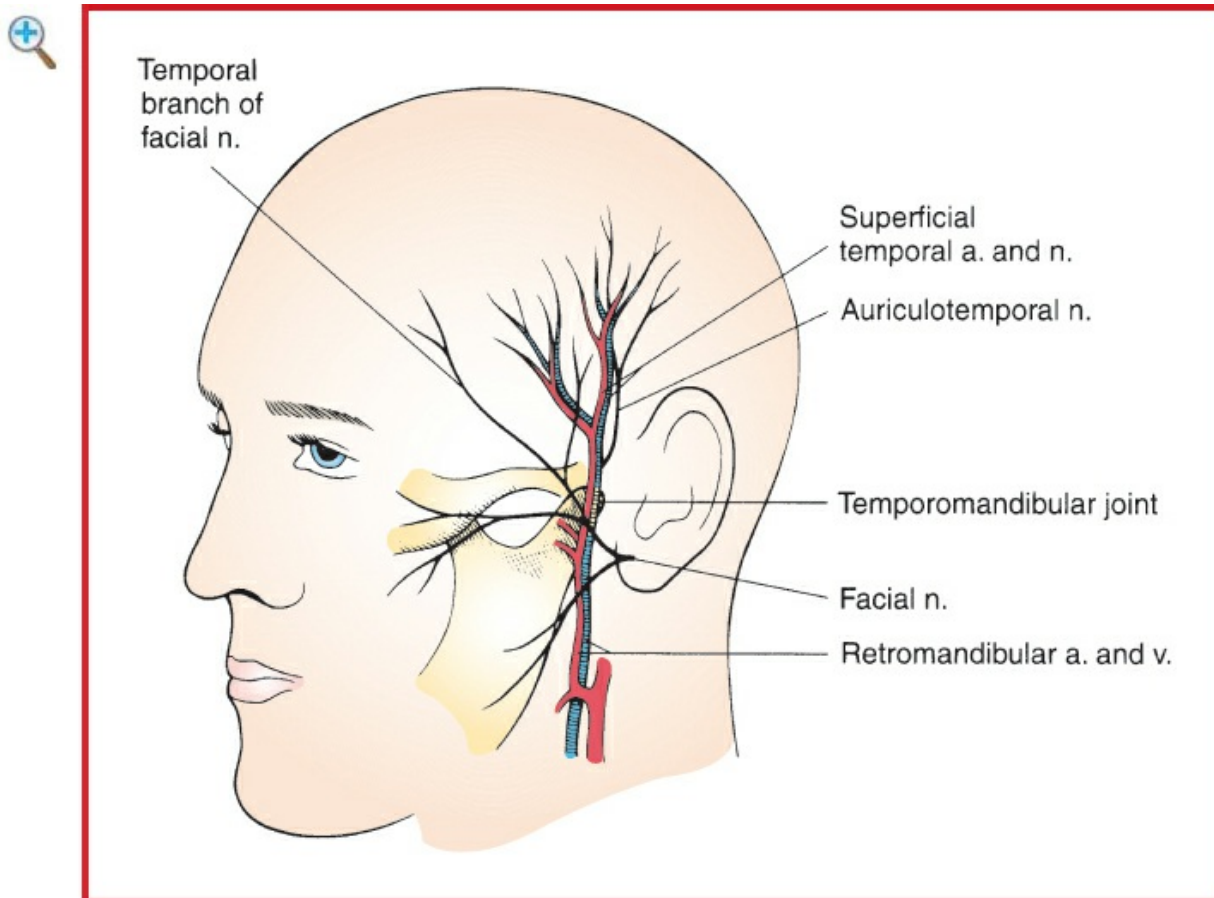


Figure 4.1 Anatomy for TMJ procedure.

Usual preop diagnosis

Internal derangement; subluxation; and ankylosis of TMJ

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Arthroscopy	Arthrotomy/Arthroplasty
Position	Supine	⇐
Incision	Preauricular	⇐
Special instrumentation	Arthroscope; laser	Power tools, endoscope, implant plates
Antibiotics	Cefazolin 2 g	⇐

Surgical time	0.5 h	1.5–3.5 h/side
EBL	Minimal	Minimal-moderate
Postop care	Outpatient procedure	24-h stay
Mortality	Minimal	⇐
Morbidity	VII nerve damage	⇐
	V nerve damage	⇐
	Hemorrhage	⇐
	Partial hearing loss	
	Ear fullness	
	Vertigo	
Pain score	5	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–40 yr
Male:Female	1:9
Incidence	20% of adult population suffers from TMJ dysfunction (TMJD)
Etiology	TMJD possibly 2° muscle spasm, bruxism, and osteoarthritis; idiopathic; trauma
Associated conditions	Psychiatric problems (typically depression); trismus; pain on opening the mouth; stress

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Dental/Oral Surgery.

Suggested Readings

1. Al-Ani Z, Gray R: TMD current concepts: 1. An update. *Dent Update* 2007; 34(5):278–80, 282–4, 287–8.
2. Furst IM, Kryshchak B, Weinberg S: The use of intra-articular opioids and bupivacaine for analgesia following temporomandibular joint arthroscopy: a prospective, randomized trial. *J Oral Maxillofac Surg* 2001; 59(9):979–83.
3. Hoffman D, Puig L: Complications of TMJ surgery. *Oral Maxillofac Surg Clin North Am* 2015; 27(1):109–24.
4. Indresano AT: Surgical arthroscopy as the preferred treatment for internal

- derangements of the temporomandibular joint. J Oral Maxillofac Surg 2001; 59(3):308–12.
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 7. Rajapakse S, Ahmed N, Sidebottom AJ: Current thinking about the management of dysfunction of the temporomandibular joint: a review. Br J Oral Maxillofac Surg 2017; 55(4):351–6.
 8. Rigon M, Pereira LM, Bortoluzzi MC, et al: Arthroscopy for temporomandibular disorders. Cochrane Database Syst Rev 2011; (5):CD006385.
 9. Tsuyama M, Kondoh T, Seto K, et al: Complications of temporomandibular joint arthroscopy: a retrospective analysis of 301 lysis and lavage procedures performed using the triangulation technique. J Oral Maxillofac Surg 2000; 58(5):500–5.

ORAL SURGERY

SURGICAL CONSIDERATIONS

Description

The most common surgeries of the oral cavity are third-molar removal, surgical extractions, apicoectomies, orthodontic exposures of teeth, osseointegrated implants, bone grafting, treatment of oral pathologic conditions, and preprosthetic surgery. **Surgical extractions** of teeth involve intraoral exposure of the roots through a mucosal incision and removal of the overlying bone with a surgical drill. Risks associated with removal of teeth in the mandible are damage to the inferior alveolar nerve (anesthetic numb lip), lingual nerve (anesthetic numb tongue), and, rarely, mandibular fracture. In the posterior maxilla, oroantral fistulas can occur and are closed with a mucoperiosteal flap. **Exposure of teeth** for orthodontic therapy involves creation of a mucoperiosteal flap and attachment of a bracket with a small gold chain on which the orthodontist can pull to integrate the tooth into the dental arch. **Grafting** to the maxilla and mandible is done for augmentation of the atrophied alveolar ridge and the maxillary sinus and in cases of cleft lip and palate, and can consist of bone and/or other materials to induce bone formation. A second team usually harvests the bone at the same time. Possible extraoral harvesting sites include the anterior or posterior iliac crest, the tibia, and the skull. **Preprosthetic surgery** of the oral soft tissue in preparation for dentures has been replaced largely by insertion of **osseointegrated implants** for retention of individual

teeth and dentures. Surgical treatment of **oral pathology** can range from removal of dentigerous cysts, with and without bone graft, to laser or surgical removal of mucosal lesions.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES				
	Dental Surgery	Dental Implants	Oral Pathology	Grafting
Position	Supine	⇐	⇐	Supine or prone
Incision	Intraoral	⇐	⇐	Intraoral and donor site
Special instrumentation	Surgical drill	Implant drill and kit	Surgical drill, laser	Special graft materials
Antibiotics	None	Penicillin 1 g	Cefazolin 2 g iv	⇐
Unique considerations	Nasotracheal intubation ±Throat pack	⇐ ⇐	⇐ ⇐	⇐ ⇐
Surgical time	0.5 h/tooth	0.5 h/implant	1–3 h	2–3 h
EBL	Minimal	⇐	⇐	Moderate
Postop care	Outpatient	⇐	⇐ or 24-h stay	24-h stay
Mortality	Minimal	⇐	⇐	⇐
Morbidity	V nerve damage Aspiration of dental debris	⇐ ⇐	⇐	Hemorrhage
Pain score	3	2	3–5	5

■ PATIENT POPULATION CHARACTERISTICS				
Age range	12–40 yr	>16 yr	All ages	>8 yr
Male:Female	1:1	⇐	⇐	⇐
Etiology	Idiopathic	Tooth loss	Various	⇐
Associated conditions	Craniofacial syndromes			

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Dental/Oral Surgery.

Suggested Readings

1. Bataineh AB: Sensory nerve impairment following mandibular third molar surgery. J Oral Maxillofac Surg 2001; 59(9):1012–7.
2. Zou L, He D, Ellis E: A comparison of clinical follow-up of different total temporomandibular joint replacement prostheses: a systematic review and meta-analysis. J Oral Maxillofac Surg 2018; 76(2):294–303.

Suggested Viewing

Links are available online to the following video:

TMJ Anatomy “The Joint” Video: <https://www.youtube.com/watch?v=nH8KG3fIoBM>

RESTORATIVE DENTISTRY

SURGICAL CONSIDERATIONS

Description

Multiple dental restorative procedures are performed under GA when there are rampant caries, and an extensive amount of dental work must be performed at one time. The second most common indication for GA is for procedures that need to be performed on mentally retarded patients who are not candidates for a local anesthetic. The actual amount of restorative dentistry is quite variable, depending on the individual case; thus, surgical time can be quite variable. Generally, blood loss is not a problem.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine
Incision	Intraoral
Special instrumentation	Dental armamentarium
Unique considerations	Nasal intubation; throat pack

Antibiotics	Penicillin × 5 d po
Surgical time	0.5–3 h
EBL	Minimal
Postop care	PACU → home
Mortality	Minimal
Morbidity	Pain Aspiration of dental debris Swelling
Pain score	1–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	2 yr to adult
Male:Female	1:1
Incidence	Unknown
Etiology	Idiopathic or congenital anomalies
Associated conditions	Mental retardation (majority), Down syndrome, seizures

ANESTHETIC CONSIDERATIONS FOR DENTAL/ORAL SURGERY

PREOPERATIVE

Most patients presenting for dental or oral surgery usually will require only local anesthesia provided by the dentist/oral surgeon. Deep sedation or GA may be required, however, for several unique patient groups: (a) young children (some with systemic diseases such as CHD and hemophilia), (b) the mentally disabled, (c) those with poorly controlled seizure disorders, (d) those presenting for TMJ procedures, and (e) those with an oral septic focus, who may be quite ill. If the patient does not fall into one of these readily identifiable categories, the reasons for GA should be ascertained. An LMA with a flexible wire-reinforced airway tube (LMA Flexible) has been used successfully for a variety of oral surgical procedures. The use of an LMA should be discussed with the surgeon in advance because its presence may interfere with the planned procedure.

Airway	Patients presenting for TMJ procedures may have problems with mouth opening (2° pain, trismus, and arthritis), making airway examination difficult. Mouth opening may not improve with GA and muscle relaxation. Nasotracheal intubation using FOL (done awake in patients with difficult airways) should be planned. Examine nares for patency; check for loose teeth.
Respiratory	Surgery should be postponed (at least 2 wk) in patients presenting with Sx of acute RTI (fever, coughing, purulent sputum, etc.). Sx of chronic respiratory disease should be sought and treated before surgery. LMA use has been reported to decrease respiratory complications in children with upper RTIs. Tests: As indicated from H&P
Cardiovascular	Patients with dysrhythmias may be sensitive to the epinephrine used in local anesthetic solutions administered intraop. As with other types of elective surgery, preexisting cardiovascular problems should be treated before surgery. Prophylactic antibiotics for endocarditis are not required in most patients; exceptions include patients with prosthetic valves, congenital heart disease, h/o infective endocarditis, or heart transplant. Endocarditis prophylaxis: amoxicillin 2 g po or cefazolin 2 g iv 30–60 min before procedure. Tests: As indicated from H&P
Neurologic	Patients with seizure disorders should be on optimal medical therapy before surgery. Discuss precipitating factors and prodromal Sx with the patient. Tests: ✓ therapeutic levels of anticonvulsant (e.g., levetiracetam = 12–46 mcg/mL; phenytoin = 10–20 mcg/mL; carbamazepine = 3–12 mcg/mL; phenobarbital = 10–40 mcg/mL)
Musculoskeletal	In addition to TMJ problems, rheumatoid arthritis is

	associated with cricoarytenoid joint immobility and cervical spine immobility/instability that may complicate intubation. Patients with prosthetic joints do not require special antibiotic prophylaxis. Other tests as indicated from H&P
Laboratory	
Premedication	Standard premedication usually is appreciated, although, in patients with limited airway access, sedation may be inappropriate. If FOL is planned, pretreatment with an antisialagogue (e.g., glycopyrrolate 4 mcg/kg) is useful. Metoclopramide (e.g., 10–20 mg iv adult) will reduce the incidence of PONV 2° swallowed blood. Avoid long-lasting sedative medications.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen (adult: 1000 mg po; peds: 10–15 mg/kg iv or po as an elixir).

INTRAOPERATIVE

Anesthetic technique

GETA (although LMAs have been used successfully for many procedures). Typically a nasotracheal intubation is requested, using an ETT 0.5–1 mm smaller than for oral intubation. In patients with difficult airways, an awake nasal FOL is indicated. For conscious sedation, dexmedetomidine (e.g., 0.5 mcg/kg loading dose over 10 min + 0.5 mcg/kg/h infusion) in combination with midazolam (e.g., 0.03 mg/kg) ± propofol (e.g., 25 mcg/kg/min) has the advantage of minimal respiratory depression.

Induction	In patients with normal airways, a standard induction with nasal intubation is appropriate. Following loss of consciousness, topical intranasal cocaine may be applied (4% on pledgets, 4 mL maximum) to shrink the nasal mucosa and for vasoconstriction. Side effects are rare but may include ↑ BP, ↑ or ↓ HR, dysrhythmias, and Sz. Other topical vasoconstrictors (e.g., 0.05% oxymetazoline) may be used; however, they are also associated with cardiovascular side effects. The well-lubricated ETT is passed through
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	the nose into the trachea, either blindly or assisted by Magill forceps under direct laryngoscopy. The ETT is often sewn to the nasal septum. The successful use of a flexible reinforced LMA in both adult and pediatric dental patients has been reported. Claimed advantages include no risk of epistaxis and no need for a throat pack, laryngoscopy, or muscle relaxation. Disadvantages include interference with the procedure, throat trauma (e.g., swelling of the epiglottis), and aspiration risk.
ERAS	Verify preincision antibiotics. Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	NB: throat packs must be removed prior to extubation. An LMA is typically removed after the patient is awake and able to follow commands.
Blood and fluid requirements	IV: 18 ga × 1 NS/LR @ 4–6 mL/kg/h See Zero Balance Fluid Management, Appendix K, p. K-2 .
Monitoring	Standard monitors
Positioning	✓ pad pressure points ✓ eyes

▼ POSTOPERATIVE

These patients may swallow blood, with consequent PONV. Rx: metoclopramide 10–20 mg iv ± ondansetron 4 mg iv.

Complications	Airway obstruction 2° retained throat pack PONV	Always check for retained throat pack in patients exhibiting symptoms of airway obstruction. Rx: ondansetron 4 mg iv
Multimodal pain management	Oral analgesics	Use nonopioid analgesics. Backup with po opioid

		analgesics
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition and mobilization.

Suggested Readings for Dental/Oral Surgery

1. Al-Jandan BA, Syed FA, Zeidan A, et al: Pharyngeal pack placement in minor oral surgery: a prospective, randomized, controlled study. *Ear Nose Throat J* 2018; 97(3):E18–21.
2. Atan S, Ashley P, Gilthroe MS, et al: Morbidity following dental treatment of children under intubation general anaesthesia in a day-stay unit. *Int J Paediatr Dent* 2004; 14(1):9–16.
3. Cummings DR, Yamashita DD, McAndrews JP: Complications of local anesthesia used in oral and maxillofacial surgery. *Oral Maxillofac Surg Clin North Am* 2011; 23(3):369–77.
4. Hanzawa A, Handa T, Kohkita Y, et al: A comparative study of oral analgesics for postoperative pain after minor oral surgery. *Anesth Prog* 2018; 65(1):24–9.
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6. Mayhew JF: Airway management for oral and maxillofacial surgery. *Int Anesthesiol Clin* 2003; 41(3):57–65.
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9. Smith MH, Lung KE: Nerve injuries after dental injection: a review of the literature. *J Can Dent Assoc* 2006; 72(6):559–64.
10. Todd DW: Anesthetic considerations for the obese and morbidly obese oral and maxillofacial surgery patient. *J Oral Maxillofac Surg* 2005; 63(9):1348–53.
11. Wakita R, Kohase H, Fukayama H: A comparison of dexmedetomidine sedation with and without midazolam for dental implant surgery. *Anesth Prog* 2012; 59(2):62–8.
12. Yasny JS: Perioperative dental considerations for the anesthesiologist. *Anesth Analg* 2009; 108(5):1564–73.
13. Yu SH, Beirne OR: Laryngeal mask airways have a lower risk of airway complications compared with endotracheal intubation: a systematic review. *J Oral Maxillofac Surg* 2010; 68(10):2359–76.